ESE 1205 Complete Solutions

Thermodynamics for Energy Engineering · KUET · past papers 2017, 2018, 2019, 2020, 2021, 2022, 2023 · answers from the course slides (Tahir Mahmud, Afsar, Sohag) and the Cengel and Moran-Shapiro textbooks they follow

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 5 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×7, 2022 ×6, 2021 ×5, 2020 ×4, 2019 ×3, 2018 ×2, 2017 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer for each block first (30 minutes), then follow this list. Psychrometry (B8) and gas mixtures (B7) are the biggest mark-earners in Part B and both have calculators; the Otto/Diesel numerical (B4) comes every year. In Part A the first law (A5, A6) and the second law (A7) are the safest. Entropy and exergy (A8, A9) are harder: do them last, using the calculators to practise.

Section A: Concepts, first law, second law, entropy, exergy

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A5 Zeroth and first law; energy balance for closed and open systems; boundary work	Very likely 100%	Medium	13%
A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs	Very likely 100%	Medium	24%
A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger)	Very likely 86%	Medium	37%
A3 Pure substance, phase rule, state postulate, T-v diagram, quality	Very likely 82%	Medium	45%

Block 3: very likely numericals · 2.0 h

Topic	Chance	Level	Covered
A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency	Very likely 100%	Hard	59%
A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency	Very likely 100%	Hard	72%

Block 4: likely · 1.6 h

Topic	Chance	Level	Covered
A1 Basic concepts: system, viewpoints, properties, equilibrium, process	Likely 79%	Easy	79%
✔ Pass line: 79% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.4 h)			
A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions	Likely 75%	Easy	86%
A4 Ideal gas and real gas: equations of state, compressibility factor	Likely 75%	Medium	96%

Only if time is left · 1.0 h

Topic	Chance	Level	Covered
A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson	Possible 46%	Hard	100%

Section B: Power cycles, refrigeration, gas mixtures, psychrometry

Block 1: very likely and easy · 0.8 h

Topic	Chance	Level	Covered
B6 Refrigeration: vapour compression and vapour absorption systems	Very likely 100%	Easy	9%
B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle	Very likely 100%	Easy	17%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B8 Psychrometry: moist-air properties, derivations, chart and numericals	Very likely 100%	Medium	38%
B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties	Very likely 100%	Medium	57%
B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals	Very likely 100%	Medium	73%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B3 Rankine cycle numericals (simple, actual, reheat, regenerative)	Very likely 96%	Hard	85%
✔ Pass line: 85% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.1 h)			

Block 4: likely · 1.5 h

Topic	Chance	Level	Covered
B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration	Likely 79%	Medium	95%
B5 Brayton (gas turbine) cycle	Likely 64%	Medium	100%

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2. **Section A: Concepts, first law, second law, entropy, exergy**: A5 Zeroth and first law; energy balance for closed and open systems; boundary work, A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs, A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger), A3 Pure substance, phase rule, state postulate, T-v diagram, quality, A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency, A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency, A1 Basic concepts: system, viewpoints, properties, equilibrium, process, A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions, A4 Ideal gas and real gas: equations of state, compressibility factor, A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson
3. **Section B: Power cycles, refrigeration, gas mixtures, psychrometry**: B6 Refrigeration: vapour compression and vapour absorption systems, B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle, B8 Psychrometry: moist-air properties, derivations, chart and numericals, B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties, B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals, B3 Rankine cycle numericals (simple, actual, reheat, regenerative), B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration, B5 Brayton (gas turbine) cycle

One-shot notes: ESE 1205 Thermodynamics

Basic words (A1)

System / surroundings / boundary. Closed (control mass: energy only), open (control volume: mass + energy), isolated (nothing). Intensive (T, p, v) vs extensive (m, V, U); specific = extensive/m.

Equilibrium = Mechanical + Thermal + Chemical + Phase. Quasi-equilibrium = infinitely slow, can be drawn as a line. Cycle: $\oint d(\text{property}) = 0$. Steady: nothing inside changes with time.

Energy, heat, work (A2)

Stored E = KE + PE + U (sensible, latent, chemical, nuclear). In transit: Q (because of ΔT) and W (anything else), plus energy with mass flow.

Q, W are path functions (δQ, area under p-V), boundary phenomena. Heat or work depends on where you draw the boundary (oven example).

Pure substance (A3)

Fixed uniform composition. Water + steam = pure; liquid air + gas air = not pure (O_2 condenses first).

Phase rule $F = C - \pi + 2$. State postulate: 2 independent intensive properties.

$x = (v - v_f)/(v_g - v_f)$; $v = v_f + x\, v_{fg}$ (same for u, h, s).

Rigid tank = constant v: vertical line on T-v; $Q = m(u_2 - u_1)$.

Gases (A4)

$Pv = RT$, $R = 8.314/M$. $Z = Pv/RT$; corresponding states: $Z = f(P_R, T_R)$. Ideal if $P \ll P_{cr}$ or $T > 2T_{cr}$.

vdW $(P + a/v^2)(v - b) = RT$, $a = 27R^2T_{cr}^2/64P_{cr}$, $b = RT_{cr}/8P_{cr}$. RK: 0.42748, 0.08664, √T. N_2 175 K: ideal 13.85, vdW 9.47, RK 10.09 MPa (true 10).

First law (A5)

0th law → T; 1st → U; 2nd → S. Cycle: $\oint \delta Q = \oint \delta W$. Closed: $Q - W = \Delta U + \Delta KE + \Delta PE$.

CV: $dE/dt = \dot{Q} - \dot{W} + \Sigma \dot{m}(h + V^2/2 + gz)_{in} - \Sigma \dot{m}(\ldots)_{out}$. SFEE: $q - w = \Delta h + \Delta V^2/2 + g\Delta z$. Flow work pv ⇒ h = u + pv.

W_b: pV^n → $(p_2V_2 - p_1V_1)/(1 - n)$; n = 1 → $p_1V_1 \ln(V_2/V_1)$; p const → $p\Delta V$.

Devices (A6)

Turbine $w = h_1 - h_2$; compressor/pump $w_{in} = h_2 - h_1$; pump $w = v\Delta p$; heat exchanger $\dot{m}_h\Delta h_h = \dot{m}_c\Delta h_c$; throttle $h_1 = h_2$; $\dot{m} = AV/v$. KE in kJ/kg = $V^2/2000$.

Key answers: compressor 119.5 kW; pump 25.8 kW, steam 0.0616 kg/s; 2-stage turbine 17.36 kg/s, 17.6 MW, 3.82 MW; turbine 2023: 872.5 kJ/kg, 5.73 kg/s.

Second law (A7)

KP: no cycle produces work from a single reservoir. Clausius: no cycle moves heat cold → hot with no other effect. Equivalent (KP violator + fridge = Clausius violator).

Carnot principles: $\eta_{irr} < \eta_{rev}$; all reversible engines equal. $(Q_L/Q_H)_{rev} = T_L/T_H$; $\eta_C = 1 - T_L/T_H$. Proof pattern: assume opposite → reverse the reversible one → one reservoir gives work.

Entropy (A8)

Clausius inequality $\oint \delta Q/T \le 0 \Rightarrow dS = (\delta Q/T)_{rev}$. $S_2 - S_1 = \int \delta Q/T + S_{gen}$; $\Delta S_{isolated} \ge 0$.

T ds = du + p dv; T ds = dh − v dp. Ideal gas: $\Delta s = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(v_2/v_1) + c_v \ln(p_2/p_1)$.

Solid/liquid: $c \ln(T_2/T_1)$.

CV: $dS/dt = \Sigma \dot{Q}/T + \Sigma \dot{m}s_{in} - \Sigma \dot{m}s_{out} + \dot{\sigma}$. $\eta_C = (h_{2s} - h_1)/(h_{2a} - h_1)$; $\eta_T = (h_1 - h_{2a})/(h_1 - h_{2s})$.

Exergy (A9) and relations (A10)

$X = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) + KE + PE$.

Flow $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$. $X_{dest} = T_0S_{gen} \ge 0$ (cannot create exergy). $W_u = W - p_0\Delta V$; $\eta_{II} = W_u/W_{rev}$.

Maxwell: $(\partial s/\partial v)_T = (\partial p/\partial T)_v$, $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$, $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$, $(\partial T/\partial p)_s = (\partial v/\partial s)_p$. $\beta = (1/v)(\partial v/\partial T)_p$, $\kappa = -(1/v)(\partial v/\partial p)_T$. Throttling cools only below max inversion T ($\mu_{JT} > 0$).

Cycles basics (B1)

Air-standard: air ideal gas in closed loop, reversible, combustion → heat in, exhaust → heat out; cold-air: c_p 1.005, c_v 0.718, γ 1.4.

Carnot vapour cycle impractical: T_H < 374 °C, wet turbine, two-phase compression. $\eta = 1 - T_0/T_m$, $T_m = (h_3 - h_2)/(s_3 - s_2)$.

Rankine (B2, B3)

1-2 pump, 2-3 boiler, 3-4 turbine, 4-1 condenser. $\eta = [(h_3 - h_4) - v_1\Delta p]/(h_3 - h_2)$. Improve: lower p_{cond}, superheat, higher p_{boiler}, reheat ($p_{RH} \approx$ ¼-⅕ p_B), regeneration.

Open FWH $y = (h_3 - h_2)/(h_6 - h_2)$; $\eta = 1 - (1 - y)(h_7 - h_1)/(h_5 - h_4)$.

Answers: 4 MPa/400 °C/27 °C: 38.2 % (actual 32.4 %); 3 MPa/355 °C/11 kPa: 33.2 %; reheat 15 MPa: p_{RH} 4 MPa, 45.0 %; FWH 600 °C: y 0.227, 46.3 %; 550 °C: y 0.234, 45.5 %.

IC engines (B4)

Otto $\eta = 1 - 1/r^{\gamma-1}$. Diesel $\eta = 1 - (1/r^{\gamma-1})(r_c{}^{\gamma} - 1)/[\gamma(r_c - 1)]$. Dual: $(r_p r_c{}^{\gamma} - 1)/[(r_p - 1) + \gamma r_p(r_c - 1)]$.

Same r: Otto > Dual > Diesel; same p_{max}: reverse.

Diesel uses higher r (air only, no knock). MEP = $w_{net}/(v_1 - v_2)$.

Answers: Otto r 4.89, 47 %; Diesel r 17: r_c 1.95, 867 kJ/kg, 62.6 %; r 16: 2.01, 884, 61.3 %, MEP 699 kPa; Dual 2019 57.3 %; 2018 62.7 %.

Brayton (B5) and refrigeration (B6)

$\eta = 1 - 1/r_p{}^{0.2857}$; $r_{bw} = w_c/w_t$. r_p 8: 543 K, 718 K, 44.8 %; with 80/85 %: 604 K, 805 K, 0.615, 27.4 %; r_p 5: 36.9 %, 176, 444.5 kJ/kg.

VCR: compressor → condenser → valve (h const) → evaporator; $COP = (h_1 - h_4)/(h_2 - h_1)$. VAR: absorber + pump + generator replace compressor; heat-operated, quiet, low COP. Ice plant: 48.5 kW, motor 21.5 kW.

Gas mixtures (B7)

$p_i = y_i p$ (Dalton), $V_i = y_i V$ (Amagat). $M_m = \Sigma y_i M_i = m/n$; $R_m = 8.314/M_m$; $c_p = \Sigma m f_i c_{p,i}$, $c_{p,i} = \gamma R/(\gamma - 1)$.

3 kg N_2 + 5 kg CO_2: y 0.485/0.515, M 36.24, R 0.2294, V 1.794 m^3, ρ 4.46, c_p 0.915, c_v 0.686. 3.5 + 5.5 kg: y 0.5, M 36, V 3.28, c_p 0.918.

Psychrometry (B8)

$\omega = 0.622 p_v/(p - p_v)$; $\varphi = p_v/p_s$; $\mu = \omega/\omega_s$; $\varphi = \mu/[1 - (1 - \mu)p_s/p]$; DPT = $T_{sat}(p_v)$; $h = 1.005t + \omega(2500 + 1.88t)$.

Carrier: $p_v = p_{wb} - (p - p_{wb})(t_d - t_w)/(1544 - 1.44t_w)$. Adiabatic saturation: h const. DBT > WBT (evaporation cools the wick). Humid summer ⇒ slow drying, sweaty.

35/31: 0.0424 bar, 0.0272, 75.3 %, 30 °C, 104.9 kJ/kg. 30/27: 0.0337, 0.0214, 79.4 %, 26 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BDC	bottom dead centre
CI	compression-ignition
CM	control mass
COP	coefficient of performance
CV	control volume
DBT	dry-bulb temperature
DPT	dew-point temperature
EOS	equation of state
FWH	feedwater heater
HP	high-pressure
IAPWS	International Association for the Properties of Water and Steam
IC	internal combustion
KE	kinetic energy
kmol	kilomole (molar mass in kilograms)
KP	Kelvin-Planck (statement)
kPa	kilopascal
kW	kilowatt
LP	low-pressure
MEP	mean effective pressure
MPa	megapascal
MW	megawatt
PE	potential energy
PMM1	perpetual-motion machine of the first kind
PMM2	perpetual-motion machine of the second kind
RH	relative humidity
RK	Redlich-Kwong
SFEE	steady-flow energy equation
SI	spark-ignition
SRK	Soave-Redlich-Kwong
TDC	top dead centre
Tk	Bangladeshi taka
TR	ton of refrigeration (3.5 kW)
VAR	vapour absorption refrigeration
VCR	vapour compression refrigeration
vdW	van der Waals
WBT	wet-bulb temperature

Section A: Concepts, first law, second law, entropy, exergy

A5 Zeroth and first law; energy balance for closed and open systems; boundary work

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q2(b)** [10] Write down the energy balance equation for a general steady-flow system and describe each term.
- **2022 Q1(a)** [10] With a proper example state the zeroth law of thermodynamics. Which property is based on this law?
- **2022 Q2(b)** [15] State the first law. For an open system with one inlet and one outlet, derive the mathematical form of the first law, considering no interactions other than mass flows and the associated volume-change work.
- **2021 Q2(a)** [10] State the zeroth and first laws. Write down the mass and energy rate balance equations for a control volume and mention what each term means.
- **2020 Q2(b)** [9] Which thermodynamic properties are based on the zeroth, first and second laws? Write the mathematical forms of the first law for a control volume and a control mass.
- **2020 Q2(c)** [15] A stationary gas is compressed without friction from 0.3 m^3, 0.105 MPa to 0.15 m^3, 0.105 MPa, while 37.6 kJ of heat leaves the gas. Calculate the change of internal energy.
- **2019 Q2(a)** [5] State the zeroth law of thermodynamics. Which thermodynamic property is based on this law?
- **2019 Q3(a)** [10] State the first law of thermodynamics. Write down the energy rate equation for an open system and the meaning of each term.
- **2019 Q2(b)** [12] 8 kg of gas expands in a flexible container with $Pv^{1.2}$ = constant from 1000 kPa, 1 m^3 to 5 kPa. If the specific internal energy decreases by 40 kJ/kg, find the heat transfer (magnitude and direction).
- **2018 Q2(b)** [8] State the first law of thermodynamics. Write down the energy rate balance equation for a control volume and describe each term.
- **2018 Q1(d)** [13] A gas in a piston-cylinder expands with pV^n = constant from p_1 = 3 bar, V_1 = 0.1 m^3 to V_2 = 0.2 m^3. Find the work for (i) n = 1.5 and (ii) n = 1.
- **2017 Q2(a)** [6] State the first law of thermodynamics. Write down the energy rate balance for a control volume and mention what each term means.

UNDERSTAND **Read this to understand; you do not need to write it**

First law for a closed system: **Q − W = ΔU** (heat in minus work out = rise in stored energy). Boundary work is the area under the process on a p-V diagram, W = ∫p dV.

CALCULATOR **Boundary work for any process pV^n = constant**

Change any number; every step is recalculated and shown with the numbers put in.

1. Final pressure: $p_2 = p_1(V_1/V_2)^n$ = **100** × pow(**0.5** / **0.1**, **1.3**) = **810.33** kPa
2. Boundary work: $W = (p_2V_2 - p_1V_1)/(1 - n)$ [n = 1: $p_1V_1 \ln(V_2/V_1)$] = abs(**1.3** - 1) < 10^{-9} ? **100** × **0.5** × ln(**0.1** / **0.5**) : (**810.3** × **0.1** - **100** × **0.5**) / (1 - **1.3**) = **-103.44** kJ

Negative W = work done **on** the gas (compression).

Before you start: the first law in one line

Energy cannot be created or destroyed. For any system: **energy in – energy out = change of energy stored**. For a closed system that becomes **Q – W = ΔU + ΔKE** (kinetic energy) **+ ΔPE** (potential energy). For a device with flow (open system) we add the energy carried in and out by the flowing mass, which gives the steady-flow energy equation.

Enthalpy h = u + pv appears in open systems because the flow work pv (the work needed to push mass in or out) always comes with the internal energy u of the flowing mass.

Zeroth law (2019 2a, 2022 1a)

"If two bodies are each in thermal equilibrium with a third body, they are in thermal equilibrium with each other." **Example:** a thermometer (third body) reads 30 °C in a cup of water A and 30 °C in water B; so A and B, when brought into contact, exchange no heat. **Property based on it: temperature**; it is the basis of every thermometer.

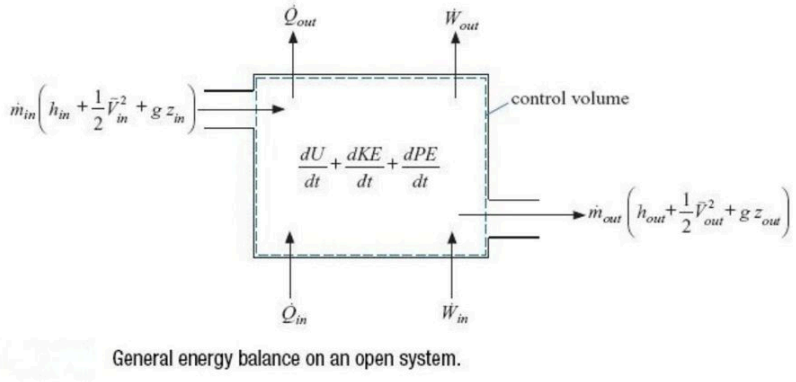
Properties from the laws (2020 2b): zeroth law → **temperature**; first law → **internal energy** (and energy); second law → **entropy**.

First law statements and mathematical forms

- **For a cycle:** "When a system undergoes a cycle, the net heat transfer equals the net work transfer": $\oint \delta Q = \oint \delta W$.
- **For a process of a control mass (closed system):** $Q - W = \Delta E = \Delta U + \Delta KE + \Delta PE$; per unit time $dE/dt = \dot{Q} - \dot{W}$.
- **For a control volume (open system), rate form:**

$$dE_{CV\ (control\ volume)}/dt = \dot{Q}_{CV} - \dot{W}_{CV} + \Sigma\dot{m}_i(h_i + V_i^2/2 + gz_i) - \Sigma\dot{m}_e(h_e + V_e^2/2 + gz_e)$$

$$\dot{m}_{in}\left(h_{in}+\frac{1}{2}V^2{}_{in}+gz_{in}\right)+\dot{Q}_{in}+\dot{W}_{in}$$
$$=\dot{m}_{out}\left(h_{out}+\frac{1}{2}V^2{}_{out}+gz_{out}\right)+\dot{Q}_{out}+\dot{W}_{out}+\frac{dU}{dt}+\frac{dKE}{dt}+\frac{dPE}{dt}$$



General energy balance on an open system.

General energy balance on an open system (course slide)

Term	Meaning (unit kW)
dE_{CV}/dt	rate of change of energy stored inside the control volume (zero at steady state)
$\dot{Q}_{CV}$	net rate of heat transfer into the CV across its boundary
$\dot{W}_{CV}$	net rate of work done by the CV other than flow work (shaft, electrical, moving boundary)
$\dot{m}_i$, $\dot{m}_e$	mass flow rates at inlets and exits (kg/s)
$h = u + pv$	enthalpy = internal energy + flow work per kg
$V^2/2$, gz	kinetic and potential energy per kg of the flowing stream (divide by 1000 to get kJ/kg)

Mass rate balance: $dm_{CV}/dt = \Sigma\dot{m}_i - \Sigma\dot{m}_e$; at steady state $\Sigma\dot{m}_i = \Sigma\dot{m}_e$; with $\dot{m} = \rho AV = AV/v$.

Derivation for an open system with one inlet and one exit (2022 2b, 15 marks)

Step (write this)	Why / what it means
Take a control mass = CV contents at time t plus the small mass Δm_i about to enter. At t + Δt it is the CV contents plus Δm_e that has just left.	Trick: the first law is known for a fixed mass, so follow a fixed mass that passes through the CV.
Control mass first law: $Q - W = E(t+\Delta t) - E(t)$	Apply Q − W = ΔE to that fixed mass.
$E(t) = E_{CV}(t) + \Delta m_i(u_i + V_i^2/2 + gz_i)$; $E(t+\Delta t) = E_{CV}(t+\Delta t) + \Delta m_e(u_e + V_e^2/2 + gz_e)$	Energy of the control mass at the two instants.
$W = W_{CV} + p_e v_e \Delta m_e - p_i v_i \Delta m_i$	Work of the control mass = useful work of the CV + **flow work**: at the exit the mass pushes the surroundings (pA × distance = p × volume = p v Δm), at the inlet the surroundings push it in.
$E_{CV}(t+\Delta t) - E_{CV}(t) = Q - W_{CV} + \Delta m_i(u_i + p_i v_i + V_i^2/2 + gz_i) - \Delta m_e(u_e + p_e v_e + V_e^2/2 + gz_e)$	Substitute and collect terms.
Divide by Δt, let $\Delta t \to 0$, put $h = u + pv$:	Rates: $\Delta m/\Delta t \to \dot{m}$, $Q/\Delta t \to \dot{Q}$.

$dE_{CV}/dt = \dot{Q}_{CV} - \dot{W}_{CV} + \dot{m}_i(h_i + V_i^2/2 + gz_i) - \dot{m}_e(h_e + V_e^2/2 + gz_e)$	The control-volume energy rate balance.
Steady flow ($dE_{CV}/dt = 0$, $\dot{m}_i = \dot{m}_e = \dot{m}$): $q - w = (h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)$	The SFEE (steady-flow energy equation) asked in 2023 2b: per kg of fluid, $q = \dot{Q}/\dot{m}$, $w = \dot{W}/\dot{m}$.

Boundary work numericals

Before you start: boundary (p dV) work

When a piston moves, the gas does work $W = \int p\,dV$ (area under the p-V curve). Polytropic process $pV^n = C$: $W = (p_2V_2 - p_1V_1)/(1 - n)$ for $n \neq 1$; for $n = 1$ (isothermal ideal gas) $W = p_1V_1\ln(V_2/V_1)$; for constant pressure $W = p(V_2 - V_1)$. Use kPa × m^3 = kJ.

2018 1d: $p_1 = 300$ kPa, $V_1 = 0.1$, $V_2 = 0.2$ m^3.

Step (write this)	Why / what it means
(i) $n = 1.5$: $p_2 = p_1(V_1/V_2)^{1.5} = 300(0.5)^{1.5} = 106.1$ kPa	pV^n = const gives the final pressure.
$W = (106.1 \times 0.2 - 300 \times 0.1)/(1 - 1.5) = (21.21 - 30)/(-0.5) =$ **17.6 kJ**	Positive: work done by the gas (expansion).
(ii) $n = 1$: $W = p_1V_1\ln(V_2/V_1) = 300 \times 0.1 \times \ln 2 =$ **20.8 kJ**	Isothermal curve lies higher, so more area, more work.

2019 2b: 8 kg gas, $pV^{1.2} = C$, $p_1 = 1000$ kPa, $V_1 = 1$ m^3, $p_2 = 5$ kPa, $\Delta u = -40$ kJ/kg.

Step (write this)	Why / what it means
$V_2 = V_1(p_1/p_2)^{1/1.2} = 1 \times 200^{0.833} = 82.7$ m^3	Final volume from the process law.
$W = (p_2V_2 - p_1V_1)/(1 - n) = (5 \times 82.7 - 1000)/(-0.2) = 2932$ kJ	Expansion work.
$\Delta U = m\Delta u = 8 \times (-40) = -320$ kJ	
$Q = \Delta U + W = -320 + 2932 =$ **+2612 kJ**	First law. Positive ⇒ 2612 kJ of heat flows **into** the gas.

2020 2c: constant pressure 105 kPa, V: 0.3 → 0.15 m^3, Q = −37.6 kJ.

Step (write this)	Why / what it means
$W = p(V_2 - V_1) = 105(0.15 - 0.3) = -15.75$ kJ	Frictionless (quasi-equilibrium) compression at constant pressure; negative = work done on the gas.
$\Delta U = Q - W = -37.6 - (-15.75) =$ **-21.85 kJ**	Internal energy decreases by 21.85 kJ.

Memory trick: "**Q minus W equals Delta E**" for everything; for flow add "**h, V-squared over 2, g z**" at each port. Laws and properties: **0 → T, 1 → U, 2 → S**.

A7 Second law: statements, heat engines, reversibility, Carnot principles and proofs

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q3(b)** [10] "The efficiencies of all reversible heat engines operating between the same two reservoirs are the same": prove it.
- **2023 Q3(a)** [7] Write down the characteristics of a heat engine and discuss its thermal efficiency.
- **2022 Q3(a)** [10] State the Clausius and Kelvin-Planck statements. With a schematic, explain their equivalence.
- **2022 Q3(b)** [8] Prove that an irreversible heat engine is always less efficient than a reversible one between the same two reservoirs.
- **2022 Q4(b)** [4] In the absence of friction and other irreversibilities, can a heat engine have 100 % efficiency? Explain.
- **2021 Q3(a)** [5] State the Clausius, Kelvin-Planck and entropy statements of the second law.
- **2020 Q3(a)** [10] "The efficiency of an irreversible heat engine is always less than that of a reversible one operating between the same two reservoirs": prove it.
- **2019 Q4(a)** [7] State the Kelvin-Planck statement of the second law and discuss its mathematical form.
- **2019 Q4(d)** [8] Short notes: (i) Carnot efficiency, (ii) irreversible process, (iii) Clausius inequality, (iv) thermal reservoir.
- **2018 Q3(b)** [4] Write down the Kelvin-Planck statement of the second law. (Tds part in A8.)
- **2018 Q4(d)** [10] Prove that $(Q_C/Q_H)_{rev\ cycle} = T_C/T_H$ (absolute temperatures).
- **2017 Q3(a)** [6] State the Kelvin-Planck statement of the second law together with the Carnot corollaries.
- **2017 Q4(a)** [17] Prove that $(Q_L/Q_H)_{rev} = T_L/T_H$ with appropriate assumptions.
- **2017 Q5(b)** [7] Prove that the efficiency of a Carnot engine between T_1 and T_2 is $(T_1 - T_2)/T_1$.

LEARN IT FROM ZERO

Second law and Carnot, from zero

The first law says energy is conserved; it does not say which way things go. Heat flows from hot to cold by itself, never the reverse. The **second law** puts this into two statements: **Kelvin-Planck** (no engine can turn all the heat it takes in into work: some must be rejected to a cold sink) and **Clausius** (no device can move heat from cold to hot without work input). Break one and you break the other.

The best possible engine between a hot reservoir T_H and a cold one T_L is the reversible **Carnot** engine: $\eta = 1 - T_L/T_H$ (kelvin!). A real engine is always less. The same reversible machine run backwards is a refrigerator (COP = $T_L/(T_H - T_L)$) or heat pump (COP = $T_H/(T_H - T_L)$).

CALCULATOR **Carnot limits and a claim checker**

Change any number; every step is recalculated and shown with the numbers put in.

1. Carnot efficiency: $\eta = 1 - T_L/T_H$ = (1 - **300** / **1000**) × 100 = **70.00** %

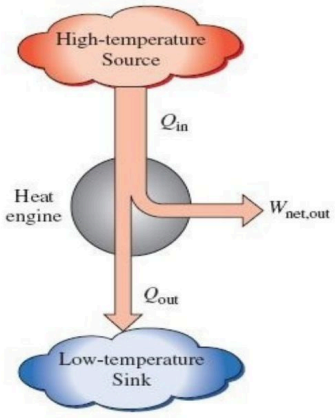
2. Maximum work: **70** / 100 × **1000** = **700.0** kJ
3. Claim: **possible (below the Carnot limit)**
4. Carnot refrigerator COP: $T_L/(T_H - T_L)$ = **300** / (**1000** - **300**) = **0.429**
5. Carnot heat pump COP: $T_H/(T_H - T_L)$ = **1000** / (**1000** - **300**) = **1.429**

Before you start: why a second law?

The first law says energy is conserved, but not *which way* processes go. Hot coffee cools in a room; it never heats up by taking energy from the cooler room, even though that would not violate the first law. The **second law** gives the direction of processes and says that heat can never be fully converted into work in a cycle.

Thermal reservoir: a body so large that it can give or take any amount of heat without its temperature changing (ocean, atmosphere, furnace). Source = high-temperature reservoir, sink = low-temperature reservoir.

Heat engine: characteristics and efficiency (2023 3a)



Part of the heat received by a heat engine is converted to work, while the rest is rejected to a sink.

A heat engine takes Q_{in} from a source, converts part into W_{net}, and rejects Q_{out} to a sink (course slide)

1. receives heat from a high-temperature source (furnace, solar, nuclear);
2. converts part of this heat to work (usually a rotating shaft);
3. rejects the remaining heat to a low-temperature sink (atmosphere, river);
4. operates on a cycle, using a working fluid.

$$\eta_{th} = W_{net,out}/Q_H = (Q_H - Q_L)/Q_H = 1 - Q_L/Q_H$$

Because Q_L can never be zero, $\eta < 1$. Typical values: car engine 25-30 %, steam plant 35-45 %, combined cycle up to 60 %.

Statements of the second law (2017 3a, 2019 4a, 2021 3a, 2022 3a)

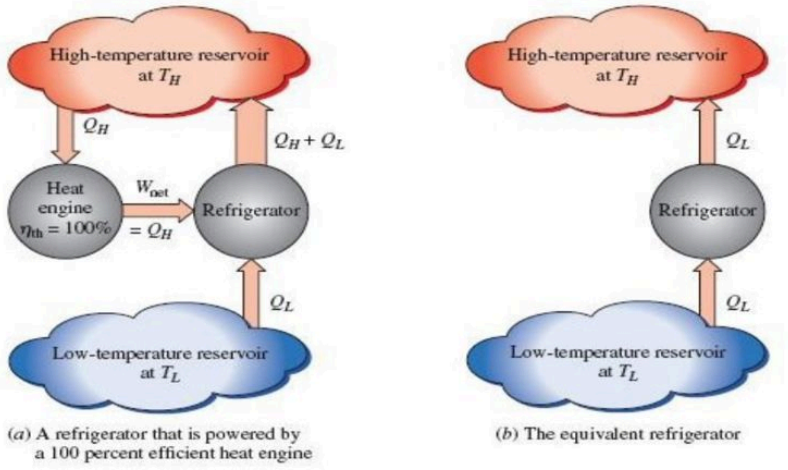
- **Kelvin-Planck:** "It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce an equal amount of work." Mathematical form: for a cycle

exchanging heat with a single reservoir, $W_{cycle} \leq 0$ (= 0 if reversible); equivalently $\eta_{th} < 100$ %.

- **Clausius:** "It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body." A refrigerator needs work input.
- **Entropy statement:** "It is impossible for any process to occur that decreases the total entropy of an isolated system": $\Delta S_{isolated} = S_{gen} \geq 0$.

Equivalence of the two statements (2022 3a)

statement, and vice versa. This can be demonstrated as follows.



A 100 %-efficient engine (violating Kelvin-Planck) driving a refrigerator gives a device that moves Q_L from cold to hot with no work: a Clausius violation (course slide)

Step (write this)	Why / what it means
Suppose an engine violates Kelvin-Planck: it takes Q_H from the hot reservoir and produces $W = Q_H$.	Assume the impossible and look at the consequence.
Use this W to drive an ordinary refrigerator that takes Q_L from the cold reservoir and delivers $Q_L + Q_H$ to the hot reservoir.	First law for the refrigerator.
Combined system: no net work; takes Q_L from cold and gives net $(Q_L + Q_H) - Q_H = Q_L$ to hot.	Draw a boundary around both devices.
This combined device violates Clausius. Hence a KP (Kelvin-Planck (statement)) violation implies a Clausius violation; the reverse argument works the same way (Clausius violator + ordinary engine ⇒ KP violator).	So the two statements are equivalent.

Reversible and irreversible processes

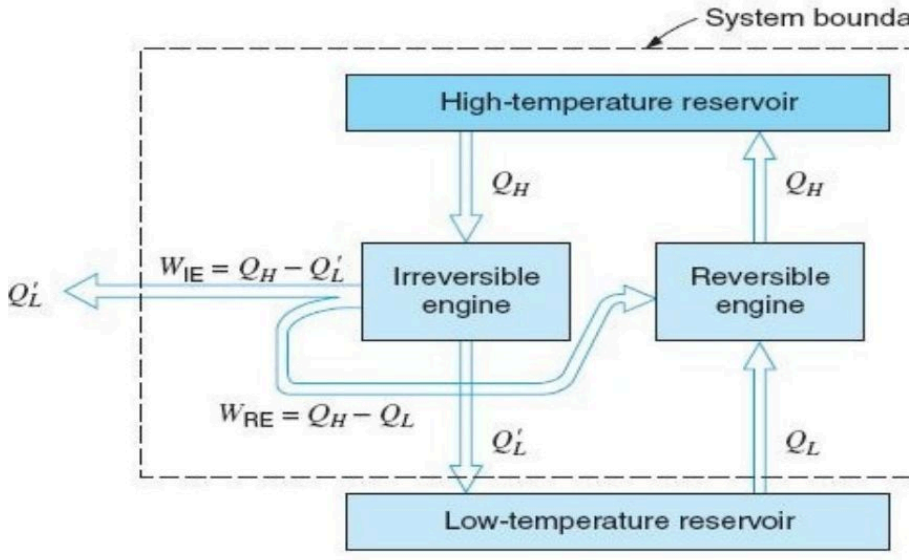
A **reversible process** can be reversed without leaving any trace on the system or surroundings. All real processes are irreversible. **Irreversibilities:** friction, heat transfer across a finite temperature difference, unrestrained expansion, mixing of different fluids, electrical resistance, inelastic deformation, chemical reaction. An **internally reversible** process has no irreversibility inside the system; **totally reversible** has none inside or at the boundary.

Can a frictionless engine be 100 % efficient? (2022 4b) No. Even with no friction and no irreversibilities, a heat engine must reject heat to a sink to complete its cycle (Kelvin-Planck). The best it can do is the Carnot efficiency $1 - T_L/T_H$, which is below 1 because $T_L > 0$ K.

Carnot principles (corollaries)

1. The efficiency of an irreversible heat engine is always less than that of a reversible one operating between the same two reservoirs.
2. The efficiencies of all reversible heat engines operating between the same two reservoirs are the same.

Proof 1: $\eta_{irr} < \eta_{rev}$ (2020 3a, 2022 3b)



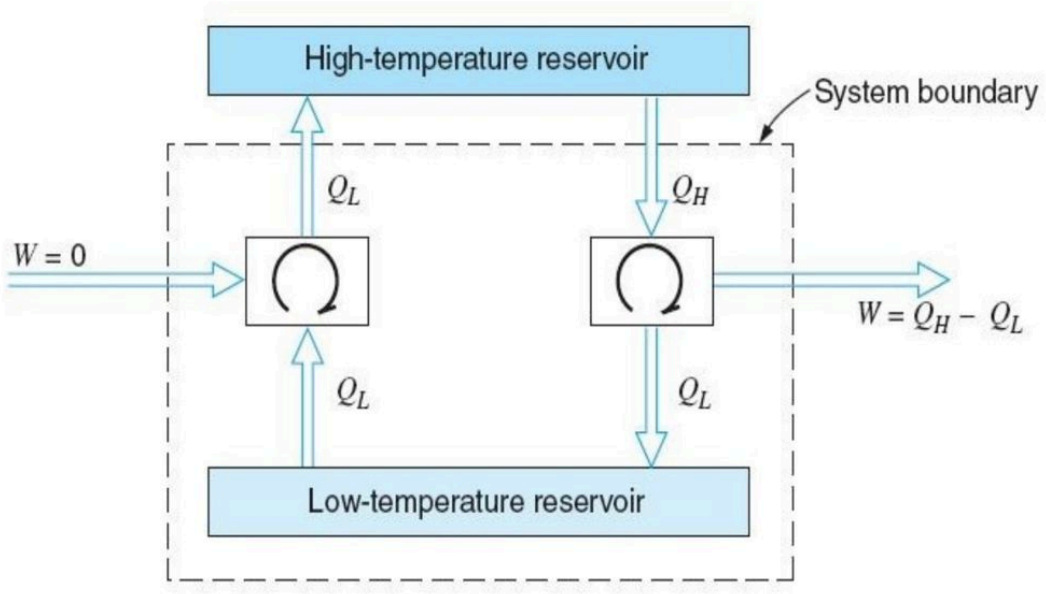
Reversible engine run backwards as a refrigerator, driven by part of the work of an irreversible engine (course slide)

Step (write this)	Why / what it means
Assume the opposite: $\eta_{irr} > \eta_{rev}$. Both receive the same Q_H from the hot reservoir.	Proof by contradiction.
Then $W_{irr} > W_{rev}$ and $Q_L' = Q_H - W_{irr} < Q_L = Q_H - W_{rev}$.	First law for each engine.
Reverse the reversible engine: it becomes a refrigerator taking W_{rev}, absorbing Q_L from the sink and delivering Q_H to the source.	Only a reversible engine can be reversed with all quantities simply changing sign.
Let the irreversible engine drive it. Combined: hot reservoir net exchange = 0; net work out = $W_{irr} - W_{rev} > 0$; heat taken from the sink = $Q_L - Q_L' > 0$.	Boundary around both devices and the hot reservoir.
The combination produces work while exchanging heat with a single reservoir (the sink): this violates Kelvin-Planck.	Contradiction.
Therefore $\eta_{irr} < \eta_{rev}$.	

Proof 2: all reversible engines between the same reservoirs have the same efficiency (2023 3b)

Step (write this)	Why / what it means
Take two reversible engines R1 and R2 between T_H and T_L; suppose $n_{R1} > n_{R2}$.	
Run R2 backwards as a refrigerator, driven by R1; both exchange the same Q_H.	Same set-up as proof 1, with R1 in place of the irreversible engine.
The combination gives net work $W_{R1} - W_{R2} > 0$ from the single low-temperature reservoir: violates Kelvin-Planck. So $n_{R1} > n_{R2}$ is impossible.	
Now suppose $n_{R2} > n_{R1}$: reverse R1 instead (it is also reversible); same contradiction.	This step is possible only because both are reversible.
Hence $\mathbf{n_{R1} = n_{R2}}$: the efficiency of a reversible engine depends only on the reservoir temperatures, $n_{rev} = f(T_H, T_L)$.	Conclusion; leads to the thermodynamic temperature scale.

Proof 3: $(Q_L/Q_H)_{rev} = T_L/T_H$ (2017 4a, 17 marks; 2018 4d)



Engines between three reservoirs used in the proof (course slide)

Step (write this)	Why / what it means
From proof 2: $Q_H/Q_L = f(T_H, T_L)$ for any reversible engine.	Assumption: reversible engines, reservoirs only.
Engine A works between T_1 and T_3; engine B between T_1 and an intermediate T_2; engine C between T_2 and T_3, using the heat B rejects. B + C together equal A.	Three reservoirs $T_1 > T_2 > T_3$.
$Q_1/Q_3 = f(T_1,T_3)$, $Q_1/Q_2 = f(T_1,T_2)$, $Q_2/Q_3 = f(T_2,T_3)$	
$Q_1/Q_3 = (Q_1/Q_2)(Q_2/Q_3) \Rightarrow f(T_1,T_3) = f(T_1,T_2)\, f(T_2,T_3)$	The left side does not contain T_2, so T_2 must cancel on the right.

So $f(T_1,T_2) = \varphi(T_1)/\varphi(T_2) \Rightarrow Q_H/Q_L = \varphi(T_H)/\varphi(T_L)$	Only a ratio form satisfies the product rule.
Kelvin chose $\varphi(T) = T$ (the thermodynamic, absolute scale): $\mathbf{(Q_L/Q_H)_{rev} = T_L/T_H}$	Temperatures must be absolute (K).

Carnot efficiency (2017 5b, 2019 4d)

Step (write this)	Why / what it means
$n = 1 - Q_L/Q_H$	Definition for any engine.
Carnot engine is reversible $\Rightarrow Q_L/Q_H = T_L/T_H$	Proof 3 (or: on T-s, $Q_H = T_H\Delta s$ and $Q_L = T_L\Delta s$ for the isothermal processes).
$\mathbf{n_{Carnot} = 1 - T_2/T_1 = (T_1 - T_2)/T_1}$	Highest possible efficiency between T_1 and T_2; increases if T_1 rises or T_2 falls.

Short notes (2019 4d): *Carnot efficiency*: maximum efficiency of any engine between two reservoirs, $1 - T_L/T_H$. *Irreversible process*: cannot be reversed without leaving a change in the surroundings (friction, finite ΔT, mixing). *Clausius inequality*: $\oint \delta Q/T \leq 0$ (see A8). *Thermal reservoir*: very large body that supplies or absorbs heat at constant temperature.

> **Memory trick: K**elvin-Planck = **K**ilowatts from one reservoir impossible (engine). **C**lausius = **C**old to hot needs work (fridge). Carnot proofs always: "assume the opposite → reverse the reversible one → one reservoir gives work → KP violated".

A6 First-law numericals for steady-flow devices (turbine, compressor, pump, heat exchanger) Pass set Very likely · 86% Medium

PAST-PAPER QUESTIONS

- **2023 Q2(c)** [15] An adiabatic steam turbine produces 5 MW. Inlet 2 MPa, 400 °C, 50 m/s, 10 m; exit 15 kPa, x = 90 %, 180 m/s, 6 m. (i) Compare Δh, ΔKE (kinetic energy), ΔPE (potential energy); (ii) work per kg; (iii) mass flow.

f each term.

te on (i) Quasi-equilibrium process, (ii) Int
namic equilibrium, (iv) Thermodynamc sy

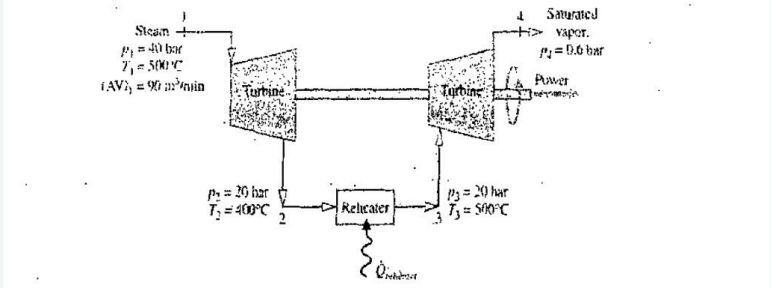
pressure exerted by 4.0 kg CO in a 0.03 m^3
Wal's and redlic-Kwong equation.

of thermodynamics. Draw T-v diagram of a
tant pressure line, saturation line, critical p

e energy balance equation for a general stea
ıc equation means

- **2022 Q2(c)** [10] Same compressor (1 bar, 290 K, 6 m/s, 0.1 m^2 → 7 bar, 450 K, 2 m/s; heat loss 180 kJ/min): find the power input, and again neglecting the kinetic energy change (1 bar = 100 kPa).
- **2021 Q2(c)** [20] Same turbine-driven pump (steam 30 bar, 400 °C → 5 bar, 180 °C; water saturated liquid at 1 bar, 50 kg/s; h_2 = 417.69 kJ/kg; Q = −2 kW; 25 m). Find the pump power and the steam mass flow.
- **2019 Q3(c)** [15] Waste heat recovery: water enters the steam generator at 290 kPa, 104 °C and leaves at 276 kPa, 160 °C, then expands in a turbine to 7 kPa, x = 90 %. Oven exhaust air enters at 182 °C, 100 kPa, 2 m^3/s and leaves at 138 °C. Power is worth Tk 7/kWh. Do you recommend the system? Support with calculations.
- **2018 Q2(c)** [15] Steam enters a first-stage turbine at 40 bar, 500 °C, 90 m^3/min, exits at 20 bar, 400 °C, is reheated at constant pressure to 500 °C and leaves the second stage as saturated vapour at 0.6 bar. Find (i) mass flow, (ii) total power, (iii) reheater heat rate.

what each term in the equation mean.
the first --stage turbine shown in figure at 40 *bar* and 500°C with volu
/min.

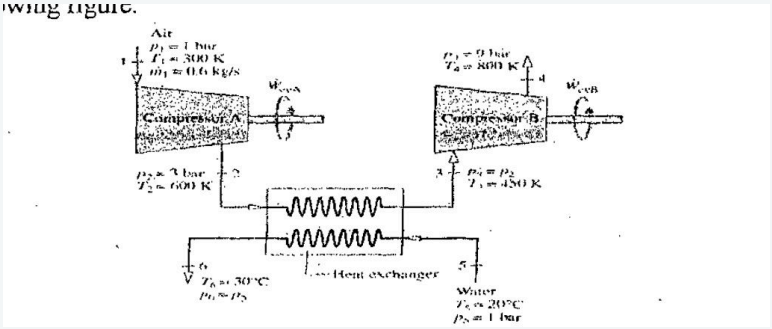


the turbine at 20 *bar* and 400°C .The steam is reheated at constant

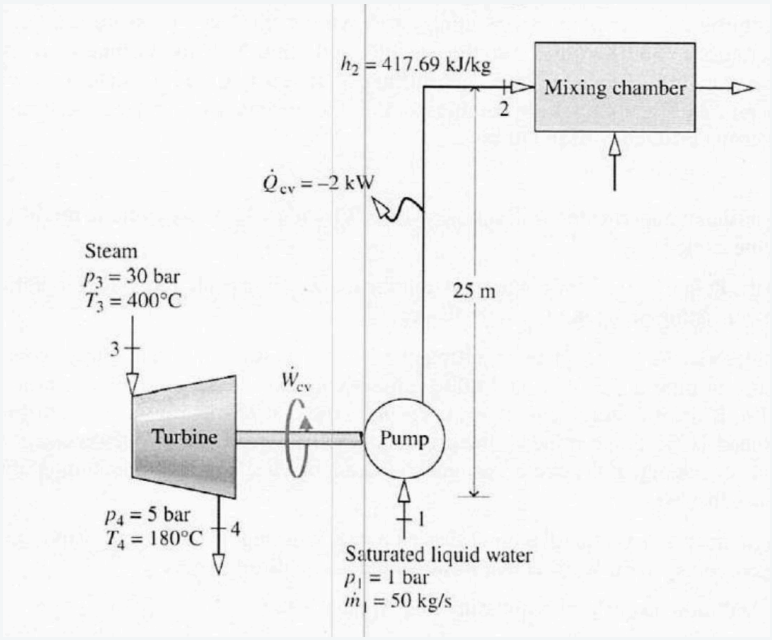
- **2018 Q3(a)** [15] Air (1 bar, 300 K, 0.6 kg/s) is compressed in compressor A to 3 bar, 600 K, cooled in a heat exchanger to 450 K, then compressed in B to 9 bar, 800 K. Cooling water enters at 20 °C, 1 bar

and leaves at 30 °C. c_p = 1.063 kJ/kg K. Find (i) total compressor power, (ii) water mass flow.

wing figure.



- **2017 Q2(b)** [12] Define a turbine and the kinds used in power generation. Air enters a compressor at steady state at 1 bar, 290 K, 6 m/s through 0.1 m^2; exits at 7 bar, 450 K, 2 m/s. Heat loss is 180 kJ/min. Using the ideal gas model, find the power input (kW).
- **2017 Q2(c)** [17] A turbine-driven pump supplies water to a mixing chamber 25 m above the pump. Heat loss from the water is 2 kW; turbine heat transfer and PE negligible; KE negligible everywhere. Find (i) the pump power and (ii) the steam mass flow through the turbine. (Figure below.)



UNDERSTAND **Read this to understand; you do not need to write it**

Steady-flow energy equation: $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2000 + g(z_2 - z_1)/1000]$. Turbine: $\dot{W}$ positive. Compressor/pump: $\dot{W}$ negative. Nozzle: no W, KE rises. Throttle: $h_1 = h_2$.

CALCULATOR **Steady-flow device with ideal-gas air (constant c_p)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Mass flow: $\dot{m} = p_1A_1V_1/(RT_1)$ = **100** × **0.1** × **6** / (0.287 × **290**) = **0.7209** kg/s
2. Enthalpy change: $\Delta h = c_p\Delta T$ = 1.005 × (**450** - **290**) = **160.80** kJ/kg
3. Kinetic energy change: (**2** × **2** - **6** × **6**) / 2000 = **-0.0160** kJ/kg
4. Power (+ out, – in): $\dot{W} = \dot{Q} - \dot{m}(\Delta h + \Delta ke)$ = **(-3)** - **0.7209** × (**160.8** + **(-0.016)**) = **-118.91** kW

2017 2(b) / 2022 2(c) compressor: about −119 kW (power input). The pack uses air-table h; constant c_p gives nearly the same.

Before you start: how to solve any device problem in five steps

1) Draw the device and the control volume; number the inlet and exit states. 2) Write what is known at each state (two properties each). 3) Get h from tables (steam) or h = c_pT / air table (ideal gas). 4) Write the steady-flow energy equation $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)]$ and drop terms that are zero. 5) Solve, then check the sign makes sense.

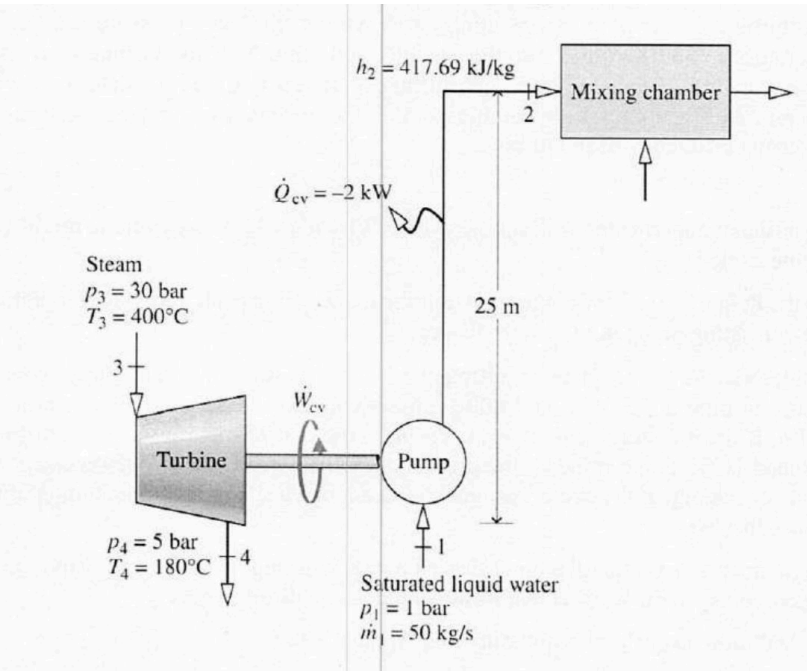
Units: $V^2/2$ in m^2/s^2 = J/kg, so divide by 1000 for kJ/kg. Heat lost by the device is negative $\dot{Q}$; power absorbed by a compressor or pump comes out negative $\dot{W}$.

Turbine (2017 2b): a rotary machine in which a fluid flowing through blades turns a shaft and produces work. Kinds used in power generation: **steam turbine** (impulse and reaction, thermal plants), **gas turbine** (Brayton cycle), **hydraulic turbine** (Pelton, Francis, Kaplan in hydro plants), **wind turbine**.

1. Air compressor (2017 2b, 2022 2c)

Step (write this)	Why / what it means
$\dot{m} = p_1A_1V_1/(RT_1) = 100 \times 0.1 \times 6/(0.287 \times 290) = 0.721$ kg/s	Mass flow from the inlet: $\dot{m}$ = AV/v and v = RT/p.
$h_1 = 290.16$, $h_2 = 451.80$ kJ/kg	Ideal-gas air table at 290 K and 450 K (constant c_p gives 1.005 × 160 = 160.8, almost the same).
$\Delta KE = (2^2 - 6^2)/2/1000 = -0.016$ kJ/kg	Tiny compared with Δh.
$\dot{Q} = -180$ kJ/min = −3.0 kW	Heat leaves the compressor.
$\dot{W} = \dot{Q} - \dot{m}[(h_2 - h_1) + \Delta KE] = -3.0 - 0.721(161.64 - 0.016)$	SFEE (steady-flow energy equation) solved for $\dot{W}$; ΔPE = 0.
$\dot{W}$ = −119.5 kW (power input 119.5 kW)	Negative sign = work done on the air.
Neglecting ΔKE: $\dot{W} = -3.0 - 0.721 \times 161.64 =$ **−119.5 kW** (difference 0.01 kW)	2022 part 2: KE change is negligible here.

2. Turbine-driven pump (2017 2c, 2021 2c)



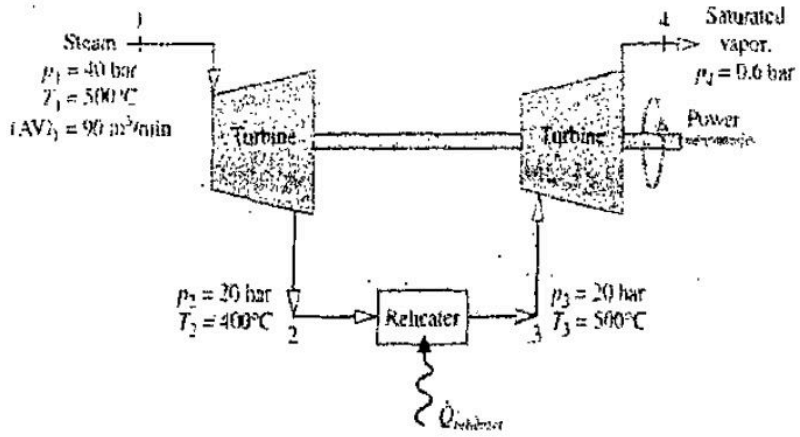
Question figure (2021 paper)

Step (write this)	Why / what it means
Pump CV (control volume): water, state 1 saturated liquid at 1 bar: h_1 = 417.46 kJ/kg; state 2: h_2 = 417.69 kJ/kg, $z_2 - z_1$ = 25 m; $\dot{Q}$ = -2 kW; $\dot{m}$ = 50 kg/s	Separate control volumes: first the pump (water side).
$\dot{W}_p = \dot{Q} + \dot{m}[(h_1 - h_2) + g(z_1 - z_2)]$	SFEE, KE neglected.
$= -2 + 50[(417.46 - 417.69) + 9.81(-25)/1000] = -2 + 50(-0.23 - 0.245)$	
$\dot{W}_p$ = -25.8 kW ⇒ pump needs 25.8 kW	(i) Answer.
Turbine CV: h_3 (30 bar, 400 °C) = 3230.9; h_4 (5 bar, 180 °C) = 2812.0 kJ/kg	Superheated steam table.
Turbine drives the pump: $\dot{W}_t = 25.8$ kW = $\dot{m}_s(h_3 - h_4)$	Adiabatic turbine, KE and PE negligible; all its power goes to the pump.
$\dot{m}_s = 25.8/(3230.9 - 2812.0) = 0.0616$ kg/s	(ii) Answer.

3. Two-stage turbine with reheat (2018 2c)

what each term in the equation mean.

the first --stage turbine shown in figure at 40 *bar* and 500°C with volu
/min.



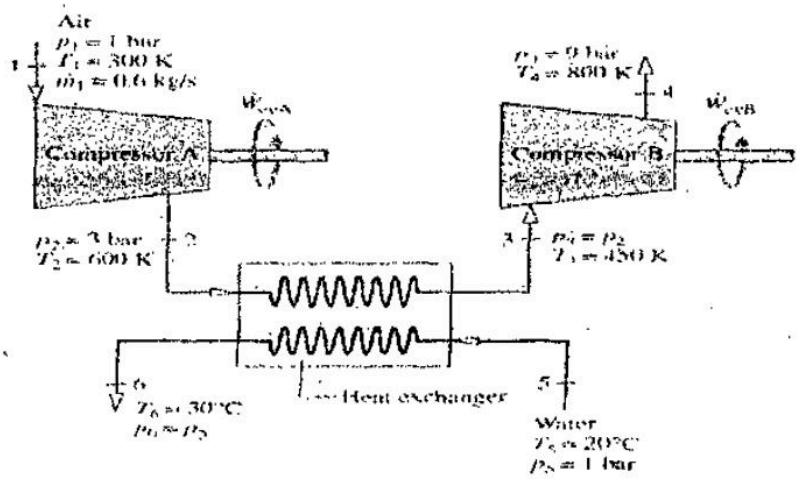
the turbine at 20 *bar* and 400°C .The steam is reheated at constant

Question figure (2018 paper)

Step (write this)	Why / what it means
State 1 (40 bar, 500 °C): v_1 = 0.08643 m^3/kg, h_1 = 3445.3; state 2 (20 bar, 400 °C): h_2 = 3247.6; state 3 (20 bar, 500 °C): h_3 = 3467.6; state 4 (sat. vapour 0.6 bar): h_4 = 2653.5 kJ/kg	Steam tables.
$\dot{m} = (AV)_1/v_1 = (90/60)/0.08643 =$ **17.36 kg/s**	(i) Volume flow ÷ specific volume.
$\dot{W} = \dot{m}[(h_1 - h_2) + (h_3 - h_4)] = 17.36(197.7 + 814.1) =$ **17,565 kW ≈ 17.6 MW**	(ii) Both stages, adiabatic, KE and PE ignored.
$\dot{Q}_{reheat} = \dot{m}(h_3 - h_2) = 17.36 \times 220.0 =$ **3,819 kW**	(iii) Reheater: no work, so $\dot{Q} = \dot{m}\Delta h$.

4. Two compressors with an intercooler (2018 3a)

wing figure.



Question figure (2018 paper)

Step (write this)	Why / what it means
$\dot{W}_A = \dot{m}c_p(T_2 - T_1) = 0.6 \times 1.063 \times (600 - 300) =$ 191.3 kW	Adiabatic compressor: power in = enthalpy rise.

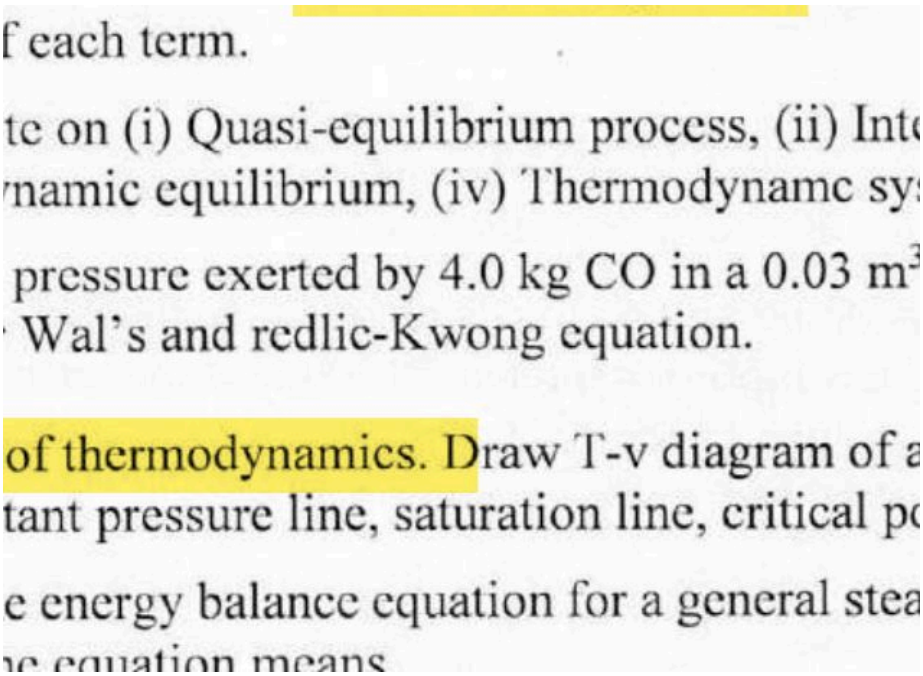
$\dot{W}_B = 0.6 \times 1.063 \times (800 - 450) = 223.2$ kW	
Total compressor power = 414.6 kW (input)	(i)
Heat exchanger: $\dot{m}_{air}c_p(T_2 - T_3) = \dot{m}_w(h_6 - h_5)$	Energy lost by air = energy gained by water (no heat loss outside, no work).
$0.6 \times 1.063 \times 150 = 95.67$ kW; $h_6 - h_5 \approx h_f$(30 °C) − h_f(20 °C) = 125.79 − 83.96 = 41.83 kJ/kg	Liquid water: use h_f at the temperature (or $c_w\Delta T = 4.18 \times 10$).
$\dot{m}_w$ = 95.67/41.83 = 2.29 kg/s	(ii)

5. Waste-heat recovery steam generator (2019 3c)

Step (write this)	Why / what it means
Air: $\dot{m}_A = p\dot{V}/(RT) = 100 \times 2/(0.287 \times 455.15) = 1.531$ kg/s	Air mass flow at the inlet (182 °C = 455.15 K).
Heat given by air: $\dot{Q} = \dot{m}_A c_p(182 - 138) = 1.531 \times 1.005 \times 44 = 67.7$ kW	Air side of the steam generator.
Water in: compressed liquid, $h_1 \approx h_f$(104 °C) = 436.1 kJ/kg; steam out (276 kPa, 160 °C, superheated): $h_2 = 2784.3$ kJ/kg	Compressed liquid approximated as saturated liquid at the same temperature.
$\dot{m}_w = 67.7/(2784.3 - 436.1) = 0.0288$ kg/s	Energy balance on the steam generator (no heat loss).
Turbine exit 7 kPa, x = 0.9: $h_3 = 163.4 + 0.9 \times 2409.1 = 2331.6$ kJ/kg	Saturated table at 7 kPa.
$\dot{W} = \dot{m}_w(h_2 - h_3) = 0.0288 \times 452.7 =$ **13.1 kW**	
Value = 13.1 kW × Tk 7/kWh = **Tk 91.5 per hour** ≈ Tk 0.73 million per year at 8000 h/yr	Assumed 8000 operating hours per year (not given).

Recommendation: the system recovers 13 kW (about 19 % of the 67.7 kW available). It is worth installing only if its capital, maintenance and operating cost can be paid back from about Tk 0.7 million per year; for a small turbine-generator set that is usually feasible within a few years, so it is **recommended** if the oven runs continuously. State this conclusion with the assumption about operating hours.

6. Adiabatic steam turbine (2023 2c)


f each term.

te on (i) Quasi-equilibrium process, (ii) Inte
namic equilibrium, (iv) Thermodynamc sy

pressure exerted by 4.0 kg CO in a 0.03 m^3
Wal's and redlic-Kwong equation.

of thermodynamics. Draw T-v diagram of a
tant pressure line, saturation line, critical po

e energy balance equation for a general stea
ic equation means

Question figure (2023 paper)

Step (write this)	Why / what it means
h_1 (2 MPa, 400 °C) = 3248.4 kJ/kg; at 15 kPa: h_f = 225.94, h_{fg} = 2372.3 ⇒ h_2 = 225.94 + 0.9 × 2372.3 = 2361.0 kJ/kg	Tables.
Δh = 2361.0 - 3248.4 = **-887.4 kJ/kg**	
$\Delta KE = (180^2 - 50^2)/2/1000$ = **14.95 kJ/kg**	
ΔPE = 9.81(6 - 10)/1000 = **-0.04 kJ/kg**	(i) Δh dominates; ΔPE is negligible, ΔKE is about 1.7 % of Δh.
$w = -(\Delta h + \Delta KE + \Delta PE)$ = -(-887.4 + 14.95 - 0.04) = **872.5 kJ/kg**	(ii) SFEE with q = 0.
$\dot{m} = \dot{W}/w$ = 5000/872.5 = **5.73 kg/s**	(iii)

Memory trick: Device shortcuts: **turbine** $w = h_{in} - h_{out}$; **compressor/pump** $w_{in} = h_{out} - h_{in}$; **heat exchanger** $\dot{m}_{hot}\Delta h_{hot} = \dot{m}_{cold}\Delta h_{cold}$; **throttle** $h_{in} = h_{out}$.

A3 Pure substance, phase rule, state postulate, T-v diagram, quality Pass set Very likely · 82% Medium

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [8] Consider (i) a mixture of liquid and gaseous water and (ii) a mixture of liquid and gaseous air. Which one is a pure substance? Explain.
- **2023 Q1(b)** [8] What is state postulate? Write down Gibbs' phase rule for a non-reacting system and mention the meaning of each term.
- **2023 Q2(a)** [10] State the first law of thermodynamics. Draw the T-v diagram of a single-component pure substance and label constant pressure line, saturation line, critical point and regions.
- **2022 Q1(b)** [4] What is phase rule? Write down the phase rule for a non-reacting system.
- **2020 Q1(b)** [6] Write down the phase rule and mention the meaning of each term.
- **2020 Q1(c)** [6] Suppose (i) a mixture of liquid and gaseous water and (ii) a mixture of liquid and gaseous air. Which one is a pure substance? Explain.
- **2019 Q1(b)** [8] Define pure substance and phase. Discuss the condition based on which air can be treated as a pure substance.
- **2019 Q4(b)** [5] State the 'state postulate' of a simple compressible system. What is its significance?
- **2019 Q1(d)** [10] A closed, rigid tank filled with water, initially at 2 MPa, quality 80 % and volume 0.5 m^3, is cooled until the pressure is 0.4 MPa. Show the process on a sketch of the T-v diagram and evaluate the heat transfer.
- **2018 Q2(a)** [7] State the zeroth law of thermodynamics. Draw the T-v diagram of a single-component pure substance and label constant pressure line, saturation lines, critical point and constant quality lines.
- **2017 Q1(c)** [7] Write down the state postulate of a simple compressible system. Define quality and derive $x = (v - v_f)/(v_g - v_f)$.

★ REMEMBER IT

- Quality x = mass of vapour / total mass (only inside the dome). Any property: $\mathbf{y = y_f + x\, y_{fg}}$. Phase rule: F = C − P + 2 (pure water at the triple point: F = 1 − 3 + 2 = 0).

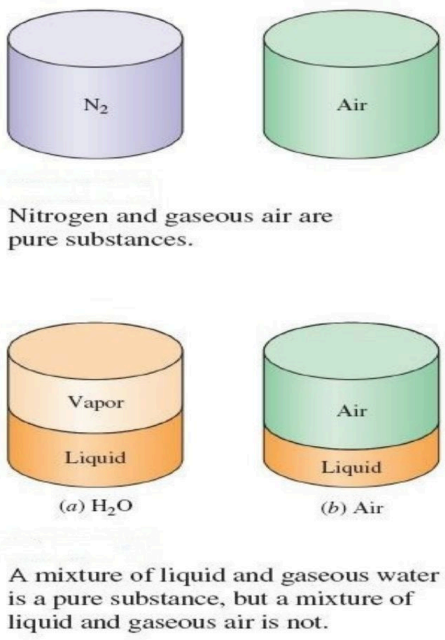
Before you start: pure substance, phases and the steam tables

Most energy machines use **water and steam**. Water can be liquid, a liquid-vapour mixture, or vapour. To find its properties (v, u, h, s) we use **steam tables**, and to use them we must know which **region** we are in. The T-v diagram is the map of those regions.

Key words: **saturation temperature** = boiling temperature at a given pressure (100 °C at 101.325 kPa, 212 °C at 2 MPa). **Quality x** = fraction of the mixture mass that is vapour (0 for saturated liquid, 1 for saturated vapour).

Pure substance and phase

Pure substance: a substance with a fixed and uniform chemical composition throughout. It may be a single element or compound (water, N_2, CO_2) or a mixture, provided the mixture is homogeneous and its composition does not change (dry air as a gas). **Phase**: a quantity of matter that is homogeneous in chemical composition and physical structure (solid, liquid, gas), separated from other phases by a boundary.



N_2 and gaseous air are pure substances; liquid + vapour water is pure, liquid + gaseous air is not (course slide)

Which mixture is a pure substance? (2020 1c, 2023 1a)

1. **Liquid water + water vapour: a pure substance.** Both phases have the same chemical composition (H_2O), so the composition is uniform even though two phases exist.
2. **Liquid air + gaseous air: NOT a pure substance.** Air is mainly N_2 and O_2, which condense at different temperatures (O_2 at −183 °C, N_2 at −196 °C). So the liquid is richer in oxygen and the vapour richer in nitrogen; composition differs between phases.

When can air be treated as a pure substance? (2019 1b) As long as it stays entirely in one phase (all gas, at ordinary temperatures), so its composition is the same everywhere and does not change. Once condensation starts, it is no longer pure.

Gibbs phase rule (non-reacting system)

```
F = C - π + 2
```

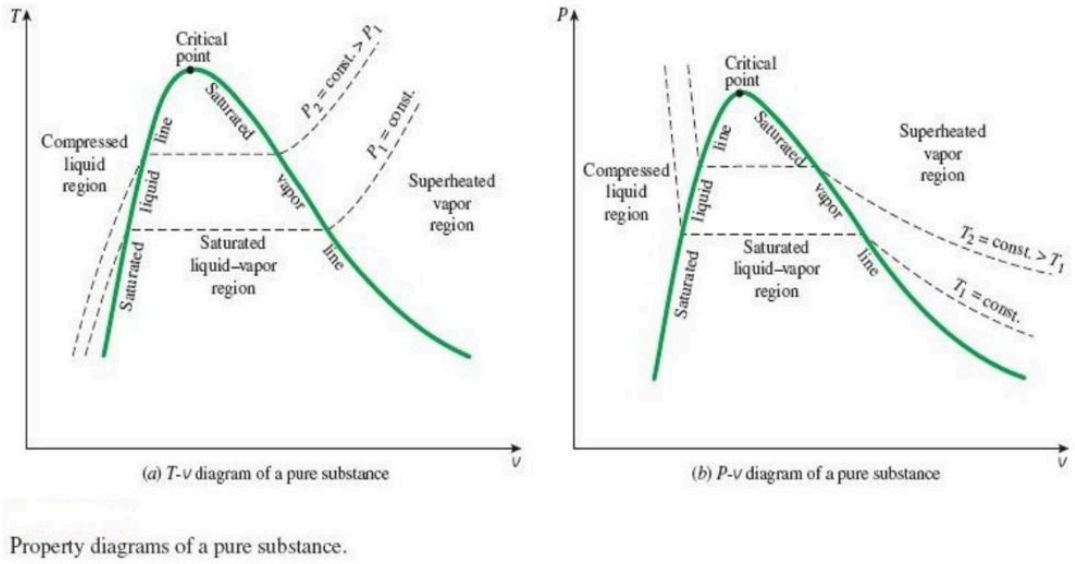
- **F** = degrees of freedom = number of independent intensive properties that must be fixed to fix the state.
- **C** = number of chemical components (distinct species).
- **π** = number of phases present.
- Water, one phase (superheated steam): F = 1 − 1 + 2 = 2 (need p and T). Water + steam: F = 1 − 2 + 2 = 1 (fix p and T is fixed automatically). Triple point: F = 1 − 3 + 2 = 0 (unique p and T).

State postulate (2017 1c, 2019 4b, 2023 1b)

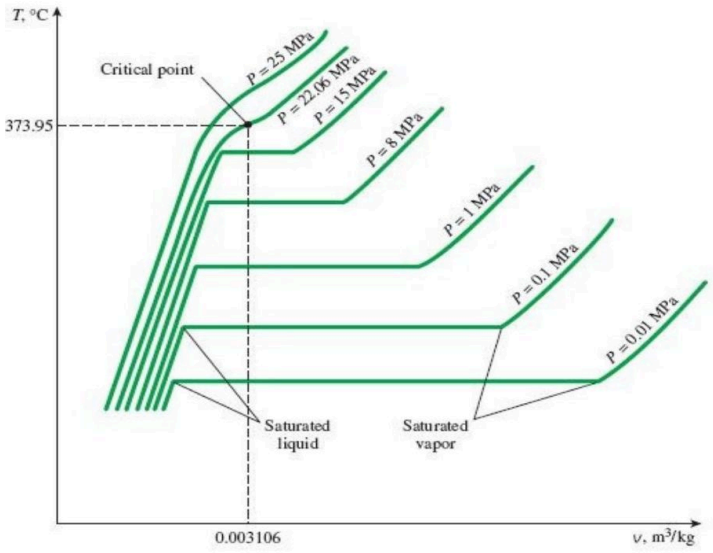
"The state of a **simple compressible system** is completely specified by **two independent, intensive properties**." A simple compressible system is one in which electrical, magnetic, gravitational, motion and surface-tension effects are absent.

Significance: (i) we only need two properties (e.g. p and T) to look up all the others in a table, which is how every steam-table problem works; (ii) the two must be *independent*: inside the wet region p and T are not independent ($T = T_{sat}(p)$), so we need p and x, or T and v, etc.; (iii) it allows property diagrams in two coordinates (T-v, p-v, T-s, h-s).

T-v diagram of a pure substance (2018 2a, 2023 2a)



T-v and p-v diagrams of a pure substance with regions labelled (course slide)



Family of constant-pressure lines on the T-v diagram of water; above the critical point there is no phase change (course slide)

- **Saturated liquid line** (left side of dome) and **saturated vapour line** (right side) meet at the **critical point** (water: 22.06 MPa, 373.95 °C, v = 0.003106 m^3/kg).
- Regions: **compressed (subcooled) liquid** left of the dome; **saturated liquid-vapour (wet) mixture** under the dome; **superheated vapour** right of the dome.
- A **constant-pressure line** rises in the liquid region, runs horizontal across the dome (T = T_{sat}), then rises again in the superheated region. Higher pressures lie higher. **Constant-quality lines** fan out inside the dome from the critical point (x = 0.25, 0.5, 0.75).

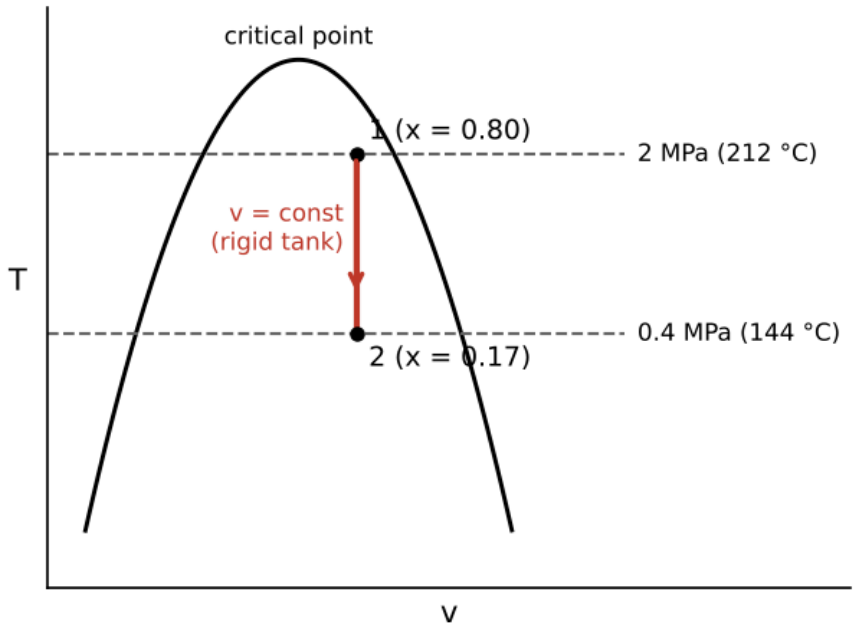
Quality and the derivation $x = (v - v_f)/(v_g - v_f)$ (2017 1c)

Step (write this)	Why / what it means
$x = m_g/m_t$, $m_t = m_f + m_g$	Definition: g = saturated vapour, f = saturated liquid, t = total.

$V = V_f + V_g \Rightarrow m_t v = m_f v_f + m_g v_g$	Volumes add; each volume = mass × specific volume.
$m_f = m_t - m_g \Rightarrow v = (1 - x)v_f + x\, v_g$	Divide by m_t and use $m_g/m_t = x$.
$v = v_f + x(v_g - v_f) = v_f + x\, v_{fg}$	Rearrange.
$\mathbf{x = (v - v_f)/(v_g - v_f)}$	Quality = how far along the horizontal line the state lies (lever rule). Same form for u, h, s.

Numerical: rigid tank cooled (2019 1d, 10 marks)

Given: rigid tank, water, $p_1 = 2$ MPa, $x_1 = 0.8$, $V = 0.5\ m^3$; cooled to $p_2 = 0.4$ MPa. **Find:** Q.



Process on the T-v diagram: rigid tank ⇒ vertical line (constant v)

Step (write this)	Why / what it means
At 2 MPa: $v_f = 0.001177$, $v_g = 0.09959\ m^3/kg$; $u_f = 906.12$, $u_{fg} = 1693.0$ kJ/kg	Saturated-water pressure table at 2 MPa (T_{sat} = 212.4 °C).
$v_1 = 0.001177 + 0.8(0.09959 - 0.001177) = 0.07990\ m^3/kg$	Specific volume of the wet mixture.
$m = V/v_1 = 0.5/0.07990 = 6.26$ kg	Mass in the tank (stays constant: closed system).
$u_1 = 906.12 + 0.8 \times 1693.0 = 2260.5$ kJ/kg	Internal energy at state 1.
$v_2 = v_1 = 0.07990$ (rigid)	Rigid tank: volume and mass fixed ⇒ v fixed. This is the second property that fixes state 2.
At 0.4 MPa: $v_f = 0.001084$, $v_g = 0.4624$; $u_f = 604.22$, $u_{fg} = 1948.9$	Pressure table at 0.4 MPa (143.6 °C). v_2 lies between v_f and v_g, so still wet.
$x_2 = (0.07990 - 0.001084)/(0.4624 - 0.001084) = 0.171$	Quality after cooling: most of the vapour has condensed.
$u_2 = 604.22 + 0.171 \times 1948.9 = 937.2$ kJ/kg	
$Q = m(u_2 - u_1) = 6.26(937.2 - 2260.5)$	First law, closed rigid system: W = 0 (no boundary movement), ΔKE (kinetic energy) = ΔPE (potential energy) = 0.

`Q = -8281 kJ ≈ -8.28 MJ`	Negative ⇒ 8.28 MJ of heat leaves the water.

Memory trick: "**Two to tango**": two independent intensive properties fix a state. Phase rule "**F = C − P + 2**" = "Freedom = Components minus Phases plus 2". Rigid tank = vertical line on T-v.

A9 Exergy: dead state, closed-system and flow exergy, exergy balance, second-law efficiency

Next to add | Very likely · 100% | Hard

PAST-PAPER QUESTIONS

- **2023 Q4(c)** [15] Show that the exergy of a flow stream is $\psi = (h-h_0) - T_0(s-s_0) + V^2/2 + gz$.
- **2022 Q4(c)** [4] What is exergy? Explain its physical significance.
- **2022 Q4(d)** [18] A piston-cylinder holds 0.05 kg steam at 1 MPa, 300 °C. It expands to 200 kPa, 150 °C doing work, losing heat to surroundings at T_0 = 25 °C, p_0 = 100 kPa. Find (i) exergy at initial and final states, (ii) exergy change, (iii) exergy destroyed, (iv) second-law efficiency.
- **2021 Q4(a)** [12] How does useful work differ from actual work? Define entropy and exergy. Show the closed-system exergy expression.
- **2021 Q4(b)** [5] Write down the exergy balance equation for an open system and the meaning of each term.
- **2021 Q3(c)** [20] Steam enters an adiabatic turbine at 6 MPa, 600 °C, 80 m/s and leaves at 50 kPa (printed "50 MPa"), 100 °C, 140 m/s. Power output 5 MW (printed "5 MPa"). Find (i) mass flow, (ii) reversible power, (iii) isentropic efficiency, (iv) second-law efficiency. Surroundings 25 °C.
- **2020 Q3(b)** [5] What is dead state? Is it possible to generate exergy? If not, why?
- **2020 Q3(c)** [15] A 200 m^3 rigid tank holds compressed air at 1 MPa, 300 K. Find the work obtainable if the environment is at 100 kPa, 300 K.
- **2019 Q4(c)** [15] Define exergy. Prove that the maximum theoretical work of a system is $W_c = (U-U_0) + p_0(V-V_0) - T_0(S-S_0) + KE$ (kinetic energy) + PE (potential energy).
- **2018 Q4(a)** [12] What is exergy? Show that the exergy of a system is given by the same expression.
- **2017 Q4(b)** [10] What is exergy? Show that the exergy of a system is $\xi = (U-U_0) + p_0(V-V_0) - T_0(S-S_0) + KE + PE$.

LEARN IT FROM ZERO

Exergy, from zero

Exergy = the **useful work** you could still get from something before it becomes just like its surroundings (the dead state, T_0 = 25 °C, p_0 = 1 atm). A tank of compressed air at room temperature has no extra heat, but it can still push a turbine: it has exergy. Hot coffee has exergy; the same energy spread through the room has none.

Closed system: $\varphi = (u - u_0) + p_0(v - v_0) - T_0(s - s_0) + V^2/2 + gz$. Flowing stream: $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$. Exergy destroyed = $T_0 S_{gen}$. Second-law efficiency = actual work / reversible work.

CALCULATOR **Exergy of a compressed-air tank at room temperature**

Change any number; every step is recalculated and shown with the numbers put in.

1. Mass of air: m = pV/RT = **1000** × **200** / (0.287 × **300**) = **2323** kg
2. Specific exergy: $\varphi = RT_0[(p_0/p - 1) + \ln(p/p_0)]$ = 0.287 × **300** × ((**100** / **1000** - 1) + ln(**1000** / **100**)) = **120.76** kJ/kg

3. Total exergy: X = mφ = **2323** × **120.8** / 1000 = **280.5** MJ

Before you start: exergy = the useful part of energy

Energy is conserved, but its **quality** is not. 1000 kJ in a furnace at 1500 K can produce much more work than 1000 kJ in warm water at 40 °C. **Exergy (availability)** is the maximum useful work that can be obtained from a system as it is brought to equilibrium with the environment.

Dead state: the state of the system when it is in complete equilibrium with the environment (T_0 = 25 °C, p_0 = 1 atm, at rest, at zero elevation, no chemical difference). At the dead state the exergy is zero.

Definitions

- **Exergy**: maximum useful work obtainable from a system at a given state in a given environment (when it is taken reversibly to the dead state).
- **Useful work vs actual work (2021 4a):** $W_u = W - W_{surr} = W - p_0(V_2 - V_1)$. When a system expands it must push back the atmosphere; that part $p_0\Delta V$ is not available for any useful purpose.
- **Reversible work** W_{rev}: maximum useful work between two states; **irreversibility** $I = X_{destroyed} = W_{rev} - W_u = T_0S_{gen}$.
- **Second-law efficiency** $\eta_{II} = W_u/W_{rev}$ (work-producing) or W_{rev}/W_u (work-consuming) = exergy recovered/exergy expended.

Is it possible to generate exergy? (2020 3b) No. Every real process destroys exergy ($X_{destroyed} = T_0S_{gen} \geq 0$) because $S_{gen} \geq 0$ by the second law. A reversible process at best conserves it. So exergy can only be transferred or destroyed, never created (unlike energy, which is always conserved).

Physical significance (2022 4c): exergy tells the true potential of an energy source; it locates and measures losses (where exergy is destroyed: combustion, heat transfer across large ΔT, throttling) and so shows where improvement is possible; it allows fair comparison of systems through η_{II}.

Derivation: exergy of a closed system (2017 4b, 2018 4a, 2019 4c, 2021 4a)

Step (write this)	Why / what it means
Let the system (U, V, S, KE, PE) go to the dead state while exchanging heat δQ only with the environment at T_0; to make it reversible, the heat passes through a reversible heat engine.	Combined system = closed system + reversible engine.
Closed system first law: $-\delta Q - \delta W = dU$ (here δQ = heat leaving the system)	Energy balance.
$\delta W = p\,dV + \delta W_{b,useful}$... write $\delta W = p_0dV + \delta W_{b,u}$	Boundary work = work against the atmosphere + useful boundary work.
Reversible engine: $\delta W_{HE} = \delta Q(1 - T_0/T) = \delta Q - T_0\delta Q/T = \delta Q - T_0(-dS)$	Carnot efficiency of the tiny engine; δQ/T = −dS for the system (reversible heat leaving).
Total useful work: $\delta W_{total,u} = \delta W_{HE} + \delta W_{b,u} = -dU - p_0dV + T_0dS$	Add and use the first-law relation to remove δQ.

Integrate from the given state to the dead state (0): $W_{total,u} = (U - U_0) + p_0(V - V_0) - T_0(S - S_0)$	
Add the kinetic and potential energies, which are fully convertible to work:	They are mechanical energy, 100 % exergy.
$X = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) + KE + PE$	This maximum work = exergy W_c of the closed system. Per kg: $\varphi = (u - u_0) + p_0(v - v_0) - T_0(s - s_0) + V^2/2 + gz$.

Derivation: exergy of a flow stream (2023 4c)

Step (write this)	Why / what it means
A flowing fluid carries flow work pv per kg; of this, p_0v must be spent pushing the atmosphere, so the useful flow work is $(p - p_0)v$.	Flow exergy = closed exergy + exergy of flow work.
$\psi = \varphi + (p - p_0)v$	Add the useful flow work to the fluid's non-flow exergy.
$= (u - u_0) + p_0(v - v_0) - T_0(s - s_0) + V^2/2 + gz + pv - p_0v$	
$= (u + pv) - (u_0 + p_0v_0) - T_0(s - s_0) + V^2/2 + gz$	The p_0v terms cancel.
$\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$	$h = u + pv$.

Exergy balance for an open system (2021 4b)

$$dX_{CV\ (control\ volume)}/dt = \Sigma(1 - T_0/T_k)\dot{Q}_k - (\dot{W} - p_0dV_{CV}/dt) + \Sigma\dot{m}_i\psi_i - \Sigma\dot{m}_e\psi_e - \dot{X}_{destroyed}$$

Terms: rate of change of exergy in the CV; exergy transfer with heat at boundary temperature T_k (heat is worth only its Carnot fraction); exergy transfer with useful work; exergy carried in and out by mass (flow exergy ψ); exergy destroyed $\dot{X}_{destroyed} = T_0\dot{S}_{gen} \geq 0$.

Numerical 1: compressed air tank (2020 3c)

Step (write this)	Why / what it means
$m = PV/(RT) = 1000 \times 200/(0.287 \times 300) = 2323$ kg	
$T = T_0 \Rightarrow u = u_0$; KE = PE = 0	Air is already at environment temperature.
$\varphi = p_0(v - v_0) - T_0(s - s_0) = RT_0(p_0/p - 1) + RT_0\ln(p/p_0)$	Ideal gas: $v = RT/p$ and $s - s_0 = -R\ln(p/p_0)$ at equal T.
$\varphi = 0.287 \times 300[(0.1 - 1) + \ln 10] = 86.1 \times 1.4026 = 120.8$ kJ/kg	
$X = m\varphi = 2323 \times 120.8 = 280{,}500$ kJ ≈ 281 MJ	Maximum work obtainable.

Numerical 2: turbine exergy and second-law efficiency (2021 3c)

Misprint: exit pressure is printed "50 MPa" and power "5 MPa"; the original problem (Cengel) has **50 kPa** and **5 MW**, used here.

Step (write this)	Why / what it means
State 1 (6 MPa, 600 °C): h_1 = 3658.8, s_1 = 7.1693; state 2 (50 kPa, 100 °C): h_2 = 2682.4, s_2 = 7.6953	Superheated table.
$w = (h_1 - h_2) - (V_2^2 - V_1^2)/2 = 976.4 - (140^2 - 80^2)/2000 = 976.4 - 6.6 = 969.8$ kJ/kg	Adiabatic SFEE (steady-flow energy equation).
(i) $\dot{m} = 5000/969.8 = 5.16$ kg/s	
$w_{rev} = \psi_1 - \psi_2 = (h_1 - h_2) - T_0(s_1 - s_2) - \Delta KE = 976.4 + 298.15 \times 0.526 - 6.6 = 1126.6$ kJ/kg	Reversible work = drop in flow exergy (T_0 = 298.15 K).
(ii) $\dot{W}_{rev} = 5.16 \times 1126.6 = 5809$ kW ≈ 5.81 MW	
Isentropic exit: s_{2s} = 7.1693 at 50 kPa: $x_{2s} = (7.1693 - 1.0912)/6.5019 = 0.935$; $h_{2s} = 340.54 + 0.935 \times 2304.7 = 2495.0$ kJ/kg	s_f, s_{fg}, h_f, h_{fg} at 50 kPa.
(iii) $\eta_T = (h_1 - h_2)/(h_1 - h_{2s}) = 976.4/1163.8 = 83.9$ %	Isentropic efficiency.
(iv) $\eta_{II} = \dot{W}/\dot{W}_{rev} = 5000/5809 = 86.1$ %	Second-law efficiency.

Numerical 3: steam expanding in a piston-cylinder (2022 4d, 18 marks)

Missing data: the amount of heat lost is not printed. The original problem (Cengel Example 8-?) states a heat loss of **2 kJ**; that value is assumed.

Step (write this)	Why / what it means
Dead state (25 °C, 100 kPa, compressed liquid ≈ sat. liquid): u_0 = 104.83, v_0 = 0.001003, s_0 = 0.3672	Properties of water at T_0.
State 1 (1 MPa, 300 °C): u_1 = 2793.7, v_1 = 0.2580, s_1 = 7.1246; state 2 (200 kPa, 150 °C): u_2 = 2577.1, v_2 = 0.9599, s_2 = 7.2810	Superheated table.
$X_1 = m[(u_1-u_0) + p_0(v_1-v_0) - T_0(s_1-s_0)] = 0.05[2688.9 + 25.7 - 2014.7] =$ **35.0 kJ**	(i) Closed-system exergy.
$X_2 = 0.05[2472.3 + 95.9 - 2061.3] =$ **25.4 kJ**	
(ii) $\Delta X = X_2 - X_1 = -9.6$ kJ	Exergy decreases.
$W_b = -Q_{out} - m(u_2 - u_1) = -2 + 0.05 \times 216.5 = 8.83$ kJ; $W_u = W_b - p_0 m(v_2 - v_1) = 8.83 - 3.51 = 5.32$ kJ	First law for the actual work, then subtract the work against the atmosphere.
(iii) $X_{destroyed} = X_1 - X_2 - W_u = 9.65 - 5.32 = 4.3$ kJ	Exergy balance (heat lost to the environment at T_0 carries no exergy). Check: $T_0 S_{gen}$ = 298.15[0.05(0.1564) + 2/298.15] = 4.3 kJ.
(iv) $\eta_{II} = W_u/(X_1 - X_2) = 5.32/9.65 = 0.55$ (55 %)	

Memory trick: Closed exergy "**U-p_0V-T_0S**": plus, plus, minus. Flow exergy: replace "U + p_0V" by H. "Exergy is like money in a leaky bag: you can spend it or lose it, never print it."

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2020 3(c) Given: tank 200 m³, air at 1 MPa and 300 K; surroundings 100 kPa, 300 K

Mass: $m = pV/(RT)$

$= 1000 \times 200 / (0.287 \times 300)$

$=$ **2323 kg**

$T = T_0$, so $u = u_0$; no KE or PE.

Specific exergy: $\varphi = RT_0[(p_0/p - 1) + \ln(p/p_0)]$

$= 0.287 \times 300[(0.1 - 1) + \ln 10] = 86.1 \times 1.4026$

$=$ **120.8 kJ/kg**

Exergy (maximum work): $X = m\varphi$

$= 2323 \times 120.8$

$=$ **281 MJ**

A8 Entropy: Clausius inequality, Tds equations, entropy balance, isentropic efficiency Pass set Very likely · 100% Hard

PAST-PAPER QUESTIONS

- **2023 Q3(c)** [18] Air is compressed adiabatically from 100 kPa, 12 °C to 800 kPa at 0.2 kg/s. Isentropic efficiency 80 %. Find (i) exit temperature, (ii) power input.
- **2022 Q3(d)** [9] Explain the increase of entropy principle.
- **2022 Q4(a)** [9] For an ideal gas derive $s_2 - s_1 = c_p \ln(v_2/v_1) + c_v \ln(p_2/p_1)$.
- **2022 Q3(c)** [8] Find the work to compress steam isentropically from 100 kPa to 1 MPa if the inlet is (i) saturated liquid, (ii) saturated vapour.
- **2021 Q3(b)** [10] State the Clausius inequality and prove $ds = (\delta Q/T)_{\text{int rev}}$. Write the entropy rate balance for a control volume and the meaning of each term.
- **2021 Q4(c)** [5] Derive the first and second Tds equations. (Real gas part in A4.)
- **2020 Q2(a)** [6] A system has only a work interaction with its surroundings. Does its entropy change? Explain.
- **2019 Q3(b)** [10] Derive the first and second Tds equations. Show that for an incompressible substance $s_2 - s_1 = \int c(T)dT/T$.
- **2018 Q3(b)** [7] Derive the first and second Tds equations. (KP statement in A7.)
- **2018 Q3(c)** [9] What is a reversible process? Show that $dS = (\delta Q/T)_{\text{int rev}}$.
- **2018 Q4(c)** [7] Write down the entropy rate balance equation for a control volume and describe each term.
- **2017 Q3(b)** [11] State the Clausius inequality and prove $dS = (\delta Q/T)_{\text{int rev}}$. Derive the first and second Tds equations.
- **2017 Q3(c)** [6] Write down the entropy rate balance equation for a control volume and the meaning of each term.
- **2017 Q3(d)** [12] Steam enters a turbine at 30 bar, 400 °C, 160 m/s and leaves as saturated vapour at 100 °C, 100 m/s. Work = 540 kJ/kg; heat transfer occurs at an outer surface temperature of 350 K. Find the entropy production per kg of steam.

LEARN IT FROM ZERO

Entropy, from zero

Entropy is the second law turned into a property. Think of it as a **measure of spread-out-ness** of energy. When heat Q enters at temperature T, entropy rises by Q/T. Friction, mixing and heat flowing across a temperature gap **create** extra entropy ($S_{gen} > 0$) that can never be destroyed. So: $\mathbf{\Delta S = \int \delta Q/T + S_{gen}}$, with $S_{gen} = 0$ only for a perfect (reversible) process.

An **isentropic** process (adiabatic and reversible) is the ideal turbine or compressor. Real machines are compared with it: compressor $\eta = w_s/w_a$, turbine $\eta = w_a/w_s$ (the ideal one is always the better one, so it goes on top for the compressor and at the bottom for the turbine).

CALCULATOR **Ideal-gas entropy change and isentropic compression of air**

Change any number; every step is recalculated and shown with the numbers put in.

1. Entropy change: $\Delta s = c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$ = 1.005 × ln(**600** / **285**) - 0.287 × ln(**800** / **100**) = **0.1514** kJ/kg K
2. Isentropic exit temperature: $T_{2s} = T_1(p_2/p_1)^{0.2857}$ = **285** × pow(**800** / **100**, 0.2857) = **516.2** K
3. Isentropic work: 1.005 × (**516.2** - **285**) = **232.4** kJ/kg
4. Actual work: $w_a = w_s/\eta$ = **232.4** / (**80** / 100) = **290.5** kJ/kg
5. Actual exit temperature: **285** + **290.5** / 1.005 = **574.1** K

Before you start: what entropy is

Entropy S (kJ/K) is a property that measures molecular disorder and, in engineering, the **loss of the ability to do work**. Heat added to a system raises its entropy; friction and other irreversibilities *generate* entropy. It is defined through reversible heat transfer: $dS = (\delta Q/T)_{\text{int rev}}$.

Practical meaning: an **isentropic** (s = constant) process is the ideal, loss-free adiabatic process. Real turbines and compressors are compared with it through the **isentropic efficiency**.

Clausius inequality and $dS = (\delta Q/T)_{\text{int rev}}$ (2017 3b, 2018 3c, 2021 3b)

$\oint (\delta Q/T)_b \leq 0$ (= 0 for an internally reversible cycle, < 0 for an irreversible cycle)

Step (write this)	Why / what it means
Consider a system receiving δQ at boundary temperature T, combined with a reversible cyclic device that supplies this δQ from a reservoir at T_R.	Standard Clausius set-up.
For the reversible device: $\delta Q_R/T_R = \delta Q/T$. Combined work $\delta W_C = T_R(\delta Q/T) - dE_C$.	Proof 3 of A7 gives Q ratio = T ratio for a reversible device.
Over a full cycle of the system: $W_C = T_R \oint \delta Q/T$ (since $\oint dE = 0$).	
The combination exchanges heat with one reservoir only, so by Kelvin-Planck $W_C \leq 0 \Rightarrow$ **$\oint \delta Q/T \leq 0$.**	$T_R > 0$.
For an internally reversible cycle the inequality must be equality (reverse it and the sign flips): $\oint (\delta Q/T)_{\text{int rev}} = 0$.	
A quantity whose cyclic integral is zero depends only on the state: it is a property. Call it entropy: **$dS = (\delta Q/T)_{\text{int rev}}$.**	Same logic as volume: $\oint dV = 0$.

Reversible process (2018 3c): a process that can be reversed leaving no change in either system or surroundings; it has no friction, no finite-ΔT heat transfer, no unrestrained expansion or mixing.

Increase of entropy principle (2022 3d)

For any process 1-2: $S_2 - S_1 \geq \int \delta Q/T$, i.e. $S_2 - S_1 = \int \delta Q/T + S_{gen}$, with $S_{gen} \geq 0$. For an **isolated system** (or system + surroundings) $\delta Q = 0$, so **$\Delta S_{isolated} = S_{gen} \geq 0$**. Entropy of the universe can

increase or stay constant (reversible) but never decrease; processes go only in the direction that increases total entropy. S_{gen} = 0 reversible, > 0 irreversible, < 0 impossible.

Only work interaction (2020 2a): the system still *contains* entropy (every substance above 0 K has entropy). Work carries no entropy, so if the work is done reversibly (e.g. slow frictionless compression with no heat) its entropy does not change. If the work involves irreversibility (paddle-wheel stirring, friction) entropy is generated inside and S increases. It can never decrease by work alone.

Tds equations (asked in 5 papers)

Step (write this)	Why / what it means
Closed simple compressible system, internally reversible process: $\delta Q_{int\ rev} - \delta W_{int\ rev} = dU$	First law in differential form.
$\delta Q_{int\ rev} = T\ dS$ (definition of entropy); $\delta W_{int\ rev} = p\ dV$ (boundary work)	
$T\ dS = dU + p\ dV$, per kg **$T\ ds = du + p\ dv$** — first Tds (Gibbs) equation	Relates only properties, so it holds for ANY process (reversible or not) between the same states.
$h = u + pv \Rightarrow dh = du + p\ dv + v\ dp \Rightarrow du + p\ dv = dh - v\ dp$	
$T\ ds = dh - v\ dp$ — second Tds equation	

Incompressible substance (2019 3b): dv = 0 and du = c dT, so ds = du/T = c(T)dT/T and $\mathbf{s_2 - s_1 = \int_{T1}^{T2} c(T)dT/T} \approx c_{avg}\ln(T_2/T_1)$.

Ideal gas (2022 4a): derive $s_2 - s_1 = c_p\ln(v_2/v_1) + c_v\ln(p_2/p_1)$.

Step (write this)	Why / what it means
$ds = du/T + (p/T)dv = c_v dT/T + R\ dv/v$	First Tds; ideal gas du = c_vdT and p/T = R/v.
$s_2 - s_1 = c_v\ln(T_2/T_1) + R\ \ln(v_2/v_1)$	Integrate with constant c_v.
$T_2/T_1 = p_2v_2/(p_1v_1) \Rightarrow \ln(T_2/T_1) = \ln(p_2/p_1) + \ln(v_2/v_1)$	From pv = RT at both states.
$s_2 - s_1 = c_v\ln(p_2/p_1) + (c_v + R)\ln(v_2/v_1)$	
$s_2 - s_1 = c_p\ln(v_2/v_1) + c_v\ln(p_2/p_1)$	$c_v + R = c_p$. (Other forms: $c_p\ln(T_2/T_1) - R\ln(p_2/p_1)$.)

Entropy rate balance for a control volume (2017 3c, 2018 4c, 2021 3b)

$$dS_{CV\ (control\ volume)}/dt = \Sigma_j\ \dot{Q}_j/T_j + \Sigma\dot{m}_i s_i - \Sigma\dot{m}_e s_e + \dot{\sigma}_{CV}$$

Term	Meaning (kW/K)
dS_{CV}/dt	rate of change of entropy inside the CV (0 at steady state)
$\dot{Q}_j/T_j$	entropy transfer with heat at the part of the boundary where the temperature is T_j
$\dot{m}_i s_i$, $\dot{m}_e s_e$	entropy carried in and out by mass flow
$\dot{\sigma}_{CV}$	rate of entropy production by irreversibilities inside (≥ 0)

Numerical: entropy production in a turbine (2017 3d)

Step (write this)	Why / what it means
$h_1 = 3230.9$, $s_1 = 6.9212$ (30 bar, 400 °C); $h_2 = 2676.1$, $s_2 = 7.3549$ (sat. vapour, 100 °C)	Steam tables.
$\dot{Q}/\dot{m} = \dot{W}/\dot{m} + (h_2 - h_1) + (V_2^2 - V_1^2)/2 = 540 + (-554.8) + (100^2 - 160^2)/2000$	Energy balance first, to find the heat loss.
$\dot{Q}/\dot{m} = 540 - 554.8 - 7.8 = -22.6$ kJ/kg	Heat leaves the turbine at $T_b = 350$ K.
$\dot{\sigma}/\dot{m} = -(\dot{Q}/\dot{m})/T_b + (s_2 - s_1) = 22.6/350 + (7.3549 - 6.9212)$	Steady-state entropy balance: $0 = (\dot{Q}/\dot{m})/T_b + s_1 - s_2 + \dot{\sigma}/\dot{m}$.
$\dot{\sigma}/\dot{m} = 0.0646 + 0.4337 = 0.498$ kJ/kg K	Positive, as it must be.

Numerical: isentropic compression of steam (2022 3c)

Step (write this)	Why / what it means
Steady-flow reversible work $w_{in} = \int v\,dp$	From the second Tds with ds = 0: dh = v dp.
(i) Saturated liquid, $v \approx v_f = 0.001043$ m^3/kg (constant): $w = v(p_2 - p_1) = 0.001043(1000 - 100) =$ **0.94 kJ/kg**	Liquids are nearly incompressible.
(ii) Saturated vapour at 100 kPa: $h_1 = 2675.0$, $s_1 = 7.3589$; at 1 MPa with $s_2 = 7.3589$: $h_2 = 3194.5$ kJ/kg	Superheated table, interpolate at s = 7.3589.
$w = h_2 - h_1 =$ **519.5 kJ/kg**	About 550 times more work than for liquid: that is why the Rankine cycle pumps liquid instead of compressing vapour.

Numerical: air compressor with 80 % isentropic efficiency (2023 3c, 18 marks)

Before you start: isentropic efficiency

Compressor: $\eta_C = w_{isentropic}/w_{actual} = (h_{2s} - h_1)/(h_{2a} - h_1)$. Turbine: $\eta_T = w_{actual}/w_{isentropic}$. First find the ideal exit state (same s), then the actual one.

Step (write this)	Why / what it means
$T_1 = 285.15$ K; $p_2/p_1 = 8$	Data (12 °C).
$T_{2s} = T_1(p_2/p_1)^{(k-1)/k} = 285.15 \times 8^{0.2857} = 516.5$ K	Ideal (isentropic) exit temperature, k = 1.4.
$T_{2a} = T_1 + (T_{2s} - T_1)/\eta_C = 285.15 + 231.4/0.8 =$ **574.4 K (301 °C)**	(i) Actual exit is hotter because friction adds heat to the air.
$\dot{W} = \dot{m}c_p(T_{2a} - T_1) = 0.2 \times 1.005 \times 289.2 =$ **58.1 kW**	(ii) Power input.
Check with air tables (variable c_p): $T_{2a} \approx 569$ K, $\dot{W} \approx 58.0$ kW	Same answer to 1 %.

Memory trick: Tds pair: "**T ds = du + p dv**; swap u→h and you swap p dv → −v dp". Entropy balance: "**change = in with heat + in with mass − out with mass + generated**".

A1 Basic concepts: system, viewpoints, properties, equilibrium, process

Pass set | Likely · 79% | Easy

PAST-PAPER QUESTIONS

- **2023 Q1(c)** [10] Write short notes on (i) quasi-equilibrium process, (ii) intensive property, (iii) thermodynamic equilibrium, (iv) thermodynamic system, (v) external property.
- **2021 Q1(c)** [5] Define continuum. Write down the Gibbs phase rule for a non-reacting system and mention the meaning of each term.
- **2021 Q2(b)** [5] What are the differences between steady state and transient analysis of a system? Explain with example.
- **2020 Q1(a)** [6] Define thermodynamic system and classify it.
- **2019 Q1(a)** [7] What does it mean by macroscopic and microscopic viewpoints of thermodynamics? Explain with example.
- **2019 Q1(c)** [10] Discuss thermodynamic equilibrium.
- **2019 Q2(d)** [6] Write the definition (with figure if necessary): (i) open system, (ii) thermodynamic process, (iii) total energy of a system.
- **2018 Q1(a)** [6] What does it mean by macroscopic and microscopic viewpoints of thermodynamics? Explain with example.
- **2018 Q1(c)** [6] Define: (i) thermodynamic open system; (ii) pure substance and phase; (iii) Gibbs phase rule for non-reacting systems.
- **2017 Q1(a)** [6] Define and write short notes on: (i) open system, (ii) thermodynamic equilibrium, (iii) quasi-equilibrium process.

★ REMEMBER IT

- System types by what crosses the boundary: **open** (mass + energy), **closed** (energy only), **isolated** (nothing).
- Intensive = does not depend on size (T, p, ρ); extensive = doubles when you double the system (m, V, E). Specific (per kg) properties are intensive.

Before you start: what thermodynamics talks about

Thermodynamics is the science of **energy and its conversion** from one form to another (heat into work in an engine, work into cooling in a fridge). Before any law can be applied we must say *what* we are studying: that chosen piece of the world is the **system**.

Everything in this topic is vocabulary. Learn each word with one example and one sketch; these short definitions are worth 5-10 marks in almost every paper.

System, surroundings, boundary and types of system

System: any quantity of matter or region of space chosen for study. **Surroundings**: everything outside it. **Boundary**: the real or imaginary surface separating the two; it may be fixed or moving.

Type	Mass crosses boundary?	Energy crosses boundary?	Example
Closed system (control mass)	No	Yes (heat and work)	Gas in a sealed piston-cylinder
Open system (control volume)	Yes	Yes	Turbine, compressor, pump, nozzle, boiler
Isolated system	No	No	Insulated, rigid, sealed flask; the universe
Adiabatic system	(either)	No heat; work possible	Well-insulated turbine

> **Side note:** "Control mass" = we follow a fixed lump of matter. "Control volume" = we watch a fixed region in space while matter flows through it. That is why the first-law equation for an open system has mass-flow terms.

Macroscopic and microscopic viewpoints (2018 1a, 2019 1a)

Macroscopic (classical) view	Microscopic (statistical) view
Looks at the overall behaviour of the matter using a few measurable quantities: p, T, V, m	Looks at the individual molecules: their positions, velocities and energies
Needs no model of molecular structure	Needs statistics and probability over a huge number of molecules
Valid when the system contains very many molecules (a continuum)	Needed for rarefied gases (very few molecules) and to explain properties
Example: pressure of gas in a tyre read on a gauge	Example: the same pressure explained as the average force of molecules hitting the wall

Continuum (2021 1c): we ignore the gaps between molecules and treat matter as continuous, so that a property such as density has a definite value at every point ($\rho = \lim \delta m/\delta V$ as δV shrinks to a small but molecule-rich volume). It fails when the mean free path of molecules is comparable to the size of the system (high vacuum).

Property, state, and kinds of property

- **Property**: any measurable characteristic of a system (p, T, V, m, U, H, S). Its value depends only on the state, not on how the system got there.
- **State**: the condition of the system described by its properties at an instant.
- **Intensive property**: independent of the size of the system: T, p, density, specific volume. If a block is cut in half, each half has the same T.
- **Extensive property**: depends on size, is additive: m, V, U, H, S.
- **Specific property**: extensive ÷ mass, e.g. $v = V/m$, $u = U/m$; it is intensive.
- **Internal property** depends on the matter inside (u, h, s); **external property** does not depend on the nature of the matter: velocity V and elevation z of the system as a whole (2023 1c).

Thermodynamic equilibrium (2019 1c, 10 marks)

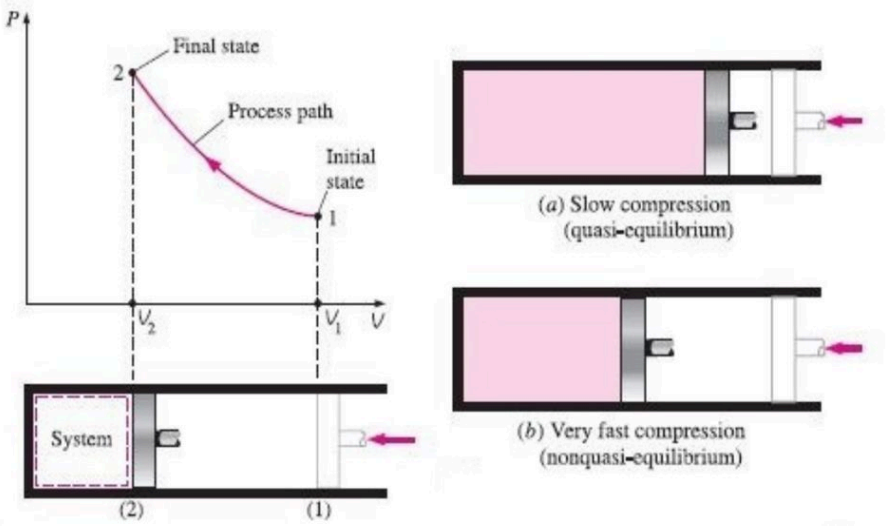
A system is in **thermodynamic equilibrium** when none of its properties change with time if it is isolated from the surroundings. It requires all of the following at once:

1. **Mechanical equilibrium**: no unbalanced force inside the system or between it and the surroundings, so pressure is uniform (apart from gravity effects).

2. **Thermal equilibrium**: no temperature difference within the system or with its surroundings across a diathermic wall.
3. **Chemical equilibrium**: no chemical reaction and no transfer of a species from one part to another.
4. **Phase equilibrium**: the mass of each phase stays constant (no net evaporation or condensation).

Only equilibrium states have single values of p, T, etc., so only they can be plotted on a property diagram and looked up in tables.

Process, path, cycle and quasi-equilibrium process



Quasi-equilibrium (slow) compression vs fast, non-quasi-equilibrium compression (course slide)

- **Process**: any change from one equilibrium state to another. **Path**: the series of states passed through.
- **Quasi-equilibrium (quasi-static) process**: carried out so slowly that the system stays infinitesimally close to equilibrium at every instant; it can be drawn as a solid line on a p-V diagram. A fast process is drawn dashed because intermediate states are not defined. Quasi-equilibrium processes are the ideal standard: they give the maximum work output from expanding devices and the minimum work input to compressors.
- **Cycle**: a series of processes whose final state equals the initial state; every property returns to its initial value ($\oint dU = 0$).
- Iso-processes: isothermal (T const), isobaric (p const), isochoric (v const), isentropic (s const), adiabatic (Q = 0).

Steady vs transient analysis (2021 2b)

Steady state (steady flow)	Transient (unsteady)
Properties at every point inside the control volume do not change with time: $dm_{CV\ (control\ volume)}/dt = 0$, $dE_{CV}/dt = 0$	Properties inside change with time
Mass in = mass out; energy in = energy out	Mass and energy stored in the CV change: $m_{in} - m_{out} = \Delta m_{CV}$
Example: turbine, pump or heat exchanger running at constant load	Example: filling a gas cylinder, discharging a tank, starting up a turbine

Memory trick: Systems: **C-O-I** = Closed (only energy), Open (mass + energy), Isolated (nothing). Equilibrium: **M-T-C-P** = "My Tea Cools Properly": Mechanical, Thermal, Chemical, Phase.

A2 Energy: energy mind map, stored vs transient energy, heat vs work, path and point functions Pass set Likely · 75% Easy

PAST-PAPER QUESTIONS

- **2022 Q2(a)** [10] "An energy transfer can be heat or work, depending on system selection": explain the statement.
- **2021 Q1(a)** [10] Why are heat and work path functions but thermodynamic properties point functions? Describe.
- **2021 Q1(b)** [10] Briefly explain the stored and transient forms of energy with the help of a schematic diagram.
- **2020 Q1(d)** [12] "An energy transfer can be heat or work, depending on system selection": explain the statement.
- **2019 Q2(c)** [12] Draw an energy mind map. Define the different terms in the map.
- **2018 Q1(b)** [10] Draw an energy mind map and describe each term in brief.
- **2017 Q1(b)** [10] Draw an energy mind map and describe each term in it.

★ REMEMBER IT

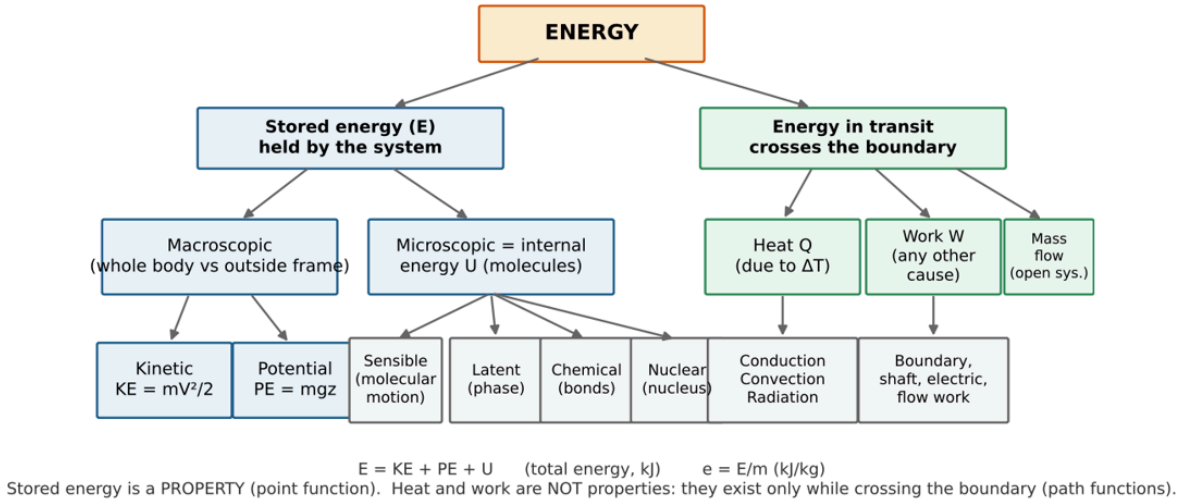
- Heat and work are **path functions** (energy crossing the boundary, inexact differentials δQ, δW). Properties (p, T, U, H, S) are **point functions** (depend only on the state, exact dU).

Before you start: energy, heat and work in plain words

Energy is the capacity to cause change. A system can *store* energy (a moving car, a lifted weight, hot water) and energy can *travel* across the boundary of a system. When energy travels because of a temperature difference we call it **heat (Q)**; when it travels for any other reason (a moving piston, a rotating shaft, an electric current) we call it **work (W)**.

Units: kJ for energy, kW (= kJ/s) for rates. Sign convention used in the course: heat *to* the system is positive, work *done by* the system is positive.

Energy mind map (asked 2017, 2018, 2019: 10-12 marks)



Energy mind map (drawn for this pack from the course classification)

Terms in the map:

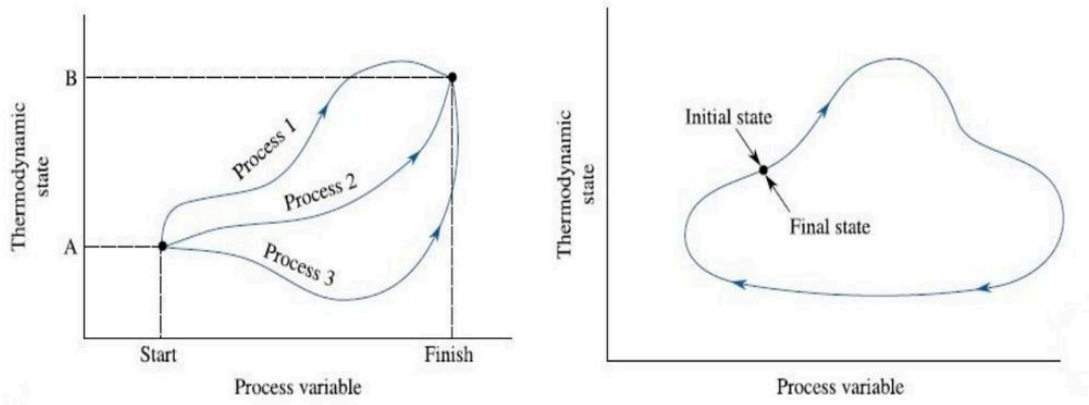
1. **Stored energy E** (property of the system) = macroscopic + microscopic energy: E = KE (kinetic energy) + PE (potential energy) + U.
2. **Kinetic energy** KE = $mV^2/2$: due to motion of the whole system relative to a reference frame.
3. **Potential energy** PE = mgz: due to elevation of the whole system in a gravity field.
4. **Internal energy U**: sum of all microscopic energies of the molecules: **sensible** (translation, rotation, vibration of molecules; shows up as temperature), **latent** (binding between molecules; released or absorbed during phase change), **chemical** (atomic bonds in molecules; released in combustion), **nuclear** (bonds inside the nucleus; released in fission/fusion).
5. **Energy in transit** (not stored, exists only while crossing the boundary): **heat** by conduction, convection and radiation; **work** as boundary (p dV) work, shaft work, electrical work, flow work; and, for open systems, energy carried by **mass flow**.

> **Side note:** A quick test: if you can say "the system *has* it", it is stored energy (a property). If you can only say "the system *received* or *gave* it", it is heat or work.

Stored and transient energy (2021 1b)

Stored forms are possessed by the system and are properties: KE, PE and U (with its sensible, latent, chemical and nuclear parts). **Transient forms** are recognised only as they cross the boundary: heat and work. Once inside, they become part of the stored energy; there is no "heat content" of a body. Sketch the mind map above with a dashed system boundary: arrows of Q and W cross the boundary, while KE, PE, U are written inside it.

Why heat and work are path functions and properties are point functions (2021 1a)

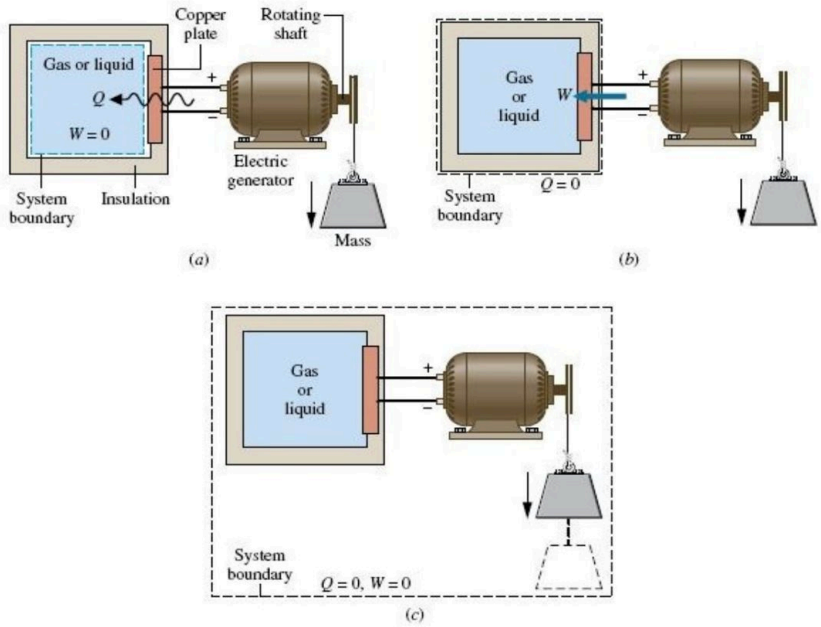


Point functions depend only on end states; path functions depend on the route (course slide)

- A **point function** (property) depends only on the state: going from state 1 to 2 by any path gives the same $\Delta V = V_2 - V_1$. Its differential is exact (dV, dU), and $\oint dV = 0$ over a cycle.
- **Work** $W = \int p\, dV$ is the area under the process curve on a p-V diagram. Different paths between the same states have different areas, so W depends on the path. Likewise Q depends on the path. Their small amounts are written δW, δQ (inexact differentials): $\int_1^2 \delta W = W_{12}$, never $W_2 - W_1$.

- Heat and work are **boundary phenomena**: they exist only while crossing the boundary, so a system never "contains" heat or work and they cannot be properties.

"An energy transfer can be heat or work, depending on the system selection" (2020 1d, 2022 2a)



The same electrical energy seen as heat (a) or work (b), or no transfer at all (c), depending on where the boundary is drawn (course slide)

Example: an electric oven (or resistance heater) heating the air inside it.

1. **Boundary drawn around the air inside the oven only** (heating element outside): the hot element is at a higher temperature than the air, so energy crosses the boundary because of a temperature difference: this is **heat transfer**.
2. **Boundary drawn around the whole oven including the heating element**: electric current (electrons) crosses the boundary, which is **electrical work**, not heat; the energy content of the oven rises because of work done on it.
3. **Boundary drawn around oven + power supply** (in the figure, generator driven by a falling weight): nothing crosses the boundary: Q = 0, W = 0; energy only changes form inside.

Therefore heat and work are not properties of the energy itself; they are labels for *how* the energy crosses a particular boundary. Always define the system first.

> **Memory trick: Heat needs ΔT at the wall; everything else is work.** Mind map = "**K-P-U** stored, **Q-W-M** moving" (kinetic, potential, internal; heat, work, mass).

A4 Ideal gas and real gas: equations of state, compressibility factor Next to add Likely · 75% Medium

PAST-PAPER QUESTIONS

- **2023 Q1(d)** [9] Determine the pressure exerted by 4.0 kg of CO in a 0.03 m^3 container at 220 K using the ideal gas law, van der Waals and Redlich-Kwong equations.
- **2022 Q1(c)** [6] Write short notes on (i) compressibility factor, (ii) principle of corresponding states.
- **2022 Q1(d)** [15] Determine the pressure of nitrogen at T = 175 K and v = 0.00375 m^3/kg using (i) the ideal gas equation of state and (ii) the Soave-Redlich-Kwong equation of state.
- **2021 Q1(d)** [10] 0.45 kg of air in a piston-cylinder undergoes the same three-process cycle (1-2 constant v, 2-3 constant T expansion, 3-1 constant p compression). State 1: 300 K, 101.325 kPa; state 2: 200 kPa. (ii) Sketch on p-v, determine T_2 and (iii) v_3 in m^3/kg.
- **2021 Q4(c)** [3] When can a real gas be treated as an ideal gas? (The Tds part of this question is answered in A8.)
- **2021 Q4(d)** [10] Predict the pressure of nitrogen at T = 175 K and v = 0.00375 m^3/kg using (i) the ideal gas equation and (ii) the van der Waals equation. Compare with the experimental value 10 MPa. For N_2: P_{cr} = 3.39 MPa, T_{cr} = 126.2 K.
- **2018 Q2(d)** [5] What is compressibility factor? When a gas is modelled as an ideal gas, what are the equations to be used?
- **2018 Q4(b)** [6] Write down the virial equation of state. How is this equation used for substances?
- **2017 Q1(d)** [11] What are the equations used when a gas is modelled as an ideal gas? 1 kg of air in a piston-cylinder undergoes a cycle: 1-2 constant specific volume, 2-3 constant temperature expansion, 3-1 constant pressure compression. At state 1, T = 300 K, p = 1.012 bar; at state 2, p = 2.0 bar. (i) Sketch the cycle on p-v, (ii) find T_2, (iii) find v_3 (M = 29).
- **2017 Q4(c)** [8] State the van der Waals equation of state and calculate the unknown constants in terms of critical pressure and temperature.
- **2017 Q8(a)** [5] Write down the comparison between ideal gases and real gases.

Before you start: ideal gas and why real gases deviate

An **ideal gas** is a model in which molecules occupy no volume and do not attract each other. Its equation of state is **Pv = RT** (or PV = mRT = nR_uT). R_u = 8.314 kJ/kmol K is universal; R = R_u/M is the gas constant of a particular gas (air: 8.314/28.97 = 0.287 kJ/kg K).

Real molecules have size (volume b) and attraction (a). These matter when the gas is **dense**: high pressure, low temperature, near the saturation dome or the critical point. The **compressibility factor Z = Pv/RT** measures the deviation: Z = 1 for an ideal gas.

Equations used when a gas is modelled as an ideal gas (2017 1d, 2018 2d)

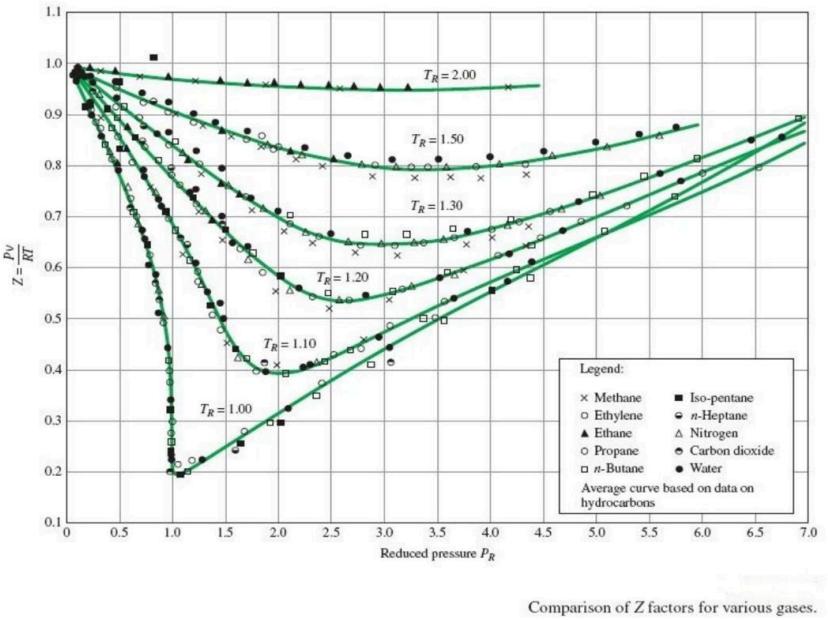
- Pv = RT, PV = mRT, PV = nR_uT, R = R_u/M, $p_1V_1/T_1 = p_2V_2/T_2$ (fixed mass).
- u = u(T) and h = h(T) only: $du = c_vdT$, $dh = c_pdT$, $c_p - c_v = R$, $\gamma = c_p/c_v$.
- Isentropic: pv^γ = const, $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma} = (v_1/v_2)^{\gamma-1}$.

When can a real gas be treated as ideal? (2021 4c) When its pressure is low compared with the critical pressure ($P_R = P/P_{cr} << 1$) and/or its temperature is high compared with the critical temperature ($T_R > 2$), i.e. far from the saturation dome; then Z ≈ 1. Air, N_2, O_2, H_2 at room conditions qualify; steam in a boiler does not.

Compressibility factor and principle of corresponding states (2018 2d, 2022 1c)

Compressibility factor $Z = Pv/(RT) = v_{actual}/v_{ideal}$. $Z = 1$: ideal; $Z < 1$: attraction dominates (gas more compressible); $Z > 1$: molecular volume dominates.

Principle of corresponding states: when normalised by their critical properties (reduced pressure $P_R = P/P_{cr}$, reduced temperature $T_R = T/T_{cr}$), all gases have approximately the same Z. So one **generalised compressibility chart** works for every gas.



Z versus reduced pressure for various gases: they collapse on one family of curves (course slide)

Ideal vs real gas (2017 8a)

Ideal gas	Real gas
Obeys $Pv = RT$ exactly, $Z = 1$	Deviates, $Z \neq 1$; needs vdW (van der Waals), RK (Redlich-Kwong), virial or Z-chart
Molecules have no volume, no intermolecular forces	Molecules have finite volume and attract each other
u, h depend on T only	u, h depend on T and p
Never liquefies	Liquefies when cooled and compressed below T_{cr}
Good model at low p, high T	Must be used at high p, low T, near saturation

van der Waals equation and its constants (2017 4c)

$(P + a/v^2)(v - b) = RT$ or $P = RT/(v - b) - a/v^2$

a/v^2 corrects for intermolecular attraction (it lowers the wall pressure); **b** corrects for the volume occupied by the molecules.

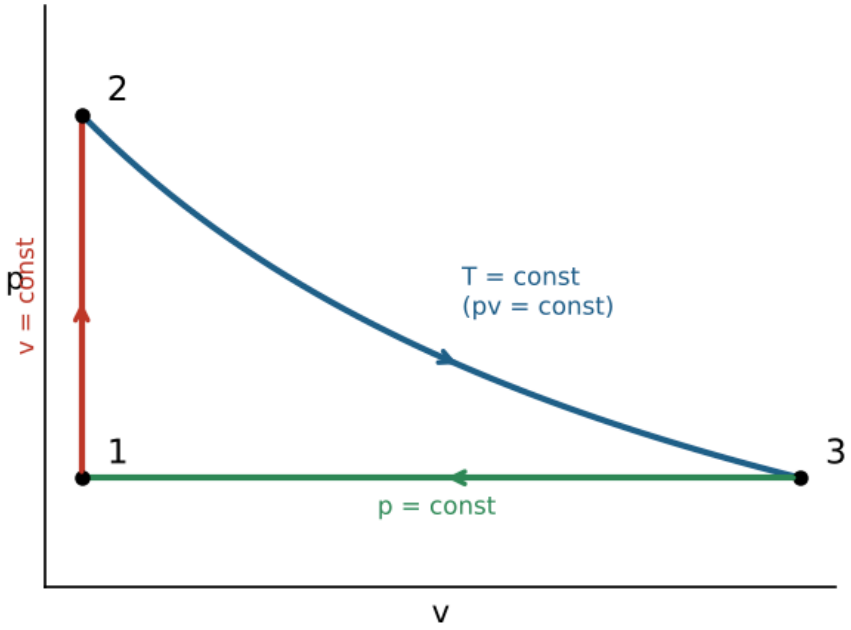
Step (write this)	Why / what it means
At the critical point the isotherm has an inflection: $(\partial P/\partial v)_T = 0$ and $(\partial^2 P/\partial v^2)_T = 0$	Experimental fact: the T_{cr} isotherm is horizontal and changes curvature at the critical point.

$-RT_{cr}/(v_{cr}-b)^2 + 2a/v_{cr}^3 = 0$ ... (1)	First derivative of $P = RT/(v-b) - a/v^2$.
$2RT_{cr}/(v_{cr}-b)^3 - 6a/v_{cr}^4 = 0$ ... (2)	Second derivative.
Divide (2) by (1): $2/(v_{cr} - b) = 3/v_{cr} \Rightarrow$ **$v_{cr} = 3b$**	Eliminates a and T.
Put in (1): $RT_{cr}/(4b^2) = 2a/(27b^3) \Rightarrow$ **$a = 27RT_{cr}b/8$**	
$P_{cr} = RT_{cr}/(2b) - a/(9b^2) = RT_{cr}/(8b) \Rightarrow$ **$b = RT_{cr}/(8P_{cr})$**	Put v = 3b into the vdW equation itself.
$a = 27R^2T_{cr}^2/(64P_{cr})$	Substitute b into $a = 27RT_{cr}b/8$.

Other equations of state (2018 4b, 2022 1d, 2023 1d)

- **Redlich-Kwong:** $P = RT/(v - b) - a/[v(v + b)\sqrt{T}]$, $a = 0.42748R^2T_{cr}^{2.5}/P_{cr}$, $b = 0.08664RT_{cr}/P_{cr}$.
- **Soave-Redlich-Kwong:** $P = RT/(v - b) - a\alpha/[v(v + b)]$, $a = 0.42748R^2T_{cr}^2/P_{cr}$, b as RK, $\alpha = [1 + m(1 - \sqrt{T_R})]^2$, $m = 0.480 + 1.574\omega - 0.176\omega^2$ (ω = acentric factor; N_2: 0.039).
- **Virial equation:** $Z = Pv/RT = 1 + B(T)/v + C(T)/v^2 + \ldots$ B, C are virial coefficients (functions of T only) found from experiment or molecular theory. Use: at low density keep only B (two-term form); the more terms kept, the higher the density for which it is accurate. It has a sound theoretical basis (B represents two-molecule interactions, C three-molecule).

Numerical 1: air three-process cycle (2017 1d, 2021 1d)



The cycle on p-v: 1-2 vertical (constant v), 2-3 hyperbola (pv = const), 3-1 horizontal (constant p)

Step (write this)	Why / what it means
$R = R_u/M = 8.314/29 = 0.2867$ kJ/kg K	Gas constant of air from the given molecular weight.
1-2 constant v: $p/T = \text{const} \Rightarrow T_2 = T_1p_2/p_1$	For fixed v, Pv = RT gives $p \propto T$.
2017: $T_2 = 300 \times 2.0/1.012 =$ **592.9 K**; 2021: $T_2 = 300 \times 200/101.325 =$ **592.2 K**	The 2017 paper prints "temperature is 2.0 bar": it means pressure.

2-3 isothermal: $T_3 = T_2$; 3-1 constant p: $p_3 = p_1$	Read the process names: they give the two properties of state 3.
$v_3 = RT_3/p_3 = 0.2867 \times 592.9/101.2 =$ **1.680 m^3/kg** (2017)	Ideal gas at state 3 (p in kPa gives v in m^3/kg).
2021: $v_3 = 0.2867 \times 592.2/101.325 =$ **1.675 m^3/kg** ($V_3 = 0.45 \times 1.675 = 0.754\ m^3$)	Check: $v_1 = 0.2867 \times 300/101.325 = 0.849\ m^3/kg$, and $v_3/v_1 = p_2/p_1 = 1.97$ as the isotherm requires.

Numerical 2: nitrogen at 175 K, v = 0.00375 m^3/kg (2021 4d, 2022 1d)

Step (write this)	Why / what it means
$R = 8.314/28.013 = 0.2968$ kJ/kg K	Gas constant of N_2.
Ideal: $P = RT/v = 0.2968 \times 175/0.00375 =$ **13,851 kPa**	Error vs 10 MPa measured: +38.5 %.
$a = 27(0.2968)^2(126.2)^2/(64 \times 3390) = 0.175\ m^6kPa/kg^2$; $b = 0.2968 \times 126.2/(8 \times 3390) = 0.00138\ m^3/kg$	vdW constants from the critical data (P_{cr} in kPa).
vdW: $P = 0.2968 \times 175/(0.00375 - 0.00138) - 0.175/0.00375^2 = 21{,}916 - 12{,}444 =$ **9,471 kPa**	Error −5.3 %: far better than ideal.
SRK (2022): $T_R = 175/126.2 = 1.387$; $m = 0.480 + 1.574(0.039) - 0.176(0.039)^2 = 0.541$; $\alpha = [1 + 0.541(1 - \sqrt{1.387})]^2 = 0.817$	Soave temperature correction (acentric factor of N_2 = 0.039 assumed: not given in the paper).
$a = 0.42748(0.2968)^2(126.2)^2/3390 = 0.1769$; $b = 0.08664 \times 0.2968 \times 126.2/3390 = 0.000957$	SRK (Soave-Redlich-Kwong) constants.
SRK: $P = 51.87/(0.00375 - 0.000957) - 0.1769 \times 0.817/[0.00375(0.00375 + 0.000957)] = 18{,}571 - 8{,}190 =$ **≈ 10,400 kPa**	Error about +4 %.

Equation	P (kPa)	Error vs 10,000 kPa
Ideal gas	13,851	+38.5 %
van der Waals	9,471	−5.3 %
Redlich-Kwong	10,087	+0.9 %
Soave-Redlich-Kwong	≈ 10,400	+4 %

Numerical 3: 4.0 kg CO in 0.03 m^3 at 220 K (2023 1d)

CO: M = 28.01, T_{cr} = 133 K, P_{cr} = 3.50 MPa (Cengel Table A-1). The slides solve the same problem with 3.7 kg at 215 K.

Step (write this)	Why / what it means
$v = 0.03/4.0 = 0.0075\ m^3/kg$; $R = 8.314/28.01 = 0.2968$ kJ/kg K	
Ideal: $P = 0.2968 \times 220/0.0075 =$ **8,707 kPa**	
vdW: $a = 27R^2T_{cr}^2/64P_{cr} = 0.1879$, $b = RT_{cr}/8P_{cr} = 0.00141$	Constants for CO.
$P = 65.30/(0.0075 - 0.00141) - 0.1879/0.0075^2 = 10{,}722 - 3{,}340 =$ **7,383 kPa**	

RK: $a = 0.42748R^2T_{cr}^{2.5}/P_{cr} = 2.195$, $b = 0.08664RT_{cr}/P_{cr} = 0.000977$	
$P = 65.30/(0.0075 - 0.000977) - 2.195/[0.0075(0.008477)\sqrt{220}] = 10{,}011 - 2{,}328 =$ **7,683 kPa**	RK is usually closest to the true value.

Memory trick: vdW constants: "**27 over 64** for a, **1 over 8** for b"; RK: "**0.427 and 0.0866**". Ideal gas always over-predicts pressure at high density because it ignores attraction.

A10 Thermodynamic relations: Maxwell relations, expansivity, compressibility, Joule-Thomson Only if time Possible · 46% Hard

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [12] Explain the significance of Maxwell relations. How do they allow entropy changes to be found where direct measurement is impossible?
- **2023 Q4(b)** [8] Short notes: (i) volume expansivity, (ii) isothermal compressibility.
- **2022 Q8(b)** [13] What is a Maxwell relation? Why do we need Maxwell relations? Express them in equation form.
- **2022 Q8(a)** [10] "A cooling effect cannot be achieved by throttling unless the fluid is below its maximum inversion temperature": explain.

★ REMEMBER IT

- Maxwell relations from **"Good Physicists Have Studied Under Very Fine Teachers"** square (G, p, H, S, U, V, F, T) or just remember the first: $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$. Joule-Thomson coefficient $\mu = (\partial T/\partial p)_h$: positive → gas cools when throttled.

Before you start: why we need relations between properties

We can measure p, v and T directly, and c_p with a calorimeter, but **u, h and s cannot be measured**. Thermodynamic relations express their changes in terms of measurable quantities. They come from mathematics: for a property z(x, y), dz = M dx + N dy is exact, so $(\partial M/\partial y)_x = (\partial N/\partial x)_y$.

Maxwell relations (2022 8b, 2023 4a)

Start from the four Gibbs equations (exact differentials of u, h, Helmholtz function a = u − Ts and Gibbs function g = h − Ts):

```
du = T ds - p dv      dh = T ds + v dp      da = -s dT - p dv      dg = -s dT + v dp
```

Applying the exactness test to each gives the **four Maxwell relations**:

$$(\partial T/\partial v)_s = -(\partial p/\partial s)_v \qquad (\partial T/\partial p)_s = (\partial v/\partial s)_p$$
$$(\partial s/\partial v)_T = (\partial p/\partial T)_v \qquad (\partial s/\partial p)_T = -(\partial v/\partial T)_p$$

Significance: (1) The last two replace entropy derivatives, which cannot be measured, by p-v-T derivatives, which can be measured or taken from an equation of state. (2) They make it possible to build steam and refrigerant tables of u, h, s from p-v-T and c_p data. (3) They give general relations for du, dh, ds, $c_p - c_v$ and the Clapeyron equation. **Example of finding Δs without measuring it:** $ds = (c_p/T)dT - (\partial v/\partial T)_p dp$; at constant T, $\Delta s = -\int (\partial v/\partial T)_p dp$, which for an ideal gas gives $-R \ln(p_2/p_1)$.

Volume expansivity and isothermal compressibility (2023 4b)

- **Volume expansivity** (coefficient of volumetric expansion) $\beta = (1/v)(\partial v/\partial T)_p$ (unit 1/K): fractional change in volume per degree temperature rise at constant pressure. For an ideal gas β = 1/T.

- **Isothermal compressibility** $\kappa = -(1/v)(\partial v/\partial p)_T$ (unit 1/kPa): fractional decrease in volume per unit pressure rise at constant temperature (positive because v falls as p rises). For an ideal gas $\kappa = 1/p$; for liquids it is tiny (nearly incompressible).
- They are measurable and appear in $c_p - c_v = vT\beta^2/\kappa$.

Throttling and the maximum inversion temperature (2022 8a)

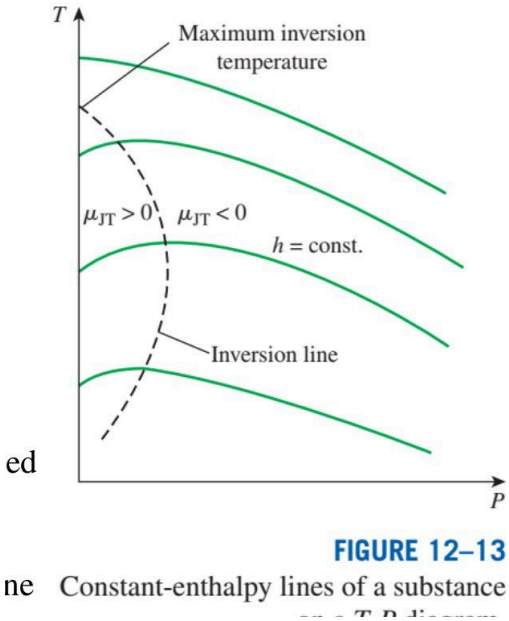


FIGURE 12–13
Constant-enthalpy lines of a substance

Constant-enthalpy lines on a T-p diagram with the inversion line (course slide)

Throttling (valve, porous plug) is a constant-enthalpy process. The **Joule-Thomson coefficient** $\mu_{JT} = (\partial T/\partial p)_h$ tells whether the temperature falls: since pressure drops ($dp < 0$), cooling ($dT < 0$) needs $\mu_{JT} > 0$. Constant-h lines on a T-p diagram rise to a maximum: the locus of these maxima is the **inversion line** ($\mu_{JT} = 0$). To the left of it (lower p) $\mu_{JT} > 0$ and throttling cools; to the right it heats. The highest temperature on the inversion line is the **maximum inversion temperature**. Above it, every constant-h line has $\mu_{JT} < 0$, so throttling at any pressure warms the gas. Hence **cooling by throttling is possible only below the maximum inversion temperature** (that is why H_2 and He, whose inversion temperatures are 202 K and 40 K, must be pre-cooled before Joule-Thomson liquefaction).

> **Memory trick:** Maxwell from Gibbs: "**T-s and p-v are partners**": each relation pairs T with s and p with v, cross-differentiated; the sign is minus when p and T sit on the same side as in "**−s dT − p dv**".

Section B: Power cycles, refrigeration, gas mixtures, psychrometry

B6 Refrigeration: vapour compression and vapour absorption systems

Next to add | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [8] Differentiate between vapour compression and vapour absorption refrigeration systems.
- **2022 Q6(b)** [12] With a neat schematic, describe the working principle of an ammonia-water absorption refrigeration system.
- **2021 Q7(a)** [12] With a neat schematic, describe the working principle of a vapour compression refrigeration system. Write its advantages and disadvantages.
- **2020 Q5(a)** [10] Write down the advantages of vapour absorption refrigeration over vapour compression refrigeration.
- **2019 Q7(a)** [10] With a neat schematic, describe the working principle of a vapour absorption refrigeration system.
- **2018 Q6(b)** [12] Why is VCR (vapour compression refrigeration) called work-operated and absorption refrigeration heat-operated? With the p-h diagram, describe the VCR mechanism.
- **2017 Q6(c)** [8] Describe the mechanism of a vapour compression refrigeration system with p-h and T-s diagrams.
- **2017 Q6(d)** [10] An ice plant produces 10 tonnes of ice per day at 0 °C from water at 20 °C. Estimate the compressor motor rating if COP (coefficient of performance) = 2.5 and overall electro-mechanical efficiency = 90 %.

★ REMEMBER IT

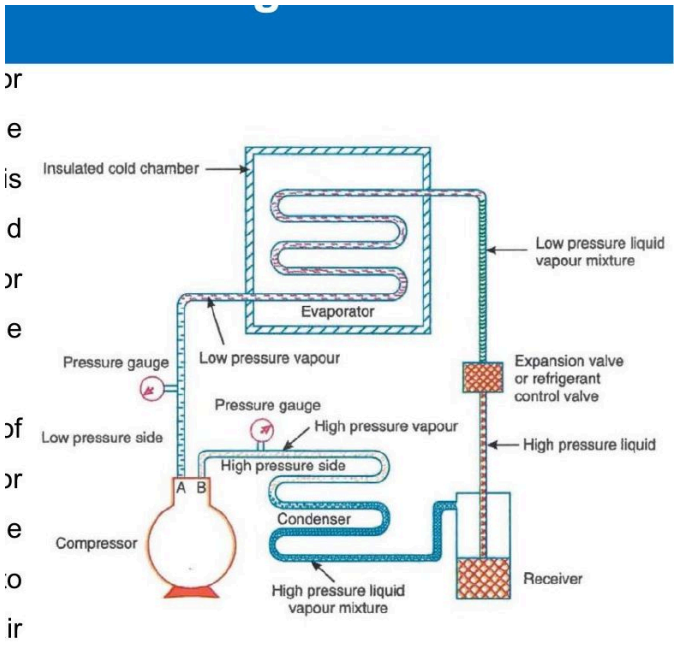
- Vapour compression: **compressor → condenser → expansion valve → evaporator**; COP = $(h_1 - h_4)/(h_2 - h_1)$. Absorption: the compressor is replaced by **absorber + pump + generator** (heat-driven; NH_3-water or LiBr-water).

Before you start: how a refrigerator moves heat uphill

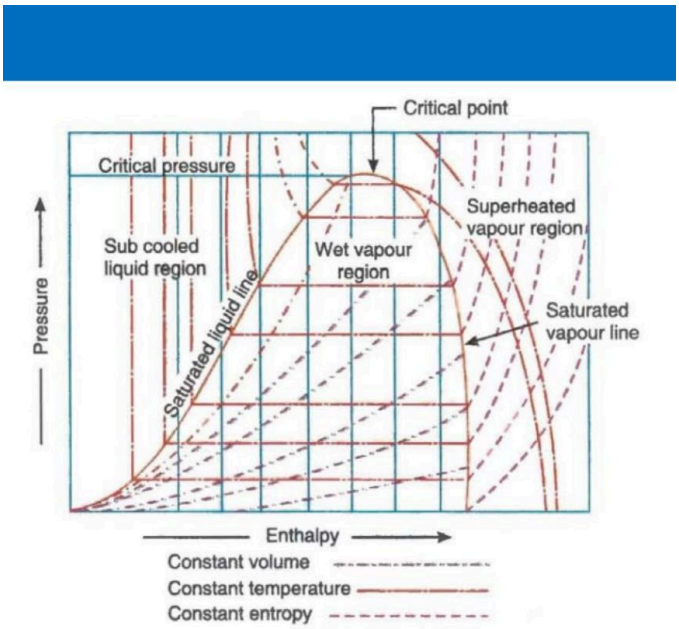
A refrigerant boils at a low temperature when its pressure is low. In the **evaporator** it boils and absorbs heat from the cold space; the vapour is then **compressed** to a high pressure so that it can condense at a temperature above the room; in the **condenser** it rejects heat to the surroundings; finally an **expansion (throttle) valve** drops its pressure again. **COP** (coefficient of performance) = refrigerating effect/work input.

Ton of refrigeration (TR) = rate of heat removal that freezes 1 short ton of water at 0 °C in 24 h ≈ 211 kJ/min = 3.517 kW.

Vapour compression refrigeration (2017 6c, 2018 6b, 2021 7a)



Ideal vapour-compression refrigeration cycle: schematic and T-s (course slide)



The same cycle on the p-h diagram (course slide)

1. **1-2 isentropic compression** in the compressor: saturated vapour at evaporator pressure → superheated vapour at condenser pressure. $w_{in} = h_2 - h_1$.
2. **2-3 constant-pressure heat rejection** in the condenser: vapour → saturated liquid. $q_H = h_2 - h_3$.
3. **3-4 throttling** in the expansion valve: $h_4 = h_3$ (vertical line on p-h); pressure and temperature drop, some liquid flashes to vapour.
4. **4-1 constant-pressure heat absorption** in the evaporator: the mixture evaporates taking heat from the refrigerated space. $q_L = h_1 - h_4$.

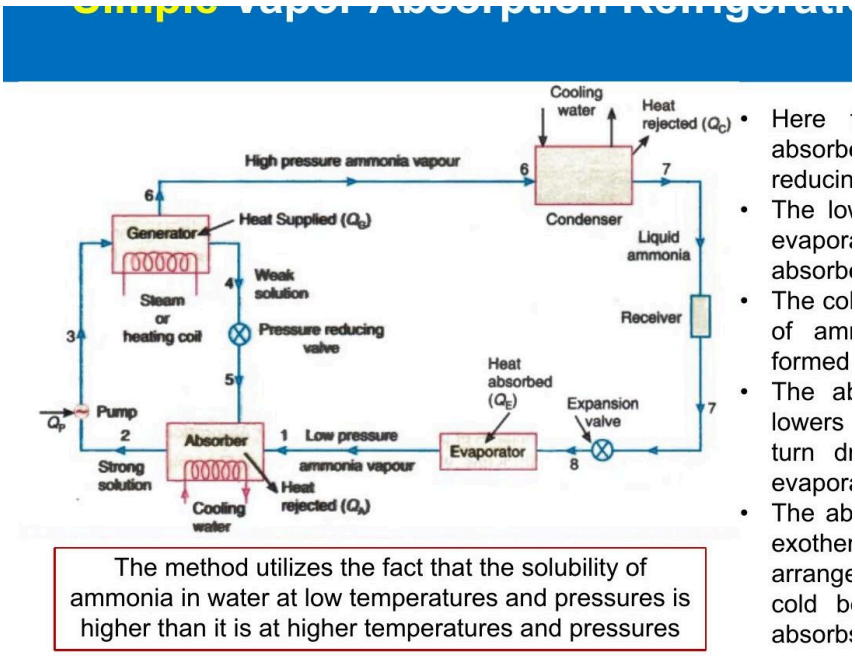
$$COP_R = q_L/w_{in} = (h_1 - h_4)/(h_2 - h_1)$$

Advantages of VCR: high COP; compact, small size for a given capacity; temperature of the evaporator can be adjusted easily; low running cost where electricity is available. **Disadvantages:**

needs high-grade energy (electricity) to drive the compressor; compressor is noisy, has many moving parts and needs maintenance; refrigerant leakage (some refrigerants harm the ozone layer or cause global warming).

Work-operated vs heat-operated (2018 6b): VCR is called **work-operated** because the refrigerant vapour is raised to the condenser pressure by mechanical work in a compressor. An absorption system is **heat-operated** because the compressor is replaced by an absorber-pump-generator set in which the pressure rise is achieved mainly by supplying **heat** to the generator; the pump work for liquid is negligible.

Vapour absorption refrigeration: ammonia-water (2019 7a, 2022 6b)



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Ammonia absorption refrigeration system (course slide)

1. Condenser, expansion valve and evaporator work exactly as in VCR, with **ammonia (NH_3)** as refrigerant and **water** as absorbent.
2. Ammonia vapour leaving the evaporator enters the **absorber**, where it dissolves in weak aqua-ammonia solution. The dissolving releases heat, which is removed by cooling water (low temperature helps absorption).
3. The **strong solution** (rich in NH_3) is pumped by a small liquid **pump** to the generator pressure (little work because liquid is pumped).
4. In the **generator**, heat from a low-grade source (steam, waste heat, solar, gas flame) drives ammonia vapour out of the solution. A **rectifier** removes water vapour so only dry ammonia goes to the condenser.
5. The **weak solution** returns from the generator to the absorber through a **pressure-reducing valve**, passing a **regenerator (heat exchanger)** that preheats the strong solution and cools the weak solution.
6. Ammonia vapour condenses in the condenser, is throttled and evaporates in the evaporator, producing the refrigeration effect.

Advantages of VAR (vapour absorption refrigeration) over VCR (2020 5a) and comparison (2023 7a)

Vapour compression (VCR)	Vapour absorption (VAR)
Driven by mechanical work (compressor, electricity)	Driven by heat (generator): waste heat, steam, solar, gas
Compressor: noisy, many moving parts, wear	Only a small pump moves: quiet, little wear, long life
High COP (2-5)	Low COP (0.3-1.2)
Performance falls at part load; liquid refrigerant may damage the compressor	Works well at part load; no damage from liquid in the evaporator outlet
Compact, lower capital cost for small units	Bulky, higher capital cost, but cheap to run where waste heat exists
Refrigerants: R-134a, R-410A, ammonia	Pairs: NH_3-water, water-LiBr

Advantages of VAR: uses low-grade or waste energy instead of electricity; few moving parts, so quiet and low maintenance; no lubrication problem; capacity is easily controlled by the generator heat input; not damaged by liquid refrigerant leaving the evaporator.

Numerical: ice plant motor rating (2017 6d)

Step (write this)	Why / what it means
Heat removed per kg of water: $q = c_w(20 - 0) + h_{if} = 4.187 \times 20 + 335 = 418.74$ kJ/kg	Cool water to 0 °C (sensible) and freeze it (latent heat of fusion 335 kJ/kg).
$\dot{Q}_L = 10{,}000 \times 418.74/(24 \times 3600) = 48.47$ kW (≈ 13.8 TR (ton of refrigeration (3.5 kW)))	10 tonnes = 10,000 kg per day.
$\dot{W} = \dot{Q}_L/COP = 48.47/2.5 = 19.39$ kW	Compressor shaft power.
Motor rating = 19.39/0.90 = 21.5 kW	Divide by the electro-mechanical efficiency.

> **Memory trick:** VCR order "**Com-Con-Ex-Ev**". VAR replaces the compressor with "**A-P-G**": Absorber, Pump, Generator (plus rectifier and heat exchanger).

B1 Power cycle basics: air-standard assumptions, gas vs vapour cycles, Carnot cycle

Next to add · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [10] Write down the classification of thermodynamic cycles. Discuss the processes of the Carnot cycle with diagrams.
- **2022 Q6(a)** [8] What is an air-standard cycle? What are its assumptions?
- **2021 Q5(a)** [8] Write down the air-standard assumptions. What are the differences between vapour cycles and gas power cycles?
- **2020 Q4(a)** [8] What is a power cycle? How does the Carnot efficiency vary with boiler and condenser pressure?
- **2020 Q4(b)** [4] Why is the Carnot cycle not used in practice?
- **2019 Q6(a)** [8] What is an air-standard cycle? Write down the air-standard assumptions.
- **2019 Q5(a)** [8] What is mean temperature of heat addition? Show that cycle efficiency is a function of it.
- **2018 Q5(a)** [7] What is an air-standard cycle? Write down the air-standard assumptions of an actual cycle.
- **2018 Q6(a)** [5] Write down the basic differences between gas power cycles and vapour cycles. Why is the Carnot vapour power cycle not used in practice?
- **2017 Q5(c)** [4] What is air-standard cycle efficiency? (Diesel derivation part in B4.)

★ REMEMBER IT

- Air-standard assumptions: air is the working fluid in a closed loop, ideal gas; combustion is replaced by heat addition from outside; exhaust by heat rejection; all processes internally reversible. Cold-air-standard: also constant c_p, c_v at 25 °C.

Before you start: what a power cycle is

A **power cycle** is a repeated series of processes in which a working fluid takes heat from a hot source, produces net work, rejects heat to a sink and returns to its starting state. Car engines, gas turbines and steam power plants all run on power cycles. A **refrigeration cycle** runs the other way: it takes work and moves heat from cold to hot.

Real cycles are complicated (combustion, friction, changing composition), so we analyse **idealised cycles** that keep the main features but are simple enough to calculate.

Classification of thermodynamic cycles (2023 5a)

- **Power cycles** (produce work) vs **refrigeration / heat pump cycles** (consume work).
- **Gas cycles**: working fluid stays a gas throughout (Otto, Diesel, Dual, Brayton, Stirling, Ericsson). **Vapour cycles**: fluid changes between liquid and vapour (Rankine, vapour-compression refrigeration).
- **Closed cycles**: the same fluid is returned to the start (steam plant, closed gas turbine). **Open cycles**: fresh fluid enters each cycle and exhaust leaves (car engine, open gas turbine).
- **Internal combustion** (fuel burnt inside the system: petrol and diesel engines, gas turbines) vs **external combustion** (heat supplied from outside through a boiler or heat exchanger: steam plant, Stirling engine).

Air-standard cycle and air-standard assumptions (2018, 2019, 2021, 2022)

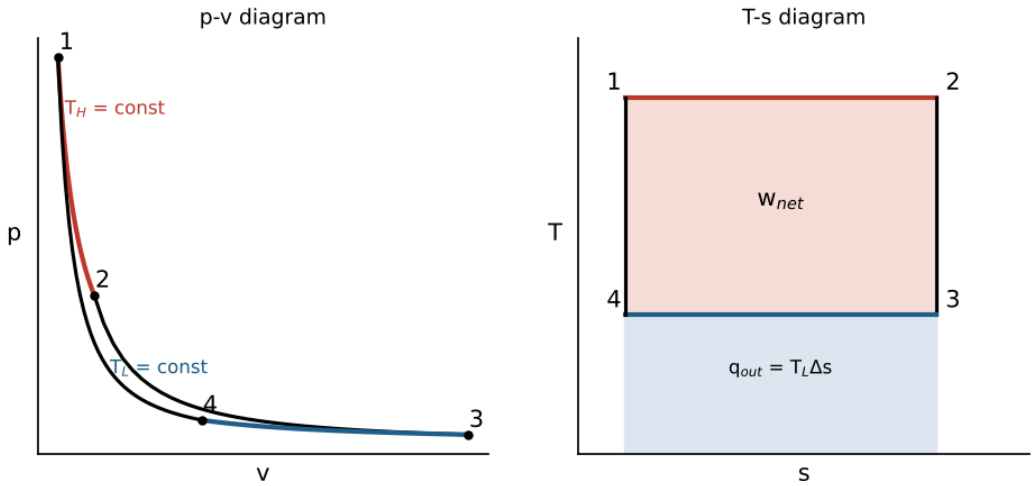
An **air-standard cycle** is the idealised model of a gas power cycle (IC engine or gas turbine) in which the working fluid is taken as air undergoing a closed cycle. Its efficiency is called the **air-standard efficiency** (2017 5c): $\eta = w_{net}/q_{in}$ of that ideal cycle, which depends only on ratios such as r and γ.

1. The working fluid is **air**, which circulates continuously in a **closed loop** and always behaves as an **ideal gas**.
2. All processes making up the cycle are **internally reversible**.
3. The **combustion process is replaced by a heat-addition process** from an external source.
4. The **exhaust process is replaced by a heat-rejection process** that restores the fluid to its initial state.
5. **Cold-air-standard assumption** (extra): specific heats are constant at their room-temperature (25 °C) values: $c_p = 1.005$, $c_v = 0.718$ kJ/kg K, $\gamma = 1.4$.

Gas power cycles vs vapour cycles (2018 6a, 2021 5a)

Gas power cycle	Vapour power cycle
Working fluid stays gas (air / combustion gases)	Fluid changes phase (water ↔ steam)
Heat added and rejected with changing temperature	Most heat added and rejected at constant temperature (boiling, condensing)
Compression of gas needs large work: back work ratio 40-80 %	Liquid is pumped: pump work only 1-2 % of turbine work
Usually internal combustion, open cycle (engines, gas turbines)	External combustion, closed cycle (steam plant)
Examples: Otto, Diesel, Dual, Brayton	Examples: Rankine, Carnot vapour cycle

Carnot cycle (2023 5a)



1-2 isothermal heat addition at T_H, 2-3 isentropic expansion, 3-4 isothermal heat rejection at T_L, 4-1 isentropic compression

Carnot cycle on p-v and T-s diagrams

1. **1-2 reversible isothermal heat addition** at T_H: gas expands slowly, taking Q_H from the source.
2. **2-3 reversible adiabatic (isentropic) expansion**: temperature falls from T_H to T_L, work is produced.
3. **3-4 reversible isothermal heat rejection** at T_L: gas is compressed, giving Q_L to the sink.

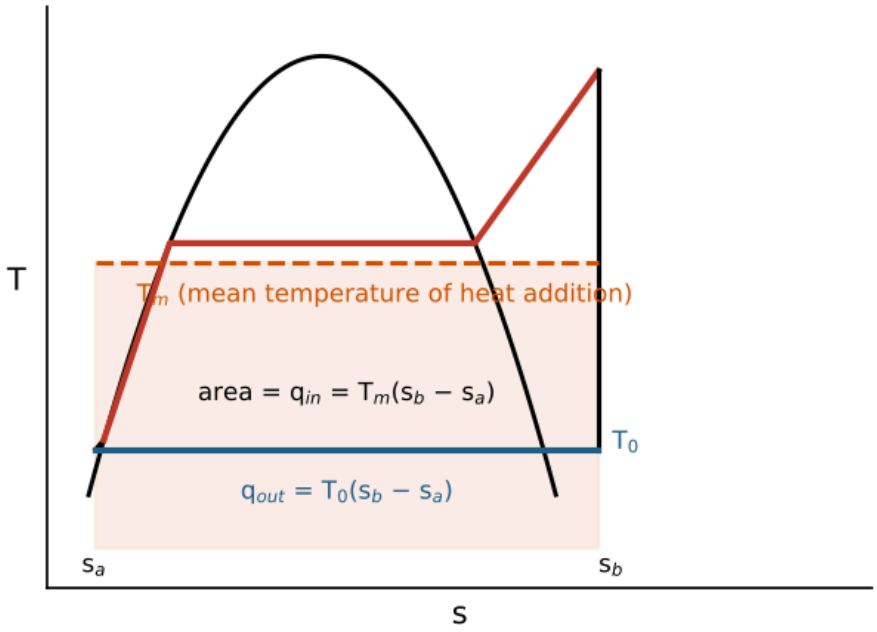
4. **4-1 reversible adiabatic compression**: temperature rises back to T_H.

On T-s it is a rectangle: $Q_H = T_H(s_2 - s_1)$, $Q_L = T_L(s_2 - s_1)$, so $\eta = 1 - T_L/T_H$, the highest efficiency possible between the two temperatures.

Why the Carnot (vapour) cycle is not used in practice (2018 6a, 2020 4b):

1. Isothermal heat addition is possible only inside the wet dome, so T_H is limited to the critical temperature (374 °C for water), which keeps the efficiency low.
2. The turbine would have to expand steam of low quality: water droplets erode the blades.
3. Compression ends at saturated liquid starting from a wet mixture: a compressor cannot handle a two-phase mixture, and stopping condensation exactly at the right quality is impossible to control.
4. Isothermal processes need very slow heat transfer (huge surface or very long time), and the net work per cycle is small compared with the size of the machine.

Mean temperature of heat addition (2019 5a, 2020 4a)



Real heat addition 2-3 replaced by an equivalent Carnot heat addition at T_m

Step (write this)	Why / what it means
In the Rankine cycle heat is added from 2 to 3 at varying temperature. Define T_m such that $q_{in} = T_m(s_3 - s_2)$.	The area under 2-3 on T-s equals the area of a rectangle of height T_m.
$T_m = (h_3 - h_2)/(s_3 - s_2)$	For constant-pressure heat addition $q = h_3 - h_2$.
Heat is rejected at constant T_0 (condenser): $q_{out} = T_0(s_3 - s_2)$	Condensation is isothermal.
$\eta = 1 - q_{out}/q_{in} = 1 - T_0/T_m$	So efficiency rises when T_m rises or T_0 falls.

Effect of boiler and condenser pressure: raising the boiler pressure raises T_{sat} and so T_m → higher η (but more moisture at turbine exit). Lowering the condenser pressure lowers T_0 → higher η (limited by the cooling-water temperature and air leakage). The same reasoning applies to the Carnot efficiency $1 - T_L/T_H$, with $T_H = T_{sat}(p_{boiler})$ and $T_L = T_{sat}(p_{condenser})$.

Memory trick: Air-standard assumptions = "**A-R-C-E**": Air ideal gas in a closed loop, Reversible, Combustion → heat in, Exhaust → heat out (+ Cold = constant c_p). Carnot = "**T-S-T-S**": isoThermal, isentropic, isoThermal, isentropic.

B8 Psychrometry: moist-air properties, derivations, chart and numericals

Pass set | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2023 Q8(a)** [10] Short notes: (i) DBT (dry-bulb temperature), (ii) WBT (wet-bulb temperature), (iii) specific humidity, (iv) relative humidity.
- **2023 Q8(b)** [10] Derive the expression for degree of saturation.
- **2023 Q8(c)** [5] Draw and mention the different parameters of a psychrometric chart.
- **2022 Q8(c)** [12] Air at 1.0132 bar, DBT 32 °C, WBT 26 °C. Find (i) p_v, (ii) φ, (iii) ω, (iv) enthalpy.
- **2021 Q8(b)** [10] Short notes: (i) psychrometry, (ii) relative humidity, (iii) moist air, (iv) wet-bulb depression, (v) dew point temperature.
- **2021 Q8(a)** [10] With the psychrometric chart, describe the change of specific humidity and relative humidity during cooling and dehumidification.
- **2021 Q8(c)** [15] Air at 1.0132 bar, DBT 30 °C, WBT 27 °C. Find (i) p_v, (ii) ω, (iii) dew point, (iv) φ, (v) μ, (vi) density, (vii) enthalpy.
- **2020 Q6(b)** [6] What are absolute humidity, relative humidity and degree of saturation?
- **2019 Q8(b)** [5] Why does a wet cloth dry slower in summer than in winter?
- **2019 Q8(a)** [5] Why is the dry-bulb temperature always greater than the wet-bulb temperature?
- **2019 Q8(c)** [9] Prove that $\varphi = \mu/[1 - (1 - \mu)(p_s/p)]$.
- **2019 Q8(d)** [16] Air at 1.0132 bar, DBT 35 °C, WBT 31 °C. Find (i) p_v, (ii) ω, (iii) φ, (iv) dew point, (v) enthalpy, (vi) specific volume.
- **2018 Q8(b)** [5] Why do we feel sweaty in summer and dry in winter?
- **2018 Q8(a)** [8] Define (i) psychrometrics, (ii) relative humidity, (iii) degree of saturation, (iv) dew point temperature.
- **2018 Q8(c)** [10] Show that the enthalpy of the mixture remains constant during an adiabatic saturation process.
- **2018 Q8(d)** [12] Air at 1.0132 bar, DBT 32 °C, WBT 24 °C. Find (i) p_v, (ii) ω, (iii) φ, (iv) μ, (v) enthalpy.
- **2017 Q7(a)** [7] What is meant by 'psychrometry'? Why does a wet cloth dry faster in winter than in summer?
- **2017 Q7(b)** [8] Distinguish between dry-bulb and wet-bulb temperature. What is degree of saturation?
- **2017 Q7(c)** [10] Derive an expression for specific humidity and show that it is a function of vapour pressure and barometric pressure.
- **2017 Q7(d)** [10] Air at DBT 30 °C, WBT 18 °C, barometer 760 mm Hg. Find (i) p_v, (ii) φ, (iii) dew point.

LEARN IT FROM ZERO

Psychrometry, from zero

Air always carries invisible water vapour. Warm air can hold more than cold air. **Relative humidity** φ says how full the air is compared with the most it could hold at that temperature (p_v/p_s). **Specific humidity** ω says how many kg of water per kg of dry air: $\omega = 0.622\ p_v/(p - p_v)$. Cool air without changing ω and φ rises until it hits 100 %: that temperature is the **dew point**, where water starts to condense (cold glass sweating).

The enthalpy per kg of dry air is $h = 1.005t + \omega(2501 + 1.88t)$ kJ/kg: the first part warms the dry air, the second evaporates and warms the water. Put any temperature and RH (relative

humidity) into the calculator; the graph below is a mini psychrometric chart.

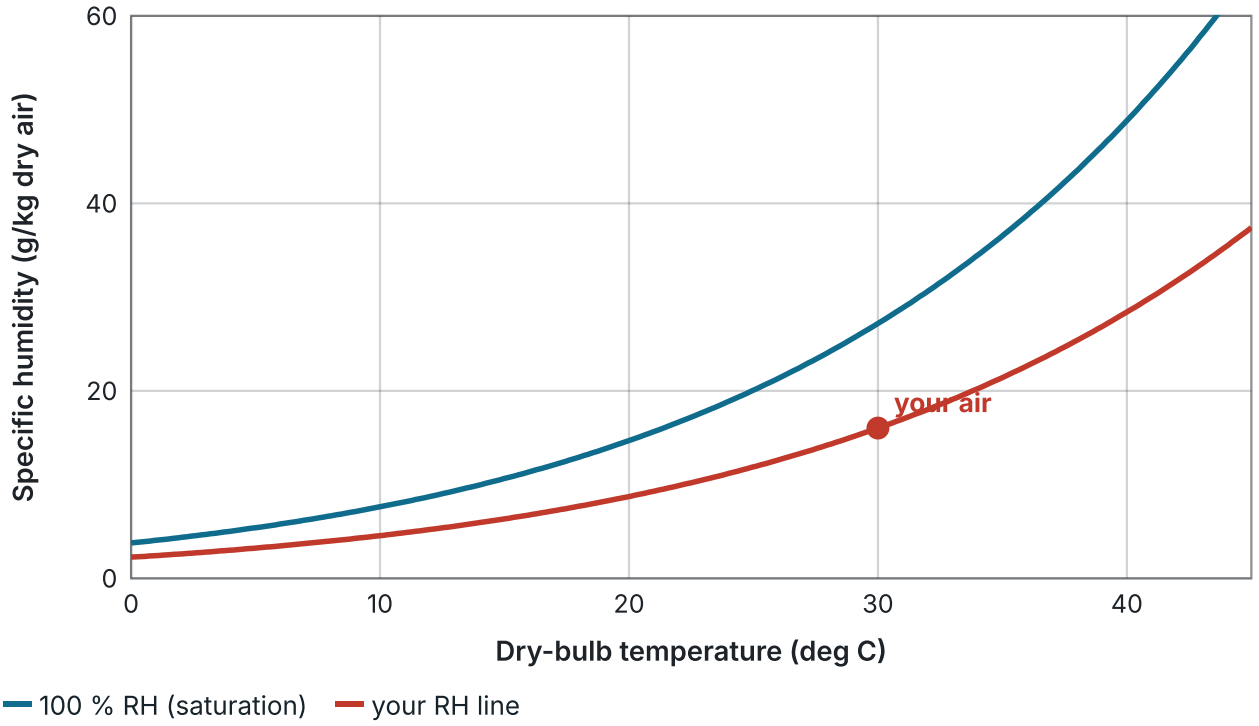
CALCULATOR Moist-air properties from temperature and relative humidity

Change any number; every step is recalculated and shown with the numbers put in.

1. Saturation pressure (Magnus formula; use the steam table in the exam): 0.61078 × exp(17.27 × **30** / (**30** + 237.3)) = **4.2429** kPa
2. Vapour pressure: $p_v = \varphi p_s$ = **60** / 100 × **4.243** = **2.5458** kPa
3. Specific humidity: $\omega = 0.622 p_v/(p - p_v)$ = 0.622 × **2.546** / (**101.3** - **2.546**) = **0.01603** kg/kg dry air
4. Enthalpy: h = 1.005t + ω(2501 + 1.88t) = 1.005 × **30** + **0.01603** × (2501 + 1.88 × **30**) = **71.15** kJ/kg dry air
5. Dew point: 237.3 × ln(**2.546** / 0.61078) / (17.27 - ln(**2.546** / 0.61078)) = **21.38** °C
6. Degree of saturation: $\mu = \omega/\omega_s$ = **0.01603** / (0.622 × **4.243** / (**101.3** - **4.243**)) = **0.5897**
7. Partial pressure of dry air: **101.3** - **2.546** = **98.779** kPa
8. Density of dry air: **98.78** / (0.287 × (**30** + 273.15)) = **1.1353** kg/m³

INTERACTIVE Mini psychrometric chart: specific humidity vs temperature

Move the sliders and watch the curve change.



ω: **16.03** g/kg

Move left from your point at constant ω until you meet the blue curve: that temperature is the dew point.

Before you start: psychrometry = the study of moist air

Air always carries some **water vapour**. Psychrometry is the study of the properties of air-water vapour mixtures, used in air conditioning, drying and cooling towers. Atmospheric air is treated as a mixture of **dry air** and **water vapour**, both ideal gases; by Dalton, $p = p_a + p_v$.

Steam table value needed: the **saturation pressure p_s at a temperature** (e.g. 30 °C → 0.04247 bar). All formulas below use p_v and p_s in the same unit as p (bar here).

Definitions (2017 7b, 2018 8a, 2020 6b, 2021 8b, 2023 8a)

- **Dry air**: air with no water vapour. **Moist air**: mixture of dry air and water vapour. **Saturated air**: moist air holding the maximum vapour possible at its temperature ($p_v = p_s$).
- **Dry-bulb temperature (DBT)**: temperature of the air measured by an ordinary thermometer, unaffected by moisture.
- **Wet-bulb temperature (WBT)**: temperature shown by a thermometer whose bulb is covered by a wet wick with air flowing over it; evaporation cools it below DBT.
- **Wet-bulb depression**: DBT − WBT; zero for saturated air.
- **Dew point temperature (DPT)**: temperature at which water vapour starts to condense when moist air is cooled at constant pressure: T_{sat} at p_v.
- **Specific humidity (humidity ratio, absolute humidity) ω**: mass of water vapour per kg of dry air: $\omega = m_v/m_a$ (kg/kg dry air).
- **Relative humidity φ**: ratio of the actual vapour mass to the maximum the air could hold at the same temperature: $\varphi = m_v/m_g = p_v/p_s$.
- **Degree of saturation μ**: ω/ω_s, the ratio of the actual specific humidity to that of saturated air at the same DBT.
- **Psychrometry (psychrometrics)**: the science of the thermodynamic properties of moist air and its processes.

Everyday questions

Why is DBT always greater than WBT? (2019 8a) At the wet bulb, water evaporates into the passing air; the latent heat of evaporation is taken from the bulb, so it cools below the air temperature. Evaporation stops only when the air is saturated (φ = 100 %), and then WBT = DBT. Since air is never more than saturated, DBT ≥ WBT.

Why does a wet cloth dry faster in winter than in summer? (2017 7a, 2019 8b) Drying depends on the difference between the vapour pressure at the cloth surface and p_v of the air, i.e. on the relative humidity. In Bangladesh the summer (monsoon) air is hot but very humid (φ 80-95 %), so it can take little extra vapour and drying is slow. Winter air is dry (low φ, large wet-bulb depression), so evaporation is fast. (In a climate where summer is dry, the opposite would hold: the answer is about humidity, not temperature alone.)

Why do we feel sweaty in summer and dry in winter? (2018 8b) The body cools by evaporating sweat. In humid summer air, φ is high, so sweat evaporates slowly and stays on the skin: we feel sticky and sweaty. In winter the air is dry (low p_v), moisture evaporates quickly from the skin and lips, so they feel dry.

Derivation: specific humidity (2017 7c)

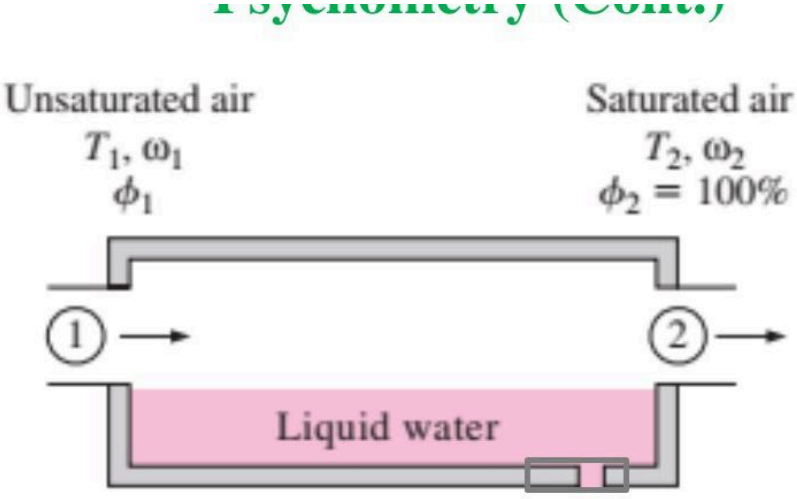
Step (write this)	Why / what it means

$\omega = m_v/m_a$	Definition.
$m_v = p_vV/(R_vT)$, $m_a = p_aV/(R_aT)$	Both are ideal gases occupying the same V at the same T (Dalton).
$\omega = (R_a/R_v)(p_v/p_a) = (0.287/0.4615)(p_v/p_a)$	$R_a = 8.314/28.97$, $R_v = 8.314/18.015$.
$\omega = 0.622\ p_v/(p - p_v)$	$p_a = p - p_v$, with p the barometric pressure. So ω depends only on the vapour pressure and the barometric pressure.

Derivation: degree of saturation (2023 8b) and φ = μ/[1 − (1 − μ)p_s/p] (2019 8c)

Step (write this)	Why / what it means
$\omega = 0.622p_v/(p - p_v)$, $\omega_s = 0.622p_s/(p - p_s)$	Saturated air at the same DBT has $p_v = p_s$.
$\mu = \omega/\omega_s = (p_v/p_s) \cdot (p - p_s)/(p - p_v) = \varphi(p - p_s)/(p - p_v)$	Degree of saturation expressed with pressures; since $p_s > p_v$, $\mu < \varphi$ slightly.
Put $p_v = \varphi p_s$: $\mu(p - \varphi p_s) = \varphi(p - p_s)$	Remove p_v.
$\mu p = \varphi(p - p_s + \mu p_s) = \varphi[p - (1 - \mu)p_s]$	Collect the φ terms.
$\varphi = \mu/[1 - (1 - \mu)(p_s/p)]$	Divide by p.

Adiabatic saturation: enthalpy stays constant (2018 8c)

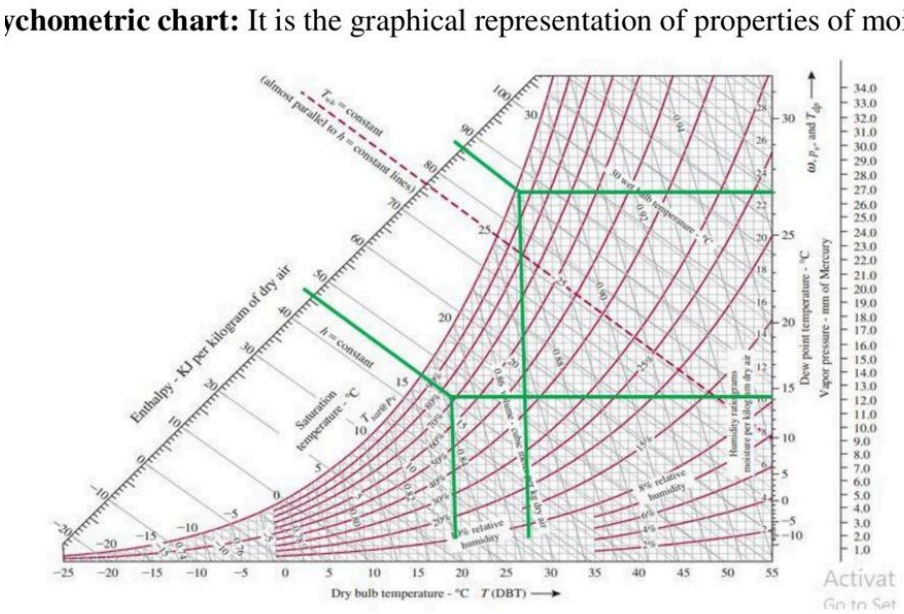


Adiabatic saturation process (course slide)

Step (write this)	Why / what it means
Unsaturated air (state 1: T_1, ω_1) flows over water in a long insulated channel and leaves saturated (state 2: T_2, $\omega_2 = \omega_s$). Make-up water at T_2 is added at the rate it evaporates.	Set-up: steady flow, adiabatic, no work.
Mass of dry air: $\dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_a$; water: $\dot{m}_a\omega_1 + \dot{m}_w = \dot{m}_a\omega_2 \Rightarrow \dot{m}_w = \dot{m}_a(\omega_2 - \omega_1)$	Mass balances.
Energy: $\dot{m}_a h_1 + \dot{m}_w h_{f2} = \dot{m}_a h_2$, h per kg dry air	SFEE (steady-flow energy equation) with Q = W = 0.
$h_1 + (\omega_2 - \omega_1)h_{f2} = h_2$	Divide by $\dot{m}_a$.
The term $(\omega_2 - \omega_1)h_{f2}$ is very small (ω difference ~0.01 × h_f ~100 kJ/kg ≈ 1 kJ/kg)	Make-up water enthalpy is negligible.
$h_1 \approx h_2$: the enthalpy of moist air is (practically) constant; the process follows a	Sensible heat lost by the air = latent heat gained by the vapour. T_2 is the adiabatic saturation

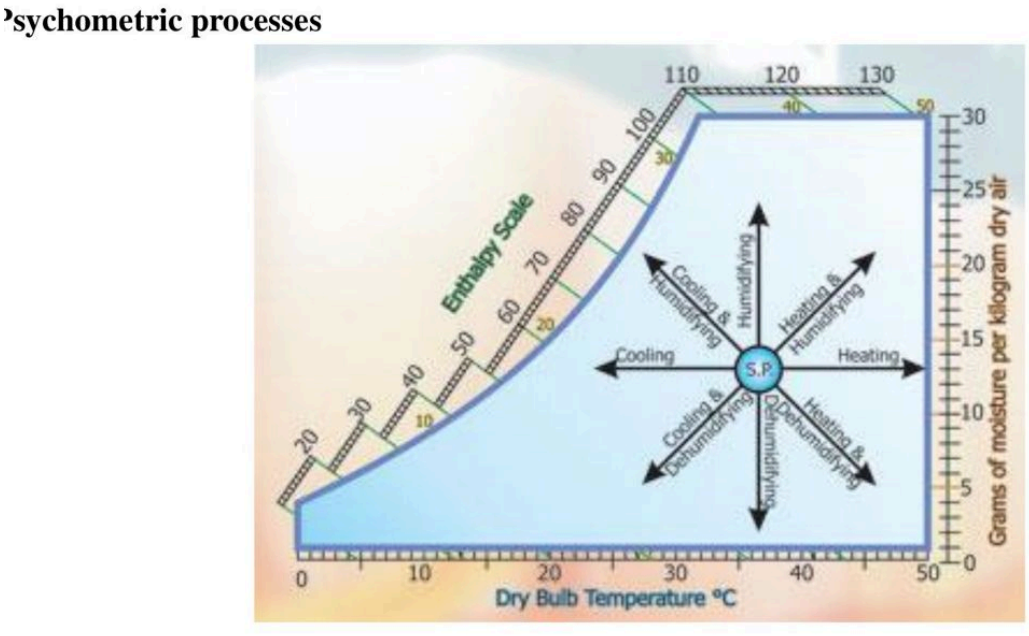
`constant-WBT line on the chart`	temperature ≈ WBT.

Psychrometric chart (2023 8c) and cooling with dehumidification (2021 8a)



Psychrometric chart: DBT on the horizontal axis, ω on the vertical axis, curved φ lines, sloping WBT/enthalpy lines and specific-volume lines (course slide)

- **Horizontal axis**: dry-bulb temperature; vertical lines are constant DBT.
- **Vertical axis (right)**: specific humidity ω; horizontal lines are constant ω and also constant dew point (read DPT (dew-point temperature) where the horizontal line meets the saturation curve).
- **Saturation curve** (φ = 100 %) on the left, with curved constant-φ lines (10 %, 20 %, …) below it.
- **Constant WBT lines** slope down to the right, nearly parallel to **constant enthalpy lines** (enthalpy scale on the left edge).
- **Constant specific-volume lines** are steeper sloping lines (m^3/kg dry air).



Air-conditioning processes on the chart (course slide)

Cooling and dehumidification: air at state 1 is cooled at constant ω (horizontal line to the left): ω stays the same while φ rises, until it reaches the saturation curve at the dew point. Further cooling

condenses water: the state moves down along the saturation curve, so **ω decreases** (water is removed) and **φ stays at 100 %**. Finally the air is usually reheated at constant ω to a comfortable temperature, which lowers φ to 40-60 %.

Carrier's equation (used in every numerical)

$$p_v = p_{wb} - (p - p_{wb})(t_{db} - t_{wb})/(1544 - 1.44\ t_{wb})$$

p_{wb} = saturation pressure at the WBT, t in °C. Then $\omega = 0.622 p_v/(p - p_v)$, $\varphi = p_v/p_s$, $\mu = \omega/\omega_s$, DPT = $T_{sat}(p_v)$, $h = 1.005t + \omega(2500 + 1.88t)$ kJ/kg dry air, $v = R_a T/(p - p_v)$.

1. DBT 35 °C, WBT 31 °C, 1.0132 bar (2019 8d)

Step (write this)	Why / what it means
p_{wb} (31 °C) = 0.04496 bar; p_s (35 °C) = 0.05628 bar	Steam table.
(i) p_v = 0.04496 - (0.96824 × 4)/(1544 - 44.64) = 0.04496 - 0.00258 = 0.0424 bar	Carrier.
(ii) ω = 0.622 × 0.0424/(1.0132 - 0.0424) = 0.0272 kg/kg dry air	
(iii) φ = 0.0424/0.05628 = 75.3 %	
(iv) DPT = T_{sat}(0.0424 bar) = 30.0 °C	Interpolate in the saturation table.
(v) h = 1.005 × 35 + 0.0272(2500 + 1.88 × 35) = 35.2 + 69.8 = 104.9 kJ/kg dry air	
(vi) v = 0.287 × 308.15/(101.32 - 4.24) = 0.911 m^3/kg dry air	Volume per kg of dry air: dry air at its own partial pressure (kPa).

2. DBT 30 °C, WBT 18 °C, 760 mm Hg (2017 7d)

Step (write this)	Why / what it means
p = 760 mm Hg = 1.01325 bar; p_{wb} (18 °C) = 0.02064; p_s (30 °C) = 0.04247 bar	
(i) p_v = 0.02064 - (0.99261 × 12)/(1544 - 25.9) = 0.02064 - 0.00785 = 0.0128 bar	
(ii) φ = 0.0128/0.04247 = 30.1 %	
(iii) DPT = T_{sat}(0.0128 bar) = 10.6 °C	

3. DBT 32 °C, WBT 24 °C (2018 8d)

Step (write this)	Why / what it means
p_{wb} (24 °C) = 0.02985; p_s (32 °C) = 0.04760 bar	
(i) p_v = 0.02985 - (0.98335 × 8)/(1544 - 34.56) = 0.02985 - 0.00521 = 0.02464 bar	
(ii) ω = 0.622 × 0.02464/0.98856 = 0.0155 kg/kg	
(iii) φ = 0.02464/0.04760 = 51.8 %	
ω_s = 0.622 × 0.0476/0.9656 = 0.03066; **(iv) μ = 0.0155/0.03066 = 0.506**	
(v) h = 32.16 + 0.0155(2560.2) = 71.9 kJ/kg dry air	

4. DBT 30 °C, WBT 27 °C (2021 8c, 15 marks)

Step (write this)	Why / what it means
p_{wb} (27 °C) = 0.03567; p_s (30 °C) = 0.04247 bar	
(i) p_v = 0.03567 - (0.97753 × 3)/(1544 - 38.88) = 0.03567 - 0.00195 = 0.0337 bar	
(ii) ω = 0.622 × 0.0337/0.9795 = 0.0214 kg/kg	
(iii) DPT = T_{sat}(0.0337 bar) = 26.0 °C	
(iv) φ = 0.0337/0.04247 = 79.4 %	
ω_s = 0.622 × 0.04247/0.97073 = 0.0272; **(v) μ = 0.0214/0.0272 = 0.787**	
(vi) ρ = $p_a/(R_aT)$ + $p_v/(R_vT)$ = 97.95/(0.287 × 303.15) + 3.37/(0.4615 × 303.15) = 1.126 + 0.024 = 1.15 kg/m^3	Density of the moist air = dry air density + vapour density.
(vii) h = 30.15 + 0.0214(2556.4) = 84.9 kJ/kg dry air	

5. DBT 32 °C, WBT 26 °C (2022 8c)

Step (write this)	Why / what it means
p_{wb} (26 °C) = 0.03363; p_s (32 °C) = 0.04760 bar	
(i) p_v = 0.03363 - (0.97957 × 6)/(1544 - 37.44) = 0.03363 - 0.0039 = 0.0297 bar	The slide example rounds to 0.03 bar.
(ii) φ = 0.0297/0.0476 = 62.5 %	
(iii) ω = 0.622 × 0.0297/0.9835 = 0.0188 kg/kg	
(iv) h = 32.16 + 0.0188(2560.2) = 80.3 kJ/kg dry air	

> **Memory trick:** Psychrometry chain: "**C-O-P-D-H**" = Carrier (p_v), Omega, Phi (and μ), Dew point, H. Remember "**0.622 and 1544**".

B7 Ideal gas mixtures: Dalton, Amagat, mole and mass fractions, mixture properties

Pass set | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2023 Q8(d)** [10] 3 kg N_2 + 5 kg CO_2 at 300 kPa, 20 °C. Find (i) mole fractions, (ii) equivalent molecular weight.
- **2022 Q7(b)** [9] State Dalton's law. What is meant by gravimetric and volumetric analysis?
- **2022 Q7(c)** [16] 5 kg N_2 + 6 kg CO_2 at 4 bar, 27 °C. Find (i) mole fractions, (ii) molecular weight, (iii) gas constant, (iv) volume and density.
- **2021 Q7(b)** [4] State Dalton's law of partial pressure and Amagat's law.
- **2021 Q7(d)** [6] Short notes: (i) gas mixture, (ii) gravimetric analysis, (iii) partial pressure of a gas mixture.
- **2021 Q7(c)** [13] Mixture of 60 % O_2 and 40 % CO_2 (mole basis) at 300 kPa, 20 °C. Find (i) mass fractions, (ii) molecular weight, (iii) gas constant, (iv) partial pressures and partial volumes, (v) volume, density, c_p, c_v (per kmol of mixture).
- **2020 Q6(a)** [6] State Dalton's law of partial pressure and Amagat's law.
- **2020 Q6(c)** [18] 3 kg N_2 + 5 kg CO_2 at 300 kPa, 20 °C. Find (i) mole fractions, (ii) molecular weight, (iii) gas constant, (iv) partial pressures, (v) density, (vi) c_p, c_v.
- **2019 Q7(b)** [4] Define: (i) gas mixture, (ii) gravimetric analysis, (iii) volumetric analysis, (iv) ideal gas.
- **2019 Q7(c)** [6] State Dalton's law of partial pressure. Show that partial pressure = mole fraction × total pressure.
- **2019 Q7(d)** [15] 3.5 kg N_2 + 5.5 kg CO_2 at 300 kPa, 200 °C. Find (i) mass and mole fractions, (ii) molecular weight, (iii) volume and density, (iv) c_p, c_v (γ CO_2 = 1.29, N_2 = 1.40).
- **2018 Q7(a)** [10] State Dalton's law. Show that $M_m = \Sigma x_i M_i$.
- **2018 Q7(b)** [15] A 300 L drum holds a gas mixture at 600 kPa, 40 °C: 35 % O_2, 65 % CH_4 by volume. To get 70 % O_2, 30 % CH_4 at the same p and T, how many kg of mixture must be bled off and what mass of O_2 added?
- **2018 Q7(c)** [10] A mixture of CH_4, N_2, CO and O_2 has partial pressures 140, 55, 70 and 15 kPa. Find the volumetric and gravimetric analysis.
- **2017 Q8(b)** [8] What is the apparent molar mass of a gas mixture? Does every molecule have this mass?
- **2017 Q8(c)** [12] For a homogeneous ideal gas mixture at T, p, V: (i) derive Dalton's law, (ii) show Σ mole fractions = 1, (iii) find the gas constant of the mixture.
- **2017 Q8(d)** [10] A mixture of 3 kg O_2, 5 kg N_2 and 12 kg CH_4. Find (i) mole fractions, (ii) average molar mass, (iii) gas constant.

UNDERSTAND **Read this to understand; you do not need to write it**

Masses → moles ($n = m/M$) → mole fractions $y = n/\Sigma n$ → mixture molar mass $M = \Sigma yM$ → $R = 8.314/M$ → then V, ρ, partial pressures $p_i = y_i p$, $c_p = \Sigma m f_i c_{p,i}$.

CALCULATOR **Gas mixture from masses**

Change any number; every step is recalculated and shown with the numbers put in.

1. Total moles: **7** / 28 + **2** / 32 + **1** / 44 + **0** / 16 = **0.33523** kmol
2. y N_2: **7** / 28 / **0.3352** = **0.7458**
3. y O_2: **2** / 32 / **0.3352** = **0.1864**
4. y CO_2: **1** / 44 / **0.3352** = **0.0678**
5. y CH_4: **0** / 16 / **0.3352** = **0.0000**
6. Apparent molar mass: M = m/n = (**7** + **2** + **1** + **0**) / **0.3352** = **29.831** kg/kmol
7. Gas constant: R = R_u/M = 8.314 / **29.83** = **0.2787** kJ/kg K
8. Mixture volume: V = nR_uT/p = **0.3352** × 8.314 × **300** / **100** = **8.361** m^3
9. Partial pressure N_2: $p_i = y_i p$ = **0.7458** × **100** = **74.58** kPa

Before you start: what a gas mixture problem asks

Air, natural gas and combustion products are **mixtures of gases**. If each component is an ideal gas, the mixture is an ideal gas too, with its own molar mass and gas constant. The whole topic is one chain: **masses → moles → mole fractions → apparent molar mass → gas constant → volume and density → specific heats**.

Moles: n = m/M (kmol). M: N_2 28, O_2 32, CO_2 44, CH_4 16, CO 28 kg/kmol. R_u = 8.314 kJ/kmol K.

Definitions (2019 7b, 2021 7d, 2022 7b)

- **Gas mixture**: a physical combination of two or more gases that do not react; each component keeps its own properties (air = N_2 + O_2 + Ar + ...).
- **Gravimetric analysis**: composition given by the mass of each component: mass fraction $mf_i = m_i/m_m$, $\Sigma mf_i = 1$.
- **Volumetric (molar) analysis**: composition given by the number of moles (or partial volume) of each component: mole fraction $y_i = n_i/n_m$; for ideal gases volume fraction = mole fraction.
- **Ideal gas**: a gas obeying Pv = RT (no intermolecular forces, negligible molecular volume).
- **Partial pressure**: the pressure a component would exert if it alone occupied the whole volume at the mixture temperature: $p_i = y_i p$.

Dalton and Amagat (2019 7c, 2020 6a, 2021 7b, 2022 7b)

- **Dalton's law of additive pressures:** the pressure of a gas mixture equals the sum of the pressures each gas would exert if it existed alone at the mixture temperature and volume: $p_m = \Sigma p_i(T_m, V_m)$.
- **Amagat's law of additive volumes:** the volume of a gas mixture equals the sum of the volumes each gas would occupy if it existed alone at the mixture temperature and pressure: $V_m = \Sigma V_i(T_m, p_m)$.

Derivations (2017 8c, 2019 7c, 2018 7a)

Step (write this)	Why / what it means
Each component alone in V at T: $p_iV = n_iR_uT$; the mixture: $pV = nR_uT$	Ideal gas law for the component and for the mixture.
Divide: $\mathbf{p_i/p = n_i/n = y_i} \Rightarrow p_i = y_ip$	Partial pressure = mole fraction × total pressure.

Add over all components: $\Sigma p_i = (\Sigma n_i)R_u T/V = nR_u T/V = p$	(i) Dalton's law follows, since $\Sigma n_i = n$.
$\Sigma y_i = \Sigma n_i/n = n/n = \mathbf{1}$	(ii) Sum of mole fractions.
$m = \Sigma m_i = \Sigma n_i M_i$; $M_m = m/n = \Sigma(n_i/n)M_i = \mathbf{\Sigma y_i M_i}$	Apparent molar mass (2018 7a).
$\mathbf{R_m = R_u/M_m} = \Sigma mf_i R_i$	(iii) Gas constant of the mixture.

Apparent molar mass (2017 8b): $M_m = m_{total}/n_{total} = \Sigma y_i M_i$: the mass of one kmol of the mixture. **No molecule has this mass**; it is only an average (air: 28.97, but every molecule is either N_2 of 28 or O_2 of 32, etc.).

Mixture specific heats

$c_{p,m} = \Sigma mf_i c_{p,i}$, $c_{v,m} = c_{p,m} - R_m$. If only γ is given: $c_{p,i} = \gamma_i R_i/(\gamma_i - 1)$. With γ N_2 = 1.40, R = 0.2968: c_p = 1.039; CO_2 γ = 1.29, R = 0.1889: c_p = 0.8405 kJ/kg K.

1. 3.5 kg N_2 + 5.5 kg CO_2, 300 kPa, 200 °C (2019 7d)

Step (write this)	Why / what it means
$n_{N2} = 3.5/28 = 0.125$; $n_{CO2} = 5.5/44 = 0.125$; $n = 0.25$ kmol	
(i) $y = 0.5$ each; $mf_{N2} = 3.5/9 = 0.389$, $mf_{CO2} = 0.611$	
(ii) $M_m = 9/0.25 = 36$ kg/kmol; $R_m = 8.314/36 = 0.2309$ kJ/kg K	
(iii) $V = mR_m T/p = 9 \times 0.2309 \times 473.15/300 = 3.28$ m³; $\rho = 9/3.28 = 2.75$ kg/m³	
(iv) $c_p = 0.389 \times 1.039 + 0.611 \times 0.8405 = 0.918$; $c_v = 0.918 - 0.2309 = 0.687$ kJ/kg K	

2. 3 kg N_2 + 5 kg CO_2, 300 kPa, 20 °C (2020 6c, 2023 8d)

Step (write this)	Why / what it means
$n_{N2} = 3/28 = 0.1071$; $n_{CO2} = 5/44 = 0.1136$; $n = 0.2208$ kmol	
(i) $y_{N2} = 0.485$, $y_{CO2} = 0.515$	
(ii) $M_m = 8/0.2208 = 36.24$ kg/kmol	2023 asks only up to here.
(iii) $R_m = 8.314/36.24 = 0.2294$ kJ/kg K	
(iv) $p_{N2} = 0.485 \times 300 = 145.6$ kPa; $p_{CO2} = 154.4$ kPa	
(v) $V = 8 \times 0.2294 \times 293.15/300 = 1.794$ m³; $\rho = 4.46$ kg/m³	
(vi) $c_p = (3/8)1.039 + (5/8)0.8405 = 0.915$; $c_v = 0.686$ kJ/kg K	

3. 5 kg N_2 + 6 kg CO_2, 4 bar, 27 °C (2022 7c)

Step (write this)	Why / what it means

n_{N2} = 0.1786, n_{CO2} = 0.1364, n = 0.3149 kmol; **(i) y = 0.567, 0.433**	
(ii) M_m = 11/0.3149 = 34.93; (iii) R_m = 0.2380 kJ/kg K	
(iv) V = 11 × 0.2380 × 300.15/400 = 1.965 m^3; ρ = 5.60 kg/m^3	

4. 60 % O_2 + 40 % CO_2 by moles, 300 kPa, 20 °C (2021 7c), basis 1 kmol

Step (write this)	Why / what it means
m_{O2} = 0.6 × 32 = 19.2 kg, m_{CO2} = 0.4 × 44 = 17.6 kg; m = 36.8 kg	Choose 1 kmol of mixture.
(i) mf_{O2} = 0.522, mf_{CO2} = 0.478; (ii) M_m = 36.8; (iii) R_m = 0.2259 kJ/kg K	
V = nR_uT/p = 8.314 × 293.15/300 = 8.124 m^3	Volume of 1 kmol.
(iv) p_{O2} = 180, p_{CO2} = 120 kPa; V_{O2} = 4.874, V_{CO2} = 3.250 m^3	Partial volume = y_iV (Amagat).
(v) V = 8.12 m^3/kmol; ρ = 36.8/8.124 = 4.53 kg/m^3	
$c_{p,O2}$ = 1.4 × 0.2598/0.4 = 0.909; $c_{p,CO2}$ = 0.8405 ⇒ **c_p = 0.876, c_v = 0.650 kJ/kg K** (32.2 and 23.9 kJ/kmol K)	γ O_2 = 1.4, CO_2 = 1.29 assumed (not given).

5. 3 kg O_2, 5 kg N_2, 12 kg CH_4 (2017 8d)

Step (write this)	Why / what it means
n = 3/32 + 5/28 + 12/16 = 0.0938 + 0.1786 + 0.75 = 1.0223 kmol	
(i) y_{O2} = 0.092, y_{N2} = 0.175, y_{CH4} = 0.734	
(ii) M_m = 20/1.0223 = 19.6 kg/kmol; (iii) R_m = 8.314/19.56 = 0.425 kJ/kg K	

6. Drum: bleed and add O_2 (2018 7b)

Step (write this)	Why / what it means
n = $pV/(R_uT)$ = 600 × 0.3/(8.314 × 313.15) = 0.06914 kmol	Total moles, unchanged at the end (same p, V, T).
Initially O_2 0.35n = 0.0242, CH_4 0.65n = 0.0449 kmol. Finally CH_4 must be 0.30n = 0.0207 kmol	Bleeding removes the mixture in its original proportions; adding O_2 adds no CH_4.
Fraction kept = 0.0207/0.0449 = 0.4615 ⇒ bled = 0.5385n = 0.0372 kmol	Only bleeding reduces CH_4.
Mass bled = 0.0372 × (0.35 × 32 + 0.65 × 16) = 0.0372 × 21.6 = 0.80 kg	
O_2 added = 0.0372 kmol × 32 = 1.19 kg	Moles added = moles bled (total n unchanged).

7. Analysis from partial pressures (2018 7c)

Step (write this)	Why / what it means
$p = 140 + 55 + 70 + 15 = 280$ kPa; $y_i = p_i/p$	Dalton.
Volumetric: CH_4 50 %, N_2 19.6 %, CO 25 %, O_2 5.4 %	
Mass per kmol: $0.5 \times 16 = 8$, $0.196 \times 28 = 5.5$, $0.25 \times 28 = 7$, $0.054 \times 32 = 1.71$; total 22.2 kg	
Gravimetric: CH_4 36.0 %, N_2 24.8 %, CO 31.5 %, O_2 7.7 %	

Memory trick: Mixture chain: "**M-N-Y-M-R-V**" = Masses, N (moles), Y (fractions), M_{mix}, R_{mix}, Volume. Mole-based things add with y; mass-based things (c_p, R) add with mf.

B4 Otto, Diesel and Dual cycles (IC engines): diagrams, derivations, comparisons, numericals

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q7(b)** [12] Describe the working principle of a four-stroke Otto cycle engine with diagrams.
- **2023 Q7(c)** [15] Air-standard Diesel: r = 16, 15 °C, 0.1 MPa, maximum temperature 1480 °C. Find (i) cut-off ratio, (ii) heat supplied per kg, (iii) efficiency, (iv) MEP (mean effective pressure).
- **2022 Q7(a)** [10] Do diesel or petrol engines operate at higher compression ratios? Why?
- **2021 Q5(b)** [15] With p-v and T-s diagrams, show that $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$ for the same compression ratio and heat rejection. Why?
- **2020 Q4(c)** [18] Air-standard Diesel cycle: r = 17, start of compression 20 °C, 100 kPa; temperature at end of constant-pressure heating 1500 °C. Find (i) cut-off ratio, (ii) heat supplied per kg, (iii) efficiency.
- **2019 Q6(b)** [7] From the derivation, show that the Diesel cycle efficiency is less than the Otto cycle efficiency for the same compression ratio.
- **2019 Q6(c)** [20] Air-standard Dual cycle: start of compression 0.1 MPa, 27 °C; end of compression 2.3 MPa; 430 kJ/kg supplied at constant volume; pressure 0.3 MPa at end of isentropic expansion. Find p and T at all points and the efficiency.
- **2018 Q5(b)** [10] Why is the dual cycle called a limited pressure cycle? Show from T-s diagrams and derivations that its efficiency lies between Otto and Diesel.
- **2018 Q5(c)** [18] Limited pressure cycle: r = 18, compression begins at 0.1 MPa, 40 °C; p_{max} = 7 MPa; heat added 1.675 MJ/kg. Find (i) heat at constant volume, (ii) heat at constant pressure, (iii) work, (iv) efficiency, (v) temperature after constant-volume heating, (vi) cut-off ratio.
- **2017 Q5(a)** [6] Draw the Diesel, Brayton and Otto cycles on p-v and T-s planes showing heat and work transfer.
- **2017 Q5(c)** [8] Derive an expression for the air-standard efficiency of the Diesel cycle in terms of cut-off ("pressure") ratio and compression ratio.
- **2017 Q5(d)** [10] In an Otto cycle the temperatures at the beginning and end of isentropic compression are 316 K and 596 K. Find the air-standard efficiency and compression ratio (γ = 1.4).

LEARN IT FROM ZERO

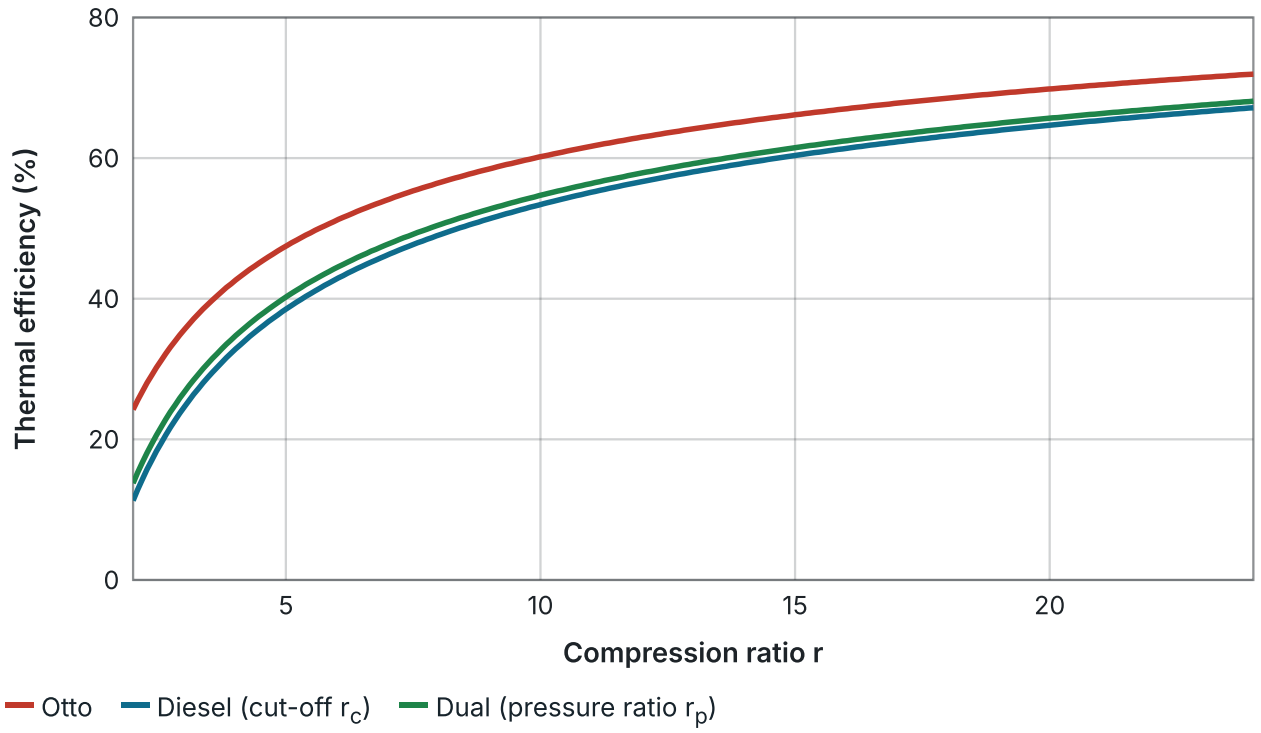
Otto, Diesel and Dual, from zero

All three cycles squeeze air (1→2, isentropic), add heat (2→3), let it push the piston (3→4, isentropic) and dump heat at constant volume (4→1). The only difference is **how the heat is added**: Otto (petrol) all at constant volume (a sudden spark explosion), Diesel at constant pressure (fuel sprayed in while the piston moves), Dual partly each.

Efficiency rises with compression ratio r. For the same r, **Otto > Dual > Diesel**, because Diesel's heat goes in while the gas is already expanding (cut-off ratio $r_c > 1$ costs efficiency). But diesels run at r = 14–22 while petrol engines are knock-limited to 8–11, so in practice the diesel is more efficient. Use the graph.

INTERACTIVE **Air-standard efficiency vs compression ratio**

Move the sliders and watch the curve change.



Same r: Otto > Dual > Diesel. Cut-off ratio near 1 makes the Diesel curve approach Otto.

Symbols used here

Symbol	Name and meaning	Unit
r	Compression ratio v_1/v_2	–
r_c	Cut-off ratio v_3/v_2 (= T_3/T_2 for Diesel)	–
r_p	Pressure ratio of constant-volume heating (Dual)	–
γ	c_p/c_v = 1.4 for air	–
c_p, c_v	1.005 and 0.718	kJ/kg K
MEP	Mean effective pressure = $w_{net}/(v_1 - v_2)$	kPa

CALCULATOR Diesel cycle calculator

Change any number; every step is recalculated and shown with the numbers put in.

1. T_2: $T_1 r^{\gamma-1}$ = **293** × pow(**17**, 0.4) = **910.0** K
2. p_2: **100** × pow(**17**, 1.4) = **5280** kPa
3. Cut-off ratio: $r_c = T_3/T_2$ = **1773** / **910** = **1.948**
4. Heat in: $c_p(T_3 - T_2)$ = 1.005 × (**1773** - **910**) = **867.3** kJ/kg
5. T_4: $T_3(r_c/r)^{0.4}$ = **1773** × pow(**1.948** / **17**, 0.4) = **745.4** K
6. Heat out: $c_v(T_4 - T_1)$ = 0.718 × (**745.4** - **293**) = **324.8** kJ/kg
7. Efficiency: (1 - **324.8** / **867.3**) × 100 = **62.55** %

8. v_1: 0.287 × **293** / **100** = **0.8409** m^3/kg

9. Mean effective pressure: (**867.3** - **324.8**) / (**0.8409** - **0.8409** / **17**) = **685** kPa

Before you start: how a reciprocating engine works

A piston moves inside a cylinder between the **top dead centre (TDC)**, where the volume is smallest (the **clearance volume**), and the **bottom dead centre (BDC)**, where it is largest. The distance between them is the **stroke**, and the volume swept is the **displacement volume**. **Compression ratio** $r = V_{max}/V_{min} = V_{BDC}$ (bottom dead centre)$/V_{TDC}$ (top dead centre)·

Petrol (spark-ignition) engines are modelled by the **Otto cycle** (heat added at constant volume), diesel (compression-ignition) engines by the **Diesel cycle** (heat added at constant pressure), and real high-speed diesel engines by the **Dual cycle** (part at constant volume, part at constant pressure). **Mean effective pressure** MEP = $w_{net}/(v_{max} - v_{min})$: the constant pressure that would give the same net work over one stroke.

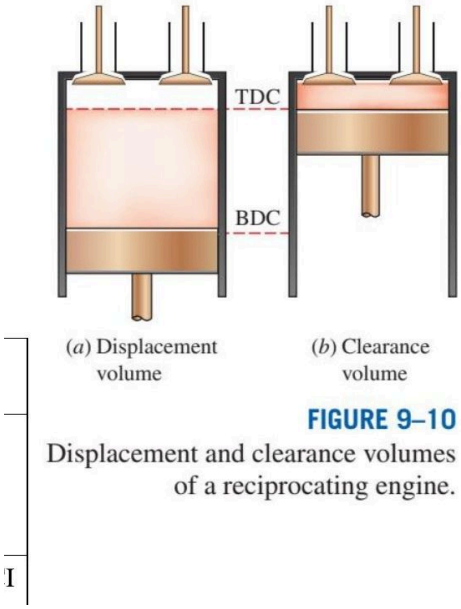
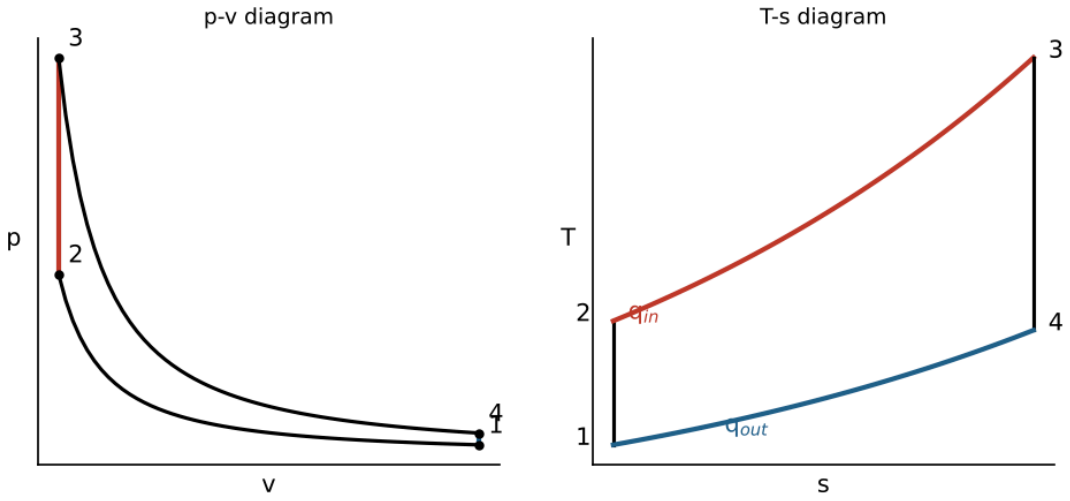


FIGURE 9–10
Displacement and clearance volumes of a reciprocating engine.

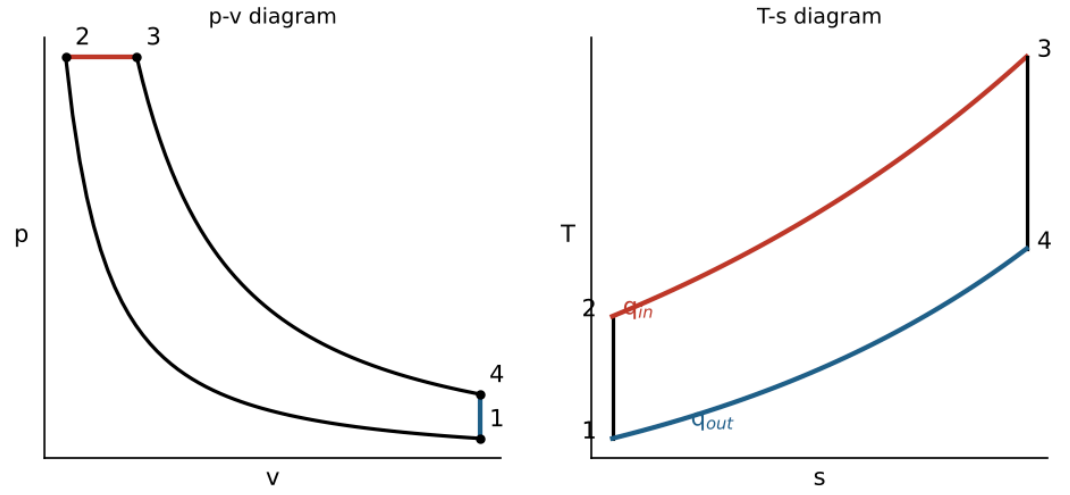
Engine nomenclature: TDC, BDC, stroke, bore, clearance volume (course slide)

Cycles on p-v and T-s (2017 5a)



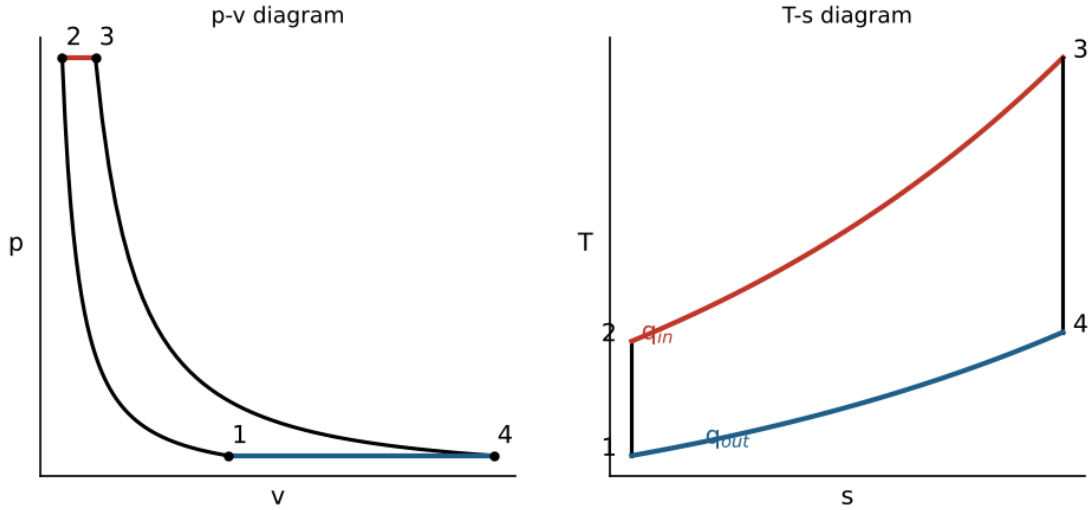
Otto: 1-2 isentropic compression, 2-3 heat added at constant v, 3-4 isentropic expansion, 4-1 heat rejected at constant v

Otto cycle: 1-2 isentropic compression, 2-3 constant-volume heat addition, 3-4 isentropic expansion, 4-1 constant-volume heat rejection



Diesel: 1-2 isentropic compression, 2-3 heat added at constant p, 3-4 isentropic expansion, 4-1 heat rejected at constant v

Diesel cycle: heat added at constant pressure (2-3)

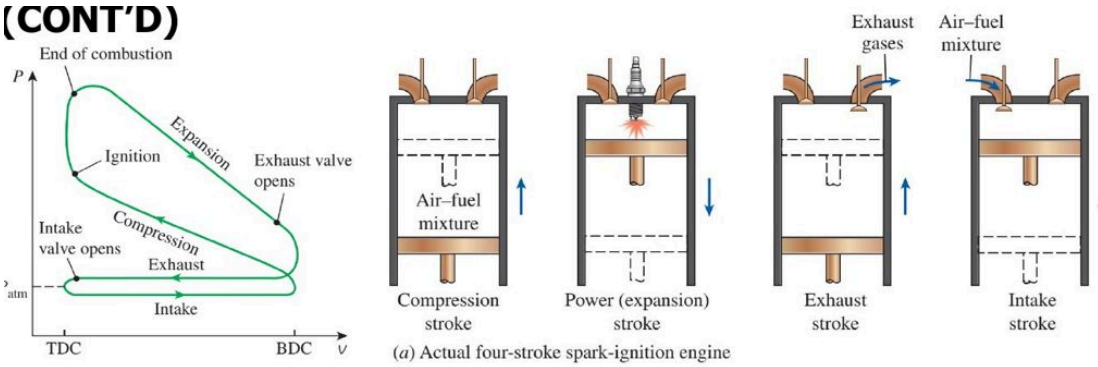


Brayton: 1-2 isentropic compression, 2-3 heat at constant p, 3-4 isentropic expansion, 4-1 heat rejected at constant p

Brayton cycle: heat added and rejected at constant pressure

Mark q_{in} on the heat-addition line and q_{out} on the rejection line; the enclosed area on p-v is w_{net} and on T-s it is $q_{net} = w_{net}$.

Four-stroke Otto (spark-ignition) engine (2023 7b)



d of expansion stroke, the exhaust valve opens and the combustion gases that are above the essure rush out of the cylinder through the open exhaust valve. This process is called **own**, and most combustion gases leave the cylinder by the time the piston reaches BDC.

still filled by the exhaust gases at a lower pressure at BDC.

moves upward one more time, purging the exhaust gases through the exhaust valve (the and down a second time, drawing in fresh air–fuel mixture through the intake valve (the

Actual four-stroke SI engine and the ideal Otto cycle (course slide)

1. **Intake (suction) stroke**: piston moves TDC → BDC with the intake valve open; fuel-air mixture is drawn in.
2. **Compression stroke**: both valves closed; piston moves BDC → TDC compressing the mixture (ideal 1-2). Just before TDC the spark plug fires.
3. **Power (expansion) stroke**: combustion raises the pressure almost at constant volume (ideal 2-3); high-pressure gases push the piston to BDC (ideal 3-4), producing work.
4. **Exhaust stroke**: exhaust valve opens (ideal heat rejection 4-1); piston moves BDC → TDC pushing out the burnt gases.
5. One power stroke per two crankshaft revolutions.

Otto cycle efficiency

Step (write this)	Why / what it means
$q_{in} = c_v(T_3 - T_2)$, $q_{out} = c_v(T_4 - T_1)$	Constant-volume processes.
$n = 1 - (T_4 - T_1)/(T_3 - T_2)$	
$T_2/T_1 = r^{\gamma-1} = T_3/T_4 \Rightarrow T_4/T_1 = T_3/T_2$	Isentropic processes between the same volumes.
$\mathbf{n_{Otto} = 1 - T_1/T_2 = 1 - 1/r^{\gamma-1}}$	Depends only on r.

Diesel cycle efficiency in terms of r and r_c (2017 5c)

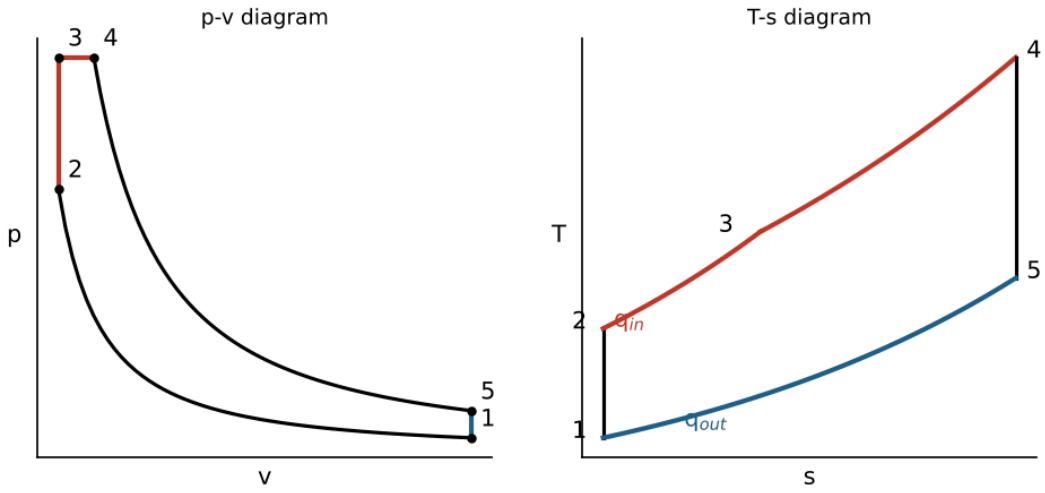
Misprint: the 2017 paper says "pressure ratio"; for the Diesel cycle it must be the **cut-off ratio** $r_c = v_3/v_2$.

Step (write this)	Why / what it means
$q_{in} = c_p(T_3 - T_2)$, $q_{out} = c_v(T_4 - T_1)$	2-3 at constant p, 4-1 at constant v.
$n = 1 - (T_4 - T_1)/[\gamma(T_3 - T_2)]$	$\gamma = c_p/c_v$.

$T_2 = T_1 r^{\gamma-1}$	1-2 isentropic.
$T_3 = T_2 r_c = T_1 r_c r^{\gamma-1}$	2-3 constant p: $T \propto v$.
$T_4 = T_3(v_3/v_4)^{\gamma-1} = T_3(r_c/r)^{\gamma-1} = T_1 r_c^{\gamma}$	3-4 isentropic, $v_4 = v_1$, $v_3/v_1 = r_c/r$.
$n = 1 - T_1(r_c^{\gamma} - 1)/[\gamma T_1 r^{\gamma-1}(r_c - 1)]$	Substitute; T_1 cancels.
$\mathbf{n_{Diesel} = 1 - (1/r^{\gamma-1}) \cdot (r_c^{\gamma} - 1)/[\gamma(r_c - 1)]}$	

Diesel less efficient than Otto for the same r (2019 6b): the bracket $(r_c^{\gamma} - 1)/[\gamma(r_c - 1)]$ is always greater than 1 for $r_c > 1$ (e.g. $r_c = 2$: $(2.639 - 1)/1.4 = 1.17$). So the Diesel efficiency subtracts a larger quantity from 1 than Otto does: $\eta_{Diesel} < \eta_{Otto}$, and the gap grows with r_c (more fuel). As $r_c \to 1$ they become equal.

Dual (limited-pressure) cycle (2018 5b, 2021 5b)



Dual: 2-3 heat at constant v, 3-4 heat at constant p (pressure limited), 4-5 isentropic expansion, 5-1 rejection at constant v

Dual cycle: heat added partly at constant volume (2-3) and partly at constant pressure (3-4)

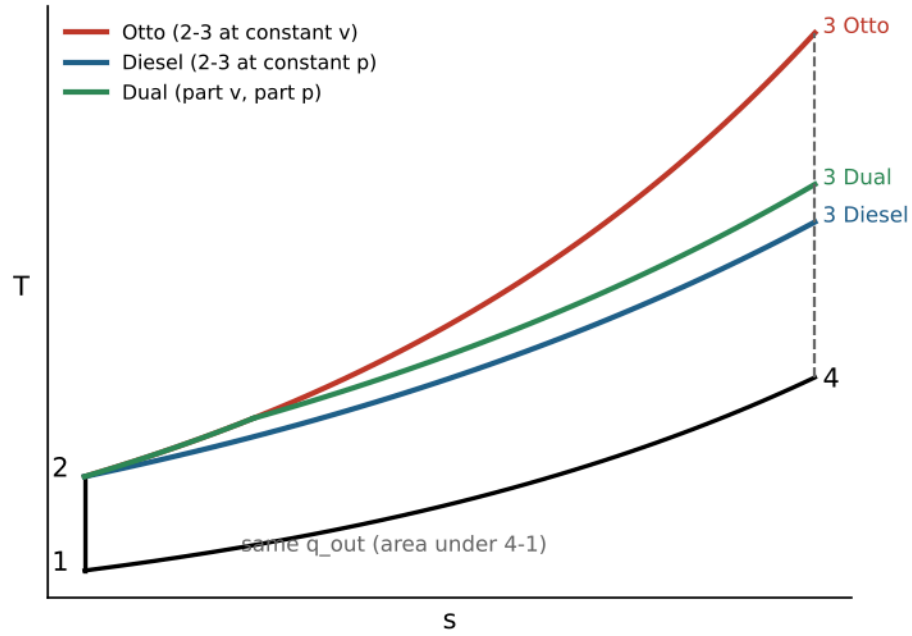
Why "limited pressure": in a real high-speed CI engine, part of the fuel burns almost instantly (at nearly constant volume) and the rest burns while the piston moves (at nearly constant pressure). The constant-pressure part limits the peak cylinder pressure to what the engine structure can stand, hence "limited pressure cycle".

Efficiency with pressure ratio $r_p = p_3/p_2$ and cut-off ratio $r_c = v_4/v_3$:

$$n_{Dual} = 1 - (1/r^{\gamma-1}) \cdot (r_p r_c^{\gamma} - 1)/[(r_p - 1) + \gamma r_p(r_c - 1)]$$

Setting $r_c = 1$ gives the Otto formula, and $r_p = 1$ gives the Diesel formula: so the Dual cycle lies between them.

Comparison: same compression ratio and same heat rejection (2021 5b, 2018 5b)



Otto (1-2-3-4), Dual (1-2-3′-3″-4) and Diesel (1-2-3‴-4) on T-s with the same r and the same heat rejection 4-1

Step (write this)	Why / what it means
All three share state 1, state 2 (same r) and the heat-rejection line 4-1, so q_{out} is the same area under 4-1.	Same starting point and same rejection.
On T-s, a constant-volume line is steeper than a constant-pressure line ($dT/ds = T/c_v > T/c_p$).	Explains why the Otto heat addition curve lies highest.
Heat added = area under the heat-addition curve from s_2 to s_4: largest for Otto, then Dual, then Diesel: $q_{in,Otto} > q_{in,Dual} > q_{in,Diesel}$.	
$n = 1 - q_{out}/q_{in}$ with equal $q_{out} \Rightarrow$ **$n_{Otto} > n_{Dual} > n_{Diesel}$**	Why: Otto adds its heat at the highest mean temperature.
For the **same maximum pressure and temperature** (a fair practical comparison) the order reverses: $n_{Diesel} > n_{Dual} > n_{Otto}$, because the Diesel engine can use a much higher compression ratio.	

Diesel or petrol: which runs at a higher compression ratio? (2022 7a) Diesel engines (r = 14-22, vs 7-12 for petrol). In a petrol engine the fuel-air mixture is compressed; too high a ratio makes it self-ignite early (**knock / detonation**), which damages the engine, so r is limited. A diesel engine compresses **only air** and needs the air hot enough (above the fuel's auto-ignition temperature) to ignite the injected fuel, so it *requires* a high r and has no knock limit; this is also why diesel engines are more efficient.

Numerical 1: Otto (2017 5d)

Step (write this)	Why / what it means
$T_2/T_1 = r^{\gamma-1} \Rightarrow r = (596/316)^{1/0.4} = 1.886^{2.5} =$ **4.89**	Compression ratio from the isentropic temperature ratio.
$n = 1 - T_1/T_2 = 1 - 316/596 = 47.0\ \%$	Otto efficiency.

Numerical 2: Diesel, r = 17 (2020 4c)

Step (write this)	Why / what it means
$T_1 = 293$ K; $T_2 = 293 \times 17^{0.4} = 910.5$ K; $p_2 = 100 \times 17^{1.4} = 5280$ kPa	Isentropic compression.
$T_3 = 1773$ K; **(i) $r_c = T_3/T_2 = 1.947$**	Constant pressure: v ∝ T.
(ii) $q_{in} = c_p(T_3 - T_2) = 1.005(862.5) = 867$ kJ/kg	
$T_4 = T_3(r_c/r)^{0.4} = 1773(0.1145)^{0.4} = 745.3$ K; $q_{out} = 0.718(745.3 - 293) = 324.7$	
(iii) n = 1 - 324.7/867 = 62.6 %	Check with formula: $1 - (1/17^{0.4})(1.947^{1.4} - 1)/(1.4 \times 0.947) = 62.6$ %.

Numerical 3: Diesel, r = 16 (2023 7c)

Step (write this)	Why / what it means
$T_1 = 288$ K; $T_2 = 288 \times 16^{0.4} = 873.5$ K; $T_3 = 1753$ K	
(i) $r_c = 1753/873.5 = 2.01$; (ii) $q_{in} = 1.005 \times 879.5 = 884$ kJ/kg	
$T_4 = 1753(2.007/16)^{0.4} = 764.2$ K; $q_{out} = 0.718(764.2 - 288) = 341.9$; $w_{net} = 542.1$ kJ/kg	
(iii) n = 542.1/884 = 61.3 %	
$v_1 = RT_1/p_1 = 0.287 \times 288/100 = 0.827$; $v_2 = v_1/16 = 0.0517$ m^3/kg	
(iv) MEP = 542.1/(0.827 - 0.0517) = 699 kPa	

Numerical 4: Dual (2019 6c, 20 marks)

Assumption: the paper says "430 kJ/kg constant heating"; taken as 430 kJ/kg added at **constant volume**, as in the original problem. States: 1 start of compression, 2 end of compression, 3 end of constant-volume heating, 4 end of constant-pressure heating, 5 end of expansion.

Step (write this)	Why / what it means
$p_2/p_1 = 23 \Rightarrow r = 23^{1/1.4} = 9.39$; $T_2 = 300 \times 23^{0.2857} = 734.8$ K	Isentropic compression.
$T_3 = T_2 + 430/0.718 = 1333.7$ K; $p_3 = 2.3 \times 1333.7/734.8 = 4.175$ MPa	Constant volume heating.
$v_5 = v_1 \Rightarrow T_5 = T_1p_5/p_1 = 300 \times 3 = 900$ K	5-1 is the constant-volume rejection.
$p_4 = p_3 = 4.175$ MPa; $T_4 = T_5(p_4/p_5)^{0.2857} = 900(13.92)^{0.2857} = 1909.6$ K	Isentropic 4-5 between known pressures.
$q_p = 1.005(1909.6 - 1333.7) = 578.8$; $q_{in} = 430 + 578.8 = 1008.8$; $q_{out} = 0.718(900 - 300) = 430.8$ kJ/kg	
n = 1 - 430.8/1008.8 = 57.3 %	

State	p (MPa)	T (K)
1	0.1	300
2	2.3	734.8
3	4.175	1333.7
4	4.175	1909.6
5	0.3	900

Numerical 5: limited-pressure cycle (2018 5c, 18 marks)

Step (write this)	Why / what it means
$T_1 = 313$ K; $T_2 = 313 \times 18^{0.4} = 995.1$ K; $p_2 = 0.1 \times 18^{1.4} = 5.72$ MPa	
(v) $T_3 = T_2 p_3/p_2 = 995.1 \times 7/5.72 = 1217.8$ K	Constant volume up to p_{max} = 7 MPa.
(i) $q_v = 0.718(1217.8 - 995.1) = 159.9$ kJ/kg	
(ii) $q_p = 1675 - 159.9 = 1515.1$ kJ/kg	
$T_4 = T_3 + q_p/c_p = 1217.8 + 1507.6 = 2725.4$ K; **(vi) $r_c = T_4/T_3 = 2.24$**	
$T_5 = T_4(r_c/r)^{0.4} = 2725.4(0.1243)^{0.4} = 1183.7$ K; $q_{out} = 0.718(1183.7 - 313) = 625.1$	
(iii) $w_{net} = 1675 - 625.1 = 1049.9$ kJ/kg; (iv) $n = 1049.9/1675 = 62.7$ %	

Memory trick: "**O-D-D**" order for the same r: Otto > Dual > Diesel; flip it for the same p_{max}. Diesel bracket $(r_c^{\gamma} - 1)/\gamma(r_c - 1) > 1$ is "the price of burning at constant pressure".

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2020 4(c) Diesel: r = 17, T_1 = 293 K, p_1 = 100 kPa, T_3 = 1773 K, cold-air standard

T_2: $T_1 r^{0.4}$

$= 293 \times 17^{0.4}$

$=$ **910.5 K**

(i) Cut-off ratio: $r_c = T_3/T_2$

$= 1773/910.5$

$=$ **1.947**

(ii) Heat in: $c_p(T_3 - T_2)$

$= 1.005 \times 862.5$

$=$ **867 kJ/kg**

T_4: $T_3(r_c/r)^{0.4}$

$= 1773 \times (0.1145)^{0.4}$

= **745.3 K**

Heat out: $c_v(T_4 - T_1)$

= 0.718 × 452.3

= **324.7 kJ/kg**

(iii) Efficiency: $1 - q_{out}/q_{in}$

= 1 − 324.7/867

= **62.6 %**

B3 Rankine cycle numericals (simple, actual, reheat, regenerative) Pass set

Very likely · 96% Hard

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [13] Same reheat problem (15 MPa, 600 °C, 10 kPa, moisture ≤ 10.4 %): reheat pressure and efficiency.
- **2022 Q5(c)** [18] Ideal regenerative cycle (printed "refrigerative") with one open FWH (feedwater heater): 15 MPa, 600 °C, 10 kPa, bleed at 1.2 MPa. Find the bleed fraction and efficiency.
- **2021 Q6(c)** [15] Ideal reheat Rankine: HP turbine inlet 15 MPa, 600 °C; condenser 10 kPa; LP exit moisture ≤ 10.4 %; reheated to 600 °C. Find (i) reheat pressure, (ii) efficiency.
- **2020 Q5(b)** [20] Ideal Rankine cycle: turbine inlet 3 MPa, 355 °C, condenser 11 kPa. Find the thermal efficiency (steam table extract given in the paper).
- **2019 Q5(d)** [16] Rankine cycle: turbine inlet 4 MPa, 400 °C; condenser at 27 °C. Find the thermal efficiency; then with turbine η = 85 % and pump η = 87 %.
- **2018 Q6(c)** [18] Ideal regenerative cycle with one open FWH: 15 MPa, 550 °C, condenser 10 kPa, bleed at 1.2 MPa. Find the bleed fraction and efficiency.

CALCULATOR **Rankine cycle from table enthalpies (read h from your steam table)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Pump work: $w_p = v_1(p_H - p_L)/\eta_p$ = **0.00101** × (**3000** - **10**) / (**100** / 100) = **3.02** kJ/kg
2. h_2 after pump: **191.8** + **3.02** = **194.83** kJ/kg
3. Turbine work: $w_t = \eta_t(h_3 - h_{4s})$ = **100** / 100 × (**3231** - **2192**) = **1038.6** kJ/kg
4. Heat added in boiler: **3231** - **194.8** = **3036.1** kJ/kg
5. Thermal efficiency: $\eta = (w_t - w_p)/q_{in}$ = (**1039** - **3.02**) / **3036** × 100 = **34.11** %
6. Back work ratio: **3.02** / **1039** = **0.0029**

Defaults: 3 MPa / 400 °C to 10 kPa. Find h_{4s} from $s_4 = s_3$: $x = (s_3 - s_f)/s_{fg}$, $h = h_f + x\,h_{fg}$.

Before you start: the Rankine numerical recipe

State 1 (pump inlet): saturated liquid at the condenser pressure: $h_1 = h_f$, $v_1 = v_f$. **Pump**: $w_p = v_1(p_2 - p_1)$, $h_2 = h_1 + w_p$. **State 3** (turbine inlet): read h_3, s_3 from the superheated table at p and T. **State 4**: $s_4 = s_3$; $x_4 = (s_4 - s_f)/s_{fg}$; $h_4 = h_f + x_4 h_{fg}$. Then $\eta = [(h_3 - h_4) - w_p]/(h_3 - h_2)$.

Keep pressure in kPa in the pump formula: $m^3/kg \times kPa = kJ/kg$.

1. Simple and actual Rankine: 4 MPa, 400 °C, condenser at 27 °C (2019 5d, 16 marks)

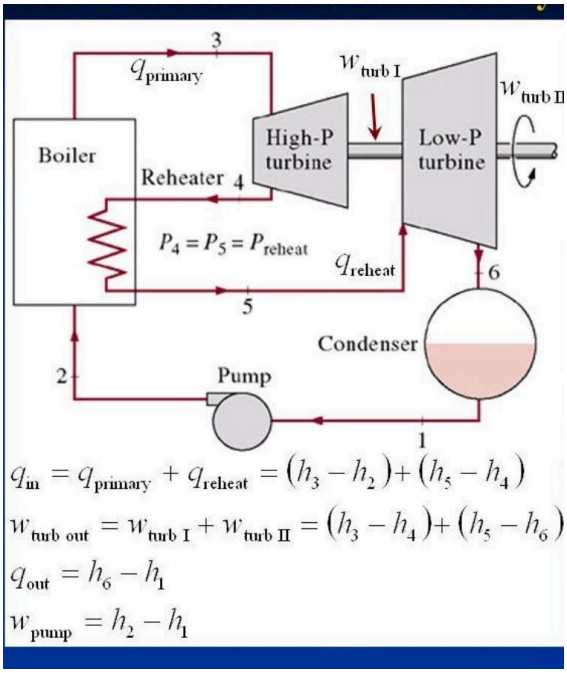
Step (write this)	Why / what it means
Condenser at T_{sat} = 27 °C ⇒ p_1 = 3.568 kPa; $h_1 = h_f$ = 113.19 kJ/kg, v_1 = 0.001003 m^3/kg; s_f = 0.3952, s_{fg} = 8.1191, h_{fg} = 2436.9	Saturated temperature table at 27 °C.
w_p = 0.001003(4000 - 3.568) = 4.01 kJ/kg; h_2 = 117.20 kJ/kg	Pump work.

h_3 = 3214.5 kJ/kg, s_3 = 6.7714 kJ/kg K	Superheated table, 4 MPa, 400 °C.
x_4 = (6.7714 - 0.3952)/8.1191 = 0.785; h_4 = 113.19 + 0.785 × 2436.9 = 2027.0 kJ/kg	Isentropic expansion.
w_t = 3214.5 - 2027.0 = 1187.5; q_{in} = 3214.5 - 117.2 = 3097.3 kJ/kg	
n = (1187.5 - 4.01)/3097.3 = 38.2 %	Ideal cycle.
Actual: w_t = 0.85 × 1187.5 = 1009.4; w_p = 4.01/0.87 = 4.61; h_{2a} = 117.80	Isentropic efficiencies: turbine gives less, pump needs more.
n_{actual} = (1009.4 - 4.61)/(3214.5 - 117.8) = 1004.8/3096.7 = 32.4 %	About 6 percentage points lower.

2. Ideal Rankine: 3 MPa, 355 °C, condenser 11 kPa (2020 5b, 20 marks)

Step (write this)	Why / what it means
At 11 kPa: T_{sat} = 47.7 °C, h_f = 199.65, v_f = 0.001011, s_f = 0.6737, s_{fg} = 7.4417, h_{fg} = 2387.5	From the table extract in the paper (interpolated between 10 and 15 kPa if needed).
w_p = 0.001011(3000 - 11) = 3.02 kJ/kg; h_2 = 202.67	
At 3 MPa, 355 °C (interpolate between 350 and 400 °C): h_3 = 3115.3 + (5/50)(3231.7 - 3115.3) = 3127.0; s_3 = 6.7428 + 0.1(6.9235 - 6.7428) = 6.7615	Linear interpolation: 355 is 1/10 of the way from 350 to 400.
x_4 = (6.7615 - 0.6737)/7.4417 = 0.818; h_4 = 199.65 + 0.818 × 2387.5 = 2153 kJ/kg	
w_t = 3127.0 - 2153.4 = 973.6; q_{in} = 3127.0 - 202.7 = 2924.3	
n = (973.6 - 3.02)/2924.3 = 33.2 %	

3. Reheat cycle with a moisture limit (2021 6c, 2023 5c)



States 1-6 of the reheat cycle (course slide)

Step (write this)	Why / what it means
10 kPa: $h_1 = h_f = 191.81$, $v_f = 0.00101$, $s_f = 0.6492$, $s_{fg} = 7.4996$, $h_{fg} = 2392.1$	Condenser state.
$w_p = 0.00101(15{,}000 - 10) = 15.14$; $h_2 = 206.95$ kJ/kg	
$h_3 = 3583.1$, $s_3 = 6.6796$ (15 MPa, 600 °C)	
Moisture 10.4 % ⇒ $x_6 = 0.896$; $s_6 = 0.6492 + 0.896 \times 7.4996 = 7.3688$; $h_6 = 191.81 + 0.896 \times 2392.1 = 2335.1$ kJ/kg	Start from the exit: the quality limit fixes state 6.
$s_5 = s_6 \approx 7.37$ at 600 °C ⇒ from the superheated table this is at **$p_5 \approx 4.0$ MPa** ($s = 7.3705$); $h_5 = 3674.9$	(i) Reheat pressure: look along the 600 °C row for s = 7.37.
State 4 (4 MPa, $s_4 = s_3 = 6.6796$): $h_4 = 3153.9$ kJ/kg ($T_4 \approx 375$ °C)	Interpolate in the 4 MPa superheated table.
$q_{in} = (3583.1 - 206.95) + (3674.9 - 3153.9) = 3376.2 + 521.0 = 3897$ kJ/kg; $q_{out} = 2335.1 - 191.81 = 2143.3$	
(ii) $\eta = 1 - 2143.3/3897 = 45.0$ %	Answer agrees with Cengel Example 10-4.

4. Regenerative cycle with one open FWH (2022 5c at 600 °C; 2018 6c at 550 °C)

Misprint: the 2022 paper prints "refrigerative"; it means **regenerative**.

Step (write this)	Why / what it means
1: 10 kPa sat. liquid $h_1 = 191.81$, $v_1 = 0.00101$; $w_{p1} = 0.00101(1200 - 10) = 1.20$; $h_2 = 193.0$	Low-pressure pump to the heater pressure 1.2 MPa.
3: 1.2 MPa sat. liquid $h_3 = 798.3$, $v_3 = 0.001138$; $w_{p2} = 0.001138(15{,}000 - 1200) = 15.71$; $h_4 = 814.0$	High-pressure pump.
2022 (600 °C): $h_5 = 3583.1$, $s_5 = 6.6796$; h_6 (1.2 MPa, $s = 6.6796$) $= 2858.8$; $x_7 = (6.6796 - 0.6492)/7.4996 = 0.804$, $h_7 = 2115.3$	Bleed and exit states.
$y = (798.3 - 193.0)/(2858.8 - 193.0) =$ **0.227**	Bleed fraction from the heater energy balance.
$q_{in} = 3583.1 - 814.0 = 2769.1$; $q_{out} = (1 - 0.227)(2115.3 - 191.81) = 1487$ kJ/kg	
$\eta = 1 - 1487/2769.1 = 46.3$ %	Cengel Example 10-5 gives the same.
2018 (550 °C): $h_5 = 3450.4$, $s_5 = 6.523$; $h_6 = 2784.4$; $x_7 = (6.523 - 0.6492)/7.4996 = 0.783$, $h_7 = 2065.3$	Same pressures, lower turbine inlet temperature.
$y = 605.3/(2784.4 - 193.0) =$ **0.234**; $q_{in} = 2636.4$; $q_{out} = 0.766 \times 1873.5 = 1435.9$	
$\eta = 1 - 1435.9/2636.4 = 45.5$ %	

> **Memory trick:** Reheat with a moisture limit: "**work backwards from the exit**" ($x_6 \rightarrow s_6 \rightarrow$ find p at 600 °C). Pump work is always tiny (1-16 kJ/kg); if you get hundreds, you used MPa instead of

> kPa wrongly or used vapour.

B2 Rankine cycle theory: ideal vs actual, efficiency improvement, superheat, reheat, regeneration Pass set Likely · 79% Medium

PAST-PAPER QUESTIONS

- **2023 Q5(b)** [12] Describe the methods of increasing the efficiency of the Rankine cycle with diagrams.
- **2023 Q6(a)** [12] Derive the expression for the efficiency of an ideal regenerative Rankine cycle.
- **2023 Q6(b)** [8] Differentiate between open and closed feedwater heaters.
- **2022 Q5(a)** [10] How can we increase the efficiency of the Rankine cycle?
- **2022 Q5(b)** [7] Why is regeneration done in the Rankine cycle? What is cogeneration?
- **2021 Q6(a)** [10] Why is steam superheated in the Rankine cycle? Why is the ideal cycle modified to the reheat cycle?
- **2021 Q6(b)** [10] With diagrams, describe the regenerative Rankine cycle with an open feedwater heater.
- **2019 Q5(b)** [6] From the T-s diagram, show the deviation of an actual Rankine cycle from the ideal one.
- **2019 Q5(c)** [5] Why is the ideal Rankine cycle modified to the reheat and regenerative Rankine cycles?
- **2017 Q6(a)** [7] Describe the Rankine cycle using T-s and h-s diagrams and give its thermal efficiency.
- **2017 Q6(b)** [8] When is reheating recommended in a steam power plant? How is the reheat pressure optimised?

★ REMEMBER IT

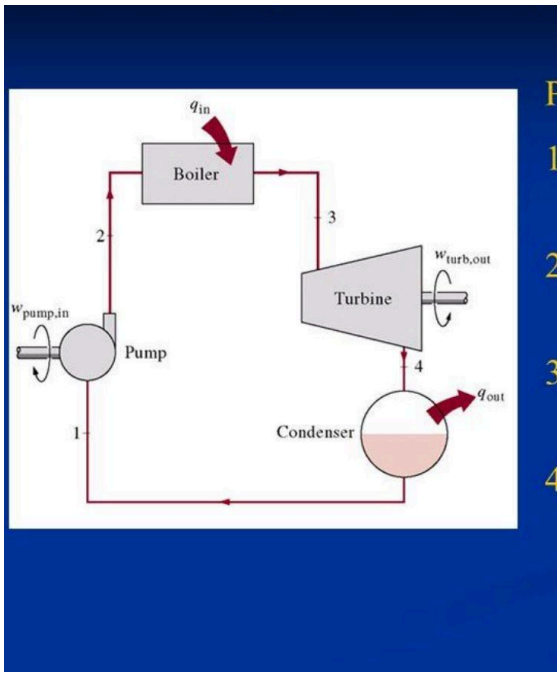
- Raise Rankine efficiency: **raise the average temperature of heat addition** (higher boiler pressure, superheat, reheat, regeneration) or **lower the condenser pressure**. Reheat also keeps the turbine exit dryer (quality > 0.9).

Before you start: the steam power plant in one picture

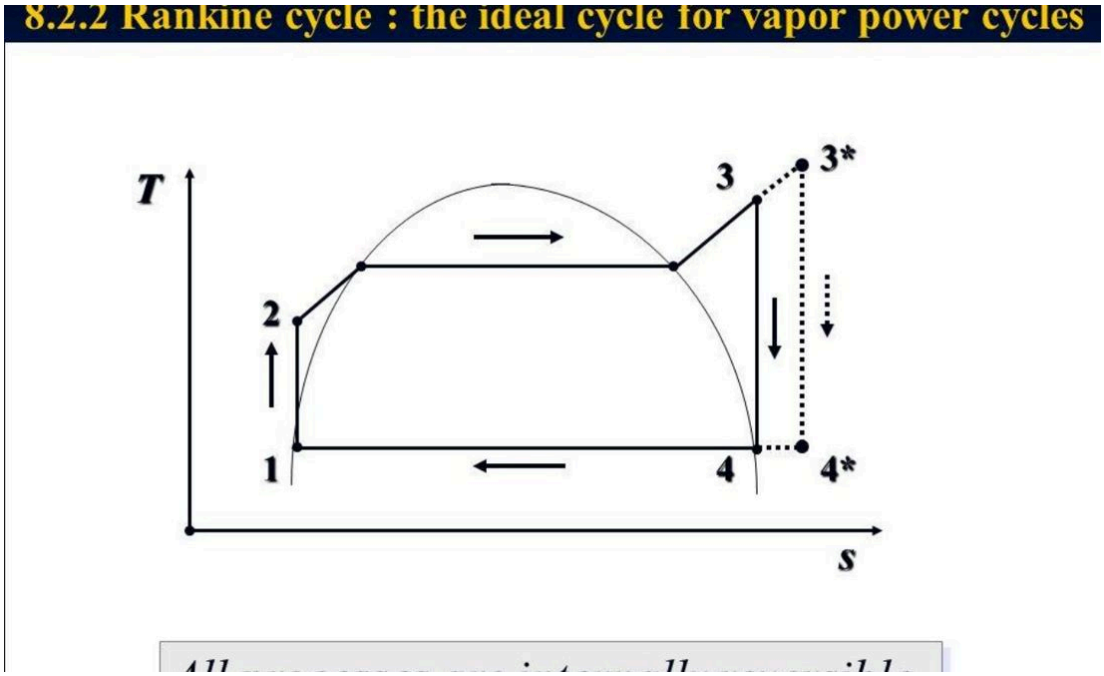
A steam power plant has four parts: a **pump** raises the pressure of water, a **boiler** turns it into high-pressure superheated steam, a **turbine** expands the steam to produce work, and a **condenser** turns the low-pressure steam back to water. The ideal model is the **Rankine cycle**.

Two numbers decide everything: the **average temperature at which heat is added** (higher is better) and the **temperature at which heat is rejected** (lower is better). Every improvement method moves one of them.

Ideal Rankine cycle (2017 6a)



Simple steam power plant (course slide)



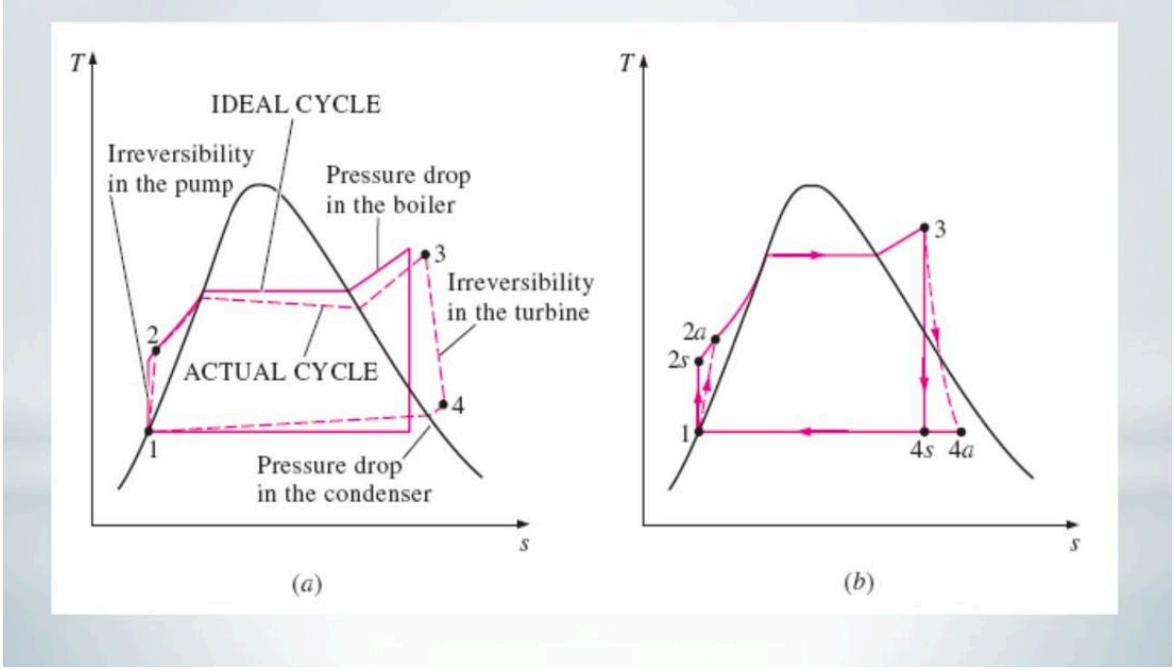
Ideal Rankine cycle on T-s (course slide)

1. **1-2 isentropic compression in the pump**: saturated liquid at the condenser pressure is pumped to the boiler pressure. $w_p = h_2 - h_1 = v_1(p_2 - p_1)$.
2. **2-3 constant-pressure heat addition in the boiler**: compressed liquid → saturated → superheated vapour. $q_{in} = h_3 - h_2$.
3. **3-4 isentropic expansion in the turbine** to the condenser pressure. $w_t = h_3 - h_4$.
4. **4-1 constant-pressure heat rejection in the condenser** to saturated liquid. $q_{out} = h_4 - h_1$.

$$n_{th} = w_{net}/q_{in} = [(h_3 - h_4) - (h_2 - h_1)]/(h_3 - h_2) = 1 - (h_4 - h_1)/(h_3 - h_2)$$

h-s (Mollier) diagram: the same cycle drawn with h vertical: 3-4 is a vertical line (isentropic) whose length is the turbine work, 1-2 is a tiny vertical line (pump work), and 2-3 and 4-1 follow constant-pressure lines. It is convenient because the work and heat appear directly as vertical distances.

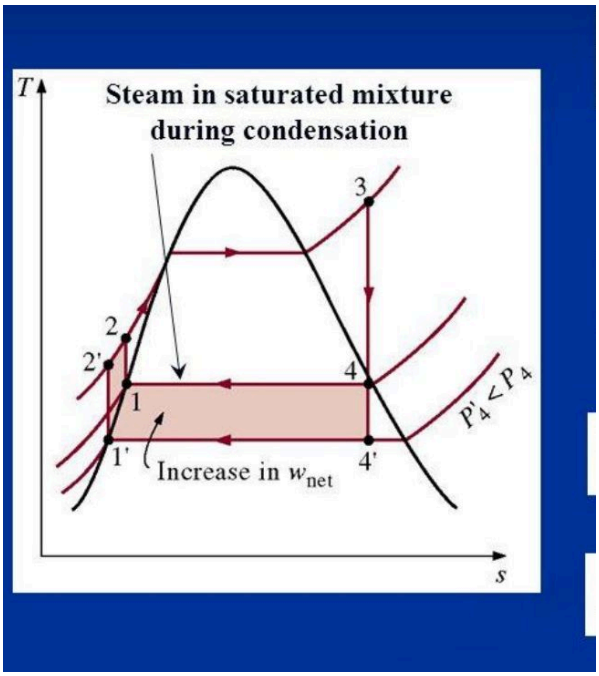
Actual vs ideal Rankine cycle (2019 5b)



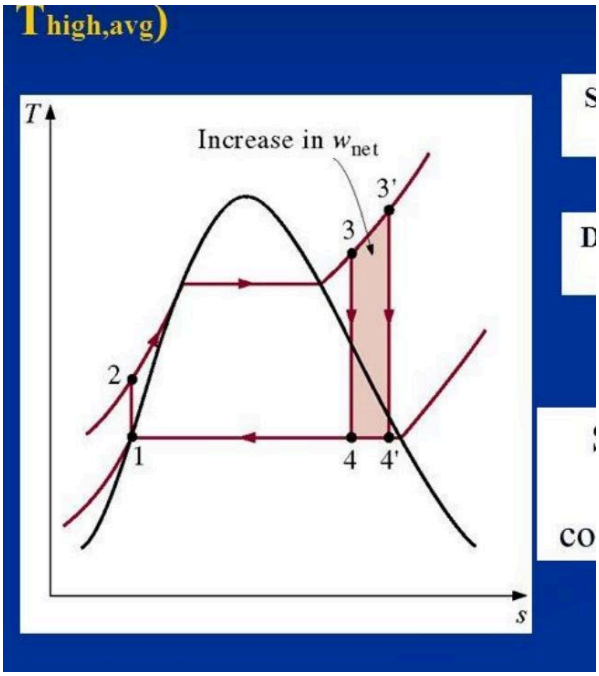
(a) Actual cycle deviates from the ideal one; (b) effect of pump and turbine irreversibilities (course slide)

- **Fluid friction**: pressure drops in the boiler, condenser and pipes, so the pump must deliver a higher pressure and the turbine receives a lower pressure.
- **Heat loss** to the surroundings from steam pipes and turbine, so more heat must be supplied in the boiler.
- **Irreversibilities in the pump and turbine**: entropy rises, so 1-2a and 3-4a slope to the right. $\eta_P = (h_{2s} - h_1)/(h_{2a} - h_1)$, $\eta_T = (h_3 - h_{4a})/(h_3 - h_{4s})$.
- Subcooling of the condensate below saturation (to avoid cavitation in the pump).

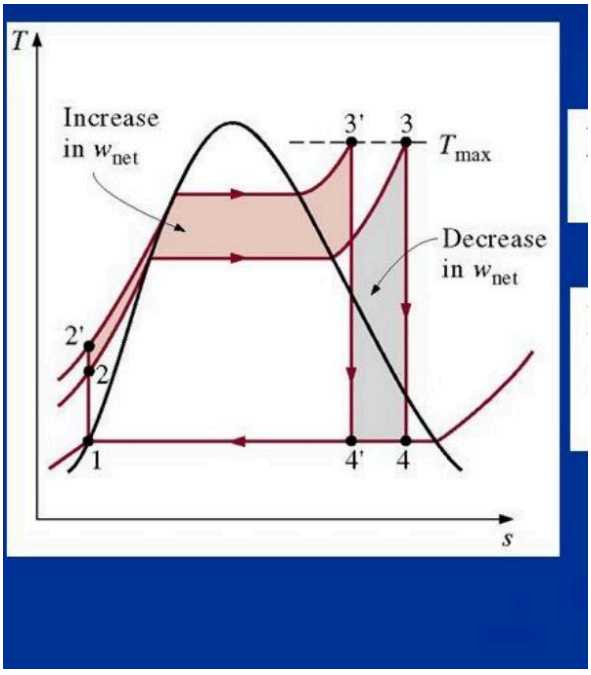
How to increase the efficiency of the Rankine cycle (2022 5a, 2023 5b)



Lowering the condenser pressure (course slide)



Superheating the steam to a higher temperature (course slide)

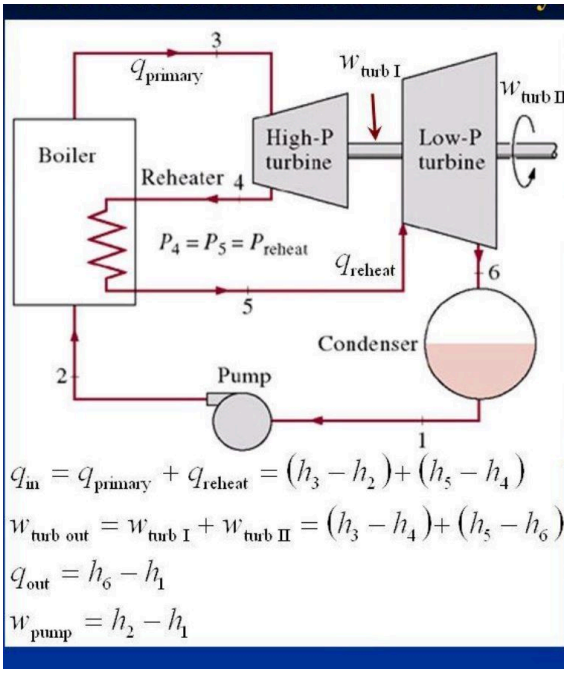


Raising the boiler pressure (course slide)

1. **Lower the condenser pressure** (lower T_{out}): the extra net work is the shaded strip at the bottom of the T-s diagram. Limit: T_{sat} must stay above the cooling-water temperature (≈ 10 °C above); side effects: air leakage into the condenser and higher moisture at the turbine exit.
2. **Superheat the steam to a higher temperature** (raises $T_{m,in}$): more work and less moisture at turbine exit. Limit: metallurgy, about 620 °C.
3. **Raise the boiler pressure** (raises T_{sat} and $T_{m,in}$): more efficient, but for a fixed turbine inlet temperature the exit moisture increases; this is solved by reheating. Modern supercritical plants run above 22.06 MPa.
4. **Reheat** the steam between turbine stages, and **regeneration** (feedwater heating with bled steam): see below.

Why superheat? Why reheat? (2019 5c, 2021 6a, 2017 6b)

Superheating raises the mean temperature of heat addition (higher η), increases the work per kg, and keeps the steam drier at the turbine exit so blades are not eroded (quality should stay above about 0.9).



Ideal reheat Rankine cycle: schematic and T-s (course slide)

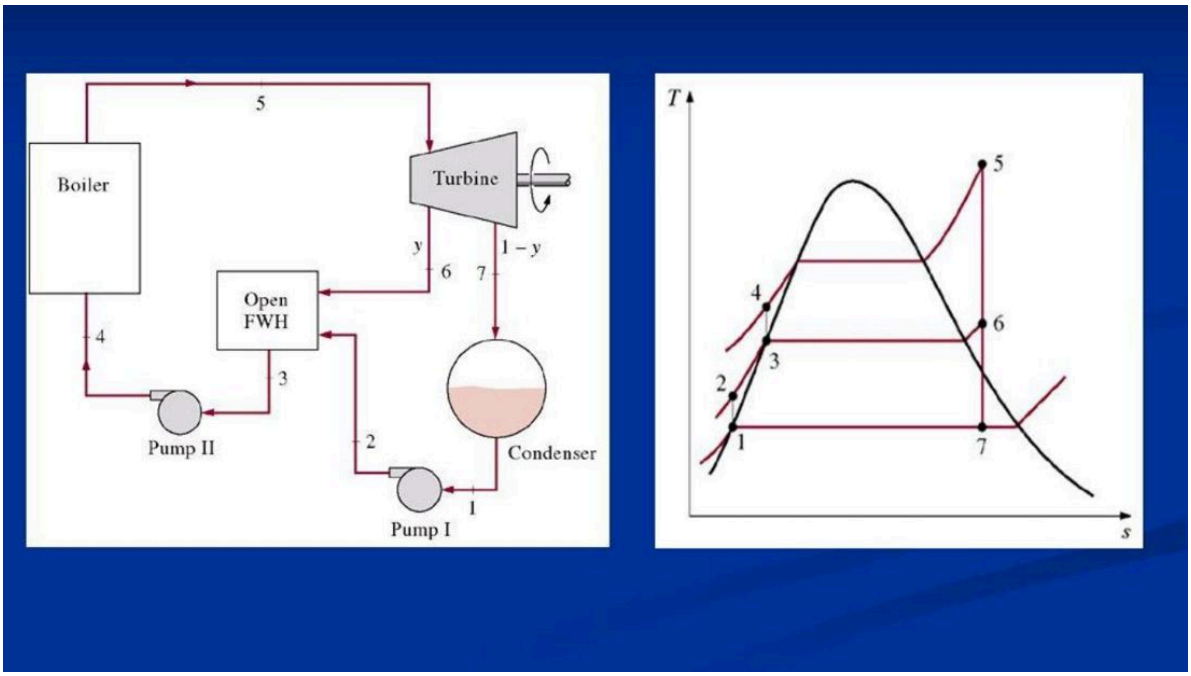
Reheat cycle: steam expands in the high-pressure turbine to an intermediate pressure, returns to the boiler to be reheated (usually to the same temperature), and then expands in the low-pressure turbine. $q_{in} = (h_3 - h_2) + (h_5 - h_4)$, $w_t = (h_3 - h_4) + (h_5 - h_6)$.

When recommended: when a high boiler pressure is used to gain efficiency and the exit moisture would otherwise exceed about 10-12 %. **Optimum reheat pressure:** about **one quarter to one fifth of the boiler pressure**. Lower reheat pressure gives drier exit steam but heat is added at a lower average temperature (efficiency may fall); higher reheat pressure adds little. In practice it is chosen to keep the exit quality at the limit (as in the 2021/2023 numerical), and single reheat raises efficiency by about 4-5 %.

Regeneration and cogeneration (2022 5b, 2021 6b)

Why regeneration: in the simple cycle the feedwater enters the boiler cold (state 2), so heat is added at a low temperature, which lowers T_m. Bleeding some steam from the turbine to heat the feedwater raises the temperature at which boiler heat starts, so η increases. It also de-aerates the water (open heater) and reduces the volume of steam the last turbine stages and condenser must handle.

Cogeneration: production of more than one useful form of energy (e.g. electricity and process heat/steam) from the same energy source; the waste heat of the power cycle is used for a process instead of being rejected, so the utilisation factor (useful work + useful heat)/fuel energy can reach 80-90 %.



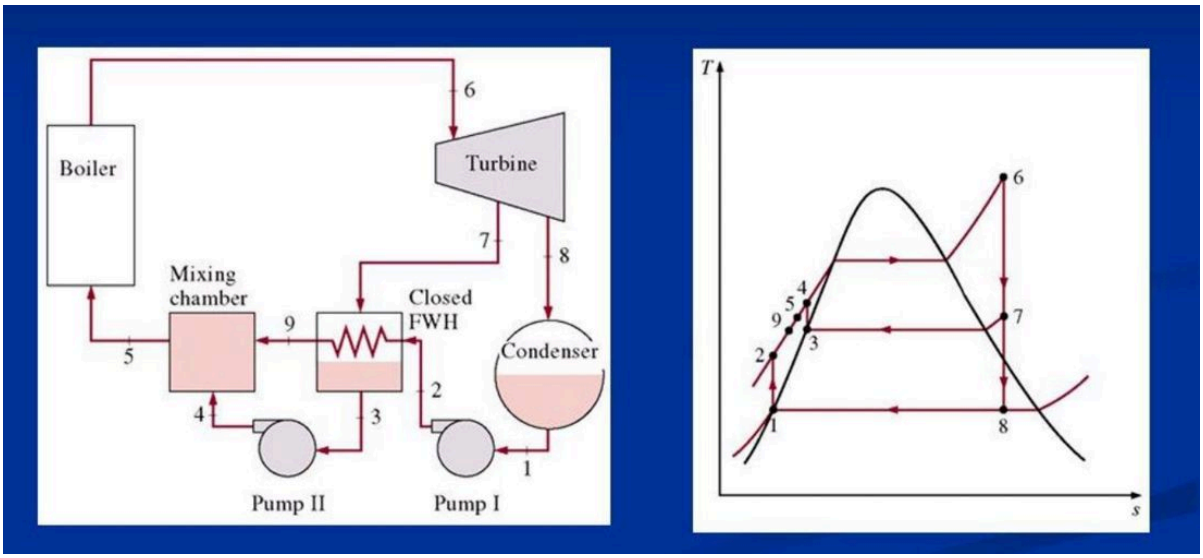
Ideal regenerative Rankine cycle with an open feedwater heater (course slide)

Open FWH (feedwater heater) working (2021 6b): after the low-pressure pump (1-2), feedwater enters a mixing chamber (open FWH) where it mixes with a fraction y of steam bled from the turbine at the heater pressure (state 6). The mixture leaves as saturated liquid at the heater pressure (state 3), and a second pump raises it to the boiler pressure (3-4). For 1 kg through the boiler, (1 − y) kg goes through the condenser.

Derivation: efficiency of the ideal regenerative cycle (2023 6a, 12 marks)

Step (write this)	Why / what it means
Mass balance on the open FWH (per kg of steam through the boiler): y kg bled at state 6, (1 - y) kg feedwater at state 2, 1 kg out at state 3.	Choose 1 kg at the boiler as the basis.
Energy balance (adiabatic, no work): $y\,h_6 + (1 - y)h_2 = h_3 \Rightarrow$ $\mathbf{y = (h_3 - h_2)/(h_6 - h_2)}$	Energy in = energy out for the mixing chamber.
$q_{in} = h_5 - h_4$	Boiler heats 1 kg from state 4 to state 5.
$q_{out} = (1 - y)(h_7 - h_1)$	Only (1 − y) kg reaches the condenser.
$w_t = (h_5 - h_6) + (1 - y)(h_6 - h_7)$; $w_p = (1 - y)(h_2 - h_1) + (h_4 - h_3)$	1 kg expands to the bleed pressure; (1 − y) kg continues. Two pumps.
$\mathbf{\eta = (w_t - w_p)/q_{in} = 1 - (1 - y)(h_7 - h_1)/(h_5 - h_4)}$	Equivalent forms. Compared with the simple cycle, q_{in} falls more than w_{net}, so η rises.

Open vs closed feedwater heaters (2023 6b)



Closed feedwater heater: bled steam and feedwater do not mix (course slide)

Open FWH (direct contact)	Closed FWH (surface type)
Bled steam and feedwater **mix** directly	Heat transferred through tubes, streams **do not mix**
Both streams at the same pressure; outlet is saturated liquid at heater pressure	Streams can be at different pressures; feedwater may stay at boiler pressure
Needs a pump after each heater	Needs no extra pump per heater (drain is trapped back or pumped forward)
Simple, cheap, good heat transfer, also de-aerates the water	More complex and costly (tube bundle), lower heat transfer (terminal temperature difference)
Feedwater leaves at the saturation temperature of the heater	Feedwater leaves slightly below the bled-steam saturation temperature

Memory trick: Rankine improvements: "**Low C, High S, High B, Re-Re**" = Lower Condenser pressure, Higher Superheat, Higher Boiler pressure, REheat, REgeneration. Bleed fraction: "**y = rise needed / rise available**" = $(h_3 - h_2)/(h_6 - h_2)$.

B5 Brayton (gas turbine) cycle Next to add Likely · 64% Medium

PAST-PAPER QUESTIONS

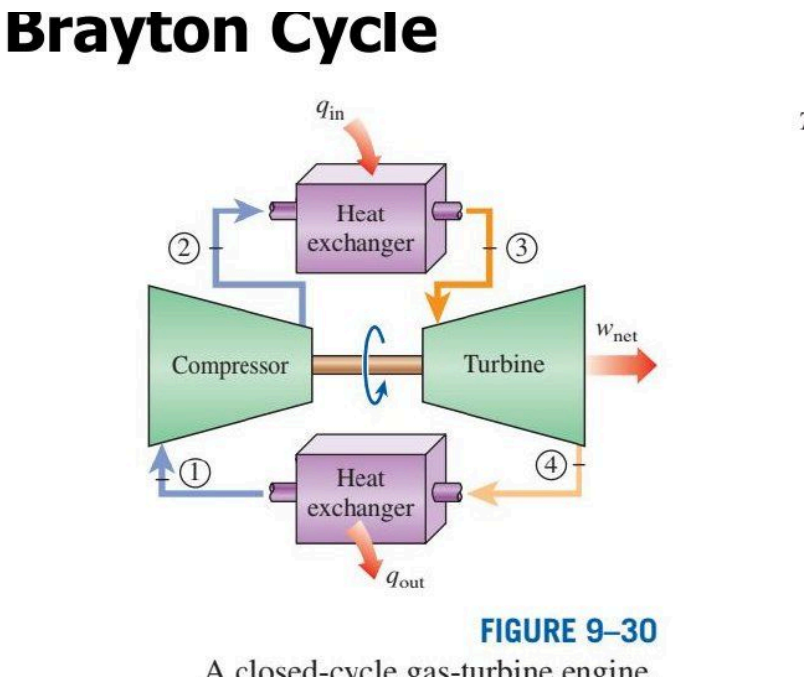
- **2023 Q6(c)** [15] Ideal Brayton cycle: pressure ratio 8, 300 K and 1300 K. Using air-standard assumptions find (i) compressor and turbine exit temperatures, (ii) efficiency.
- **2022 Q6(c)** [15] Air-standard Brayton cycle between 300 K and 1200 K and pressure limits 101 kPa and 505 kPa. Find (i) efficiency, (ii) compressor work, (iii) turbine work (kJ/kg).
- **2021 Q5(c)** [12] Brayton cycle: pressure ratio 8, 300 K compressor inlet, 1300 K turbine inlet; η_C = 80 %, η_T = 85 %. Find (i) exit temperatures of compressor and turbine, (ii) back work ratio, (iii) thermal efficiency.

★ REMEMBER IT

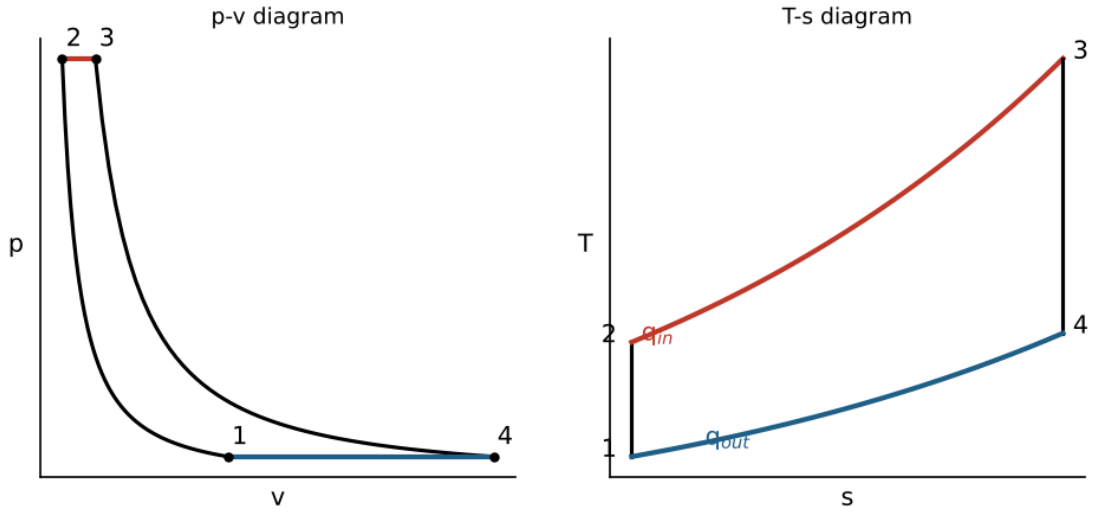
- Brayton: $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$; write $x = r_p^{0.2857}$ once; $T_2 = T_1 x$, $T_4 = T_3/x$.

Before you start: the gas turbine

A gas turbine has a **compressor**, a **combustion chamber** and a **turbine** on one shaft. Air is compressed, fuel is burned at constant pressure, and the hot gases expand through the turbine, which drives the compressor and produces net power. The ideal model is the **Brayton cycle**. **Back work ratio** $r_{bw} = w_{compressor}/w_{turbine}$: in gas turbines it is large (40-80 %), so component efficiencies matter a lot.



Open gas-turbine engine (course slide)



Brayton: 1-2 isentropic compression, 2-3 heat at constant p, 3-4 isentropic expansion, 4-1 heat rejected at constant p

Ideal Brayton cycle on p-v and T-s

$\eta_{Brayton} = 1 - 1/r_p^{(\gamma-1)/\gamma}$ $T_{2s}/T_1 = T_3/T_{4s} = r_p^{(\gamma-1)/\gamma}$

1. Ideal Brayton: r_p = 8, 300 K, 1300 K (2023 6c)

Step (write this)	Why / what it means
$r_p^{0.2857} = 8^{0.2857} = 1.811$	Cold-air-standard, γ = 1.4.
$T_2 = 300 \times 1.811 = 543.4$ K; $T_4 = 1300/1.811 = 717.7$ K	(i) Exit temperatures.
$\eta = 1 - 1/1.811 = 44.8$ %	(ii) With Cengel's air tables (variable c_p): 544.3 K, 770.1 K, 42.6 %.

2. Actual Brayton: η_C = 80 %, η_T = 85 % (2021 5c)

Step (write this)	Why / what it means
$T_{2a} = T_1 + (T_{2s} - T_1)/\eta_C = 300 + 243.4/0.80 =$ **604.3 K**	Compressor needs more work than ideal.
$T_{4a} = T_3 - \eta_T(T_3 - T_{4s}) = 1300 - 0.85 \times 582.3 =$ **805.0 K**	(i) Turbine gives less work than ideal.
$w_c = 1.005 \times 304.3 = 305.8$; $w_t = 1.005 \times 495.0 = 497.5$ kJ/kg	
(ii) $r_{bw} = 305.8/497.5 = 0.615$	61.5 % of the turbine work drives the compressor.
$q_{in} = 1.005(1300 - 604.3) = 699.2$; **(iii) $\eta = (497.5 - 305.8)/699.2 = 27.4$ %**	Cengel Example 9-6 with air tables: 598 K, 853 K, 0.592, 26.6 %.

3. Brayton between 101 and 505 kPa, 300-1200 K (2022 6c)

Step (write this)	Why / what it means
$r_p = 505/101 = 5$; $5^{0.2857} = 1.584$	
$T_2 = 300 \times 1.584 = 475.1$ K; $T_4 = 1200/1.584 = 757.7$ K	
(i) $\eta = 1 - 1/1.584 = 36.9$ %	
(ii) $w_c = 1.005(475.1 - 300) = 176.0$ kJ/kg	

(iii) w_t = 1.005(1200 - 757.7) = 444.5 kJ/kg	Net work 268.5 kJ/kg; check: 268.5/(1.005 × 724.9) = 36.9 %.

Memory trick: Brayton = "**one number, one formula**": everything comes from $r_p^{0.2857}$. Multiply T_1 by it, divide T_3 by it.